I have been crafting my pixel art skills for years and years and years. So, today in a single video, I'm going to teach you everything you need to get started with and actually not suck at pixel art, especially when it comes to making pixel art for games. We're going to start at ground zero with software and setup and then transition into line work and colors and all the fun stuff. First things first, you need a place to actually do the pixel art. So, you need a software. A sprite is the go-to for me and many other pixel artists because it has super nice UI. You can do animations really nicely. It's got sprite sheets, tile sets, a lot of things built for making games. It is $20 technically, but because it's open source, you can find it online for free pretty easily. But you didn't hear that from me. What? Who told you that? Libre Sprite is kind of the like go-to recompiled version. It's just an older version of a sprite. It's basically the same thing. Try that out. See if you like it, and then eventually, you know, support the developers, be good. But if you're not feeling a sprite, you can usually configure Photoshop for pixel art pretty easily. Or you can go to this website called Piscll. Nice. So now you have a place to actually draw things. But we're actually not going to be picking the colors to draw with ourselves. In fact, right now the goal is to avoid swatching or mixing our own colors completely. And that's because we're going to be automating the entire process using AI. I'm just kidding. A lot of pixel art tutorials out there are going to try and teach you color theory and how to mix and shift your colors and all that. My advice is to not really worry about that right now because you don't have to. Instead, you can find a palette online that someone else made that's designed to look good so that you can actually focus on the fundamentals and get good at drawing and then eventually if you want to learn color the 3, you can do that. So, this website called Lowspec is going to be our best friend. They've got all kinds of color palettes. You can sort by the number of colors that you want and it's all specifically for pixel art, which is really good because pixel art palettes usually need more contrast than traditional palettes. And right now, we're just looking for a palette that has at least three shades per color. And that's because when we shade our pixel art, we're going to want three colors per material as a general rule. We're going to talk about that later. But as you're browsing low-spec or you're you're looking at my screen as eyebrows lowspec, there's some really cool stylized palettes here that are really good for game jams in my opinion when you're you need a strong identity. But for now, we just want a nice big general palette. The one that I used is linked in the description. If you're an A-brite, then there's actually a bunch of pre-made palettes in A-prite that you can download. You can just click this button and then I would recommend something like this one. Or you can find one on lowspec, import it, and then set it to be your default palette by clicking these three lines and then save as default palette. Perfect. Now we have all the tools we need to actually draw something. My workflow when it comes to doing pixel art is to start with the outline of what I'm doing, then fill in that outline with solid base colors and then shade it. That could mean starting with construction lines for your outline and then using those to map out the proportions of what you're drawing. But another approach is to map out the basic shapes that make up whatever you're drawing and then you can mold that into your outline, which gives you a really nice silhouette. Whatever strategy speaks to you, try it out. But when you do start working, you might notice that your pixels look pretty thick. These lines look a little bit chunky, especially when we're trying to use them to make a nice clean outline. That's because we're using something called doubles. Doubles are the extra pixels that are generally unnecessary and they make your pixel art look really sharp because it's a bunch of right angles. When we get rid of our doubles, our pixel art is a lot more readable and it's a lot cleaner. In a sprite, you can actually toggle this setting called pixel perfect. And now all of your lines will not automatically have doubles on them. Now, sometimes we do want doubles in order to bring more structure to a certain shape or as a style choice. It usually comes across as really retro and bold, but you got to keep it intentional. If it's inconsistent, then it's just going to look messy. And speaking of messy lines, look at this. Now, this line doesn't have any doubles, but it does look really jagged, especially when we look at it from afar. So, how do we fix it? Great question, subscriber, because I know you're subscribed, right? Well, we want to make sure that each pixel here is smoothly transitioning to the next one. An easy way to do that is to count your pixels and repeat the same segment length on either side of a curve, usually keeping it symmetrical. Like this one follows a nice little pattern, so it looks really nice. It's buttery smooth. And even if you don't have a big curve, if it's a small line, you can still apply this principle to make it look cleaner. It's really all about just gradually building up or down the number of pixels that you're using to make your line. Whenever I'm drawing a curve, even if it's barely a curve, I start with the two opposite ends of what I'm joining, and then fill in the pixels to make them meet in the middle. So, now you know how to make smooth and clean pixel art without using doubles and without having jagged ugly lines. Now, remember how I said earlier we want three colors per material when we're coloring things in? That's because we have a base color, a shadow, and then a highlight. Now, if you're using a palette, ideally you already have those, so you don't have to worry about mixing it yourself, like I said earlier. But if you do want to level up your coloring even more, consider shading or highlighting with a different color than your base. For example, if we look at the portrait of Penny from Stardy Valley, who is the most boring NPC in the entire game, by the way. Sorry, not sorry. Her base hair color is orange because she's a ginger, but then the highlights are yellow and the shadows are purple. It isn't just light orange, dark orange, and that makes the colors look a lot more vivid. So, if you go to Google and then search color shading guide, you're going to get a bunch of examples of what colors to shade what. Okay, I know that I said I wasn't going to talk about color theory, but I feel like I need to give you a little bit better of an explanation than that. You totally can just Google it, but the reason that this works is because of analogous colors. And those are just any colors that are next to each other on the color wheel. So, if I'm coloring with red and that's my base color, I can highlight with orange and then I can shade with purple because those are analogous colors of red. And if I'm doing green, I'm going to highlight with yellow and then I'll shade with blue. But no matter what, I like to start by doing my base color and then I do my shadows and then I add my highlights. But wait a second, Juniper, this is ugly. You are so right, Queen. And that's because I'm shading without thinking about where my light source is coming from. So instead, I'm just kind of wrapping shadows around the outline of my art, which doesn't look good, and it kind of looks like a pillow, which is why this is called pillow shading. It's really easy to do on accident if you're not thinking about a light source, because how else do you know where to put the shadows? So, if I take this same piece of art, but now I decide that the light is going to be coming from this angle. Now, I kind of have an idea of where I should put my shadows. So before you really start shading anything, you should ask yourself where the light is coming from. And if you're making art for a game, you probably want to keep that angle of light consistent across all of your sprites. Now, a lot of pixel art games are top down or usually drawn in a 3/4 view or an oblique view. That means you can usually see the top of what you're drawing as well as the front of it. So the light source in these games is usually coming from the top of the screen and then casting forward and down. So, there's light on the front of everything, but it's ultimately most well-lit on the top. This kind of thinking is also what helps you think about perspective when you're drawing. If you're drawing a table and a chair for your game, you want to make sure that the angle of light is the same for both of those. But you also got to ask yourself, how much of each object should I see? Even if my chair is shorter than my table, I should still see the same amount of the top of the chair as the top of the table. Unless, of course, the table is more narrow than the chair, and I want to show that. All of those details are what makes pixel art really shine. And if you're watching this because you want to make pixel art for games, then think about today's sponsor, Game Maker. Game Maker is a game development engine designed to help you make highquality 2D games quickly, whether they're pixel art or not. This is a game called The King is Watching, and it was made in Game Maker. It's a rogike kingdom builder that came out about a month ago. All of this beautiful pixel art, the graphics, the UI, the gameplay, it's all powered by Game Maker. And the same thing goes for this co-op island survival game called Tinkerlands. It's fast, it's efficient. You're not going to be spending 10 years following tutorials on how to navigate the engine. Cuz look, I've been using Game Maker for a long time now. Back in the day, 4 years ago, I started with one of the many Game Maker tutorials that they have published on their channel, like this RPG one. I taught myself pixel art along the way. I joined their Discord server where I got all the help that I needed and here we are cuz you single-handedly all on your lonesome can release a game using Game Maker. So download Game Maker for free by the way using my link in the description. So when you're shading, you want to think about light source. You want to think about perspective. But shading doesn't even necessarily mean that one color makes up all of your shadows. There's a lot of cool ways you can shade or even manipulate the outline of your piece to exaggerate light. Dithering, for example, is when you create kind of a pattern with two colors to give the illusion of a third color. It's usually like a checker pattern between two colors, but it doesn't have to be. This is how they shaded things on older consoles when they literally didn't have space to add another color. But nowadays, it can be a really cool stylistic choice. It can show texture, or it can break up a really big area of shadow. You've also got anti-aliasing as an option, which kind of smooths or blends lines. You grab a color that's in between two colors you're wanting to blend together and place those pixels along the corners, only the corners of your piece. You're kind of creating a gradient that almost looks like a shadow, but one that is very soft and very subtle. Anti-aliasing kind of makes pixel art look less like pixel art since you're removing a lot of the sharpness. It's really up to you whether you like it or not. It definitely takes some getting the hang of, but in my opinion, I think it gives art a really nice polished look. Now, another option when you're coloring is to change the way you're coloring your outline. Whenever I make pixel art, I usually start with like a solid black outline around whatever I'm drawing. But by the end of it, it's almost always recolored somehow or I just got rid of it. Because first off, your outline doesn't have to be black. When you're looking at Stardew Valley, you're going to notice that everything has a different color outline, and it's pretty much just a darker version of whatever the object is. Plants have dark green outlines. wood has dark brown or dark orange outlines. Or you can look at something like Omorei where they use this really bright purple outline and that gives it a really dreamlike and iconic style. In a game environment, you can also play around with an inconsistent outline style in order to draw focus to different things. Look at how the player in Celeste has an outline, but nothing else in the background does. The house still has shading. And also, look at how it uses brown, but then the shadows are purple. That's that's the color thing I was talking about earlier. But looking at the big picture, only the interactable objects have an outline. Even the ground and the platforms don't have outlines because that would make the characters stand out less. It's a similar story with AK's a lot. But this time, the characters have outlines and so does everything else in the world except the player, the items, and the NPCs. They have a black outline and everything else is a dimmer, darker brown. So, keeping things the same, like lighting and perspective, is good, but you have a lot more freedom with outline and color, and you can actually use it to guide the player and give your game an identity. It's definitely worth it to look at some of your favorite pixel art games and try and pay attention to what they're outlining and how they're guiding your eyes to the right things and not making things too busy. But that's just that's some food for thought if you're making a game. But anyway, another way that you can color your outline is by shading it depending on where the light is coming from. The side of an object that's closest to the light could be a lot lighter and then, you know, vice versa for the shadowed side. When it comes to creating an art style, though, you can also think about the size of your outline. You can look at how the earlier version of Tinkerlands used to look with a really thick outline or Hyperlight Drifter, which has no outline. The art in Hyper Light Drifter is so pretty, but it also must have been so hard because to not use an outline means that first off, your character has to be super readable in the environment still, which usually means making them significantly lighter or darker than their surrounding space. And also, you need to have really defined shapes so that it's still readable without a clear outline. Anyway, so far we've talked about clean line work, perspective, shading, colors, everything to consider when you're making an art style. So, now it's time for a bunch of miscellaneous tips and tricks that I don't really know how to categorize, but I think they're important nonetheless. Tip number one, if your art looks bad, and it's not because of the line work, take a look at your colors. Do you have more than three colors per material? And if so, is it necessary? I did a video forever ago. God, I I look like a fetus, but I did a video a while ago where I critiqued the pixel art that you guys sent in, and a common problem was having way too many colors. The beauty of pixel art comes from its boldness and from its simplicity. So using too many shades of the same color usually just makes something look really noisy and harder to look at. So unless you're doing that like super intentionally, probably not great. Tip number two, use references. Same as traditional art, right? If you're making a pirate, you probably want to look at some pictures of pirates. However, we do live in 2025, unfortunately, which means if I'm drawing a horse and then I search horse, I get to play the game of is this a real horse? So, my recommendation to circumvent that is to go to Pinterest, which in my opinion has a lot less AI for now. Tip number three, zoom out. Speed paints of me doing pixel art are genuinely schizophrenic because every 3 seconds I am zooming in and out of the canvas. But the reason I do that is because it matters a lot more how something will look from far away. And it can look very different up close versus far away. You want to try and make sure that you're refreshing your eyes with the size that you'll actually be seeing something on screen, especially if it's a small character in a game. In a sprite, you can actually press F7 and that'll give you a little preview window. And then in that window, you're able to zoom out. So you don't have to be doing it manually every 5 seconds like me. Tip number four, you can't do art just because you want to get good at it. Nuh-uh. You got to do it for the love of the game. Same thing with making YouTube videos, honestly, or making a game. You can't do it because you want the output of having a finished product because then you're not going to have any motivation when you start learning and suck and realize you suck because you just started learning. Having goals, great. Expecting high things of yourself, fantastic. But realistically, you will not put in all the time that it takes to get to that point unless you can also learn to like making art, not just having made good art. But all of that being said, you got it. You're going to be fine. Hopefully something helped you out in this video. And if it did, maybe subscribe. That would that would be pretty cool. Or you can join my game development Discord server if you want a little community where you can share your stuff. And seriously, thank you so much for watching. I hope you have a lovely day and I'll see you in the next one. Byebye. [Music]

it's been four years since i picked up pixel art for the first time so let's talk about how you can become a talented pixel artist too i've learned quite a lot over the years but i don't want you guys spending four years getting to where i am when you learn a new skill you usually don't understand how to maximize your potential so sometimes your talents can develop very slowly if you aren't following the right advice or just don't know how to progress efficiently let's fix that and get you guys excited about starting or progressing your already existing journey on becoming an artist for my starting pixel artist i would like to begin with understanding how pixel art actually is created the fundamentals if you will pixel art is very different from most other art styles as you are not necessarily drawing with a brush or stroking a canvas you are placing down blocks square by square on a much smaller more restricted canvas it gives pixel art a slightly more technical approach where as long as you are understanding the concepts you can almost immediately replicate and start creating with those fundamental techniques for understanding pixel art we need to understand the principle of basic shapes and form you should start out by breaking real world objects down into shapes for example turning a bed into a square or a berry into a circle shapes are found everywhere so again try to envision these things straight away in your mind before bringing it onto a canvas beginners should also understand how outlining objects work in my last pixel art video i talked about the different kinds of outlines you might find in different pixel art styles but not on actually how to outline a sprite try and understand where you need to smooth off your pixel edging and where you should keep hard edges to keep things simple when you're using sharp and rigid objects you might find some hard edging other than that smoothing it off is much better i see way too many artists double outlining their edging where if they just were to get rid of all of the corner pieces it would make the image feel much more rounded and less flat we are trying to create 3d images on a 2d surface so the more ways we can fake depth the better our drawing is going to look when starting pixel art you should stick to 8x8 or 16x16 size canvases for your sprite work this will make the amount of space you have to work with and your artistic decision making very beginner friendly causing your art to be less challenging and actually achievable detail is the hardest part in any art style so do yourself a favor and don't stress yourself with the hard stuff in the early stages you'll progress faster mastering the basic concepts over failing to make masterpieces right off the bat and making yourself feel unaccomplished okay so you have a decent understanding of what pixel art is about and how to form basic shapes into tiny sprites my best tip i can give to intermediates is making sure that while imitating what other professional pixel artists do to achieve certain effects and styles you like make sure to find references from the real world as well when i was trying to get better at pixel art i would just follow tutorial after tutorial on how to create something i needed to know how to do but i ended up just basically copying their work without actually understanding what they were doing or why always question the artist why are they shading this object in the way that they are how are they casting light onto the object why are they a better artist than you are now i understand some people might take this as bad advice as every artist has their own style and no one is better than anybody else as it is all up to personal interpretation but to be honest when you are starting a new skill having mentors and people you look up to is not a bad thing when you get to an expert level of understanding you can branch off and do your own thing but any skill takes time to manifest so just don't take it to heart if somebody's style is objectively a bit better than yours find inspiration not jealousy a great technique for intermediate artists to start picking up would be to look into techniques like dithering and anti-aliasing these are techniques you need to know when starting to get into bigger canvases and adding more detail into your work as i said earlier being a good artist is all about faking something that isn't there making you feel there is a rounded aspect to your flat drawing or adding darker areas without actually using more colors to your art are perfect examples of this the main purpose of dithering is using the technique of pixel patterns to fake shading and darken an image in a smoother transitional way for anti-aliasing this is the technique of using pixels to help smooth out a curved object think of anti-aliasing as like a way to create in-between pixels without actually creating in-between pixels these can be used in lighter pixels when doing highlights and in darker pixels when shading this is just a technique that will really make your image feel detailed and properly shaped for those advanced curved objects here's some more example work i've done at an intermediate level of quality again just because something is made at an easier level of detail doesn't necessarily make it better or worse it's just a scale i'm using to distinguish easy from challenging tactics and skills needed to accomplish the work finally we have the pro tips for those of you looking to really bring your art to the next level and maybe even get paid as a professional in the field for those of you that are interested i actually do work from time to time as a freelancer it's one of my jobs i take on being a self-employed developer when i'm in need of some extra income so remember getting good at a skill is worth the grind as basically all of my skills allow me to find work and pay like a professional i never went to school for anything i'm doing in my life at this point so i mean you can do it too if you have the patience and the determination to succeed get educated but also remember that your actual experience and work is much more valuable when trying to find work in the internationally connected society of contract work and self-employment anyway for those of you looking to break into the gaming industry a super important skill is learning how to animate your pixel art for game characters particle effects and all the other elements that go into creating video game art i'll use an example of this little robot character i made a couple months back to start you guys off in the right direction animation is definitely not an expertise of mine but i mean at the end of the day you aren't aiming for perfection as a pixel artist your primary job is to portray a visual across to someone so they can interpret it as what you want them to see if you look at this role for example each frame of the animation looks pretty crappy but when you throw them all together it shows the interpretation of your character leaping onto the ground and into a rolling ball objective complete animation will require an entire video all on its own but just try looking into the concepts of squash and stretch to make your animations feel fluid and less robotic also try to just draw your keyframes first and fill in the extra frames when you have your main poses picked these are basic things you've probably heard many times but they are still super important so just make sure to keep it in mind next is on the topic of materials using different techniques to shade or highlight different materials will help you get the information across that something is smooth rough sharp furry or whatever other texture you can think of you get the idea my final tip for today is on understanding perspective perspective is everything when it comes to truly fooling people into believing your image has depth this piece i did a few weeks ago is a great example of me trying to have a consistent perspective and viewing point use grids and draw lines from points of interest to keep your perspective consistent and believable perspective and sizing is super key to having something feel realistic if you want to learn about proper color techniques different types of outlines and even more general tips for pixel artists i have linked my previous how to pixel art video in the carded video and in the description below the video is kind of blowing up right now and is definitely still extremely relevant for people looking to learn pixel art i wanted to make sure i didn't overlap too many tips and advice so just check out the other video if you need some more help like the video to help promote it to more aspiring artists out there and subscribe if you are enjoying the style of content feel free to check out my game i'm releasing on steam and hopefully consoles later this year anyone can become great if they put their all into their efforts don't accept anything less than your best and then shoot higher

I need to learn pixel art fast, so I watched the only pixel art guide you need and 15 minutes later made this. Okay, dang. I really wanted my game to have pixel art, but I wanted stuff that looks like this. And I'm beginning to think it probably takes longer than 15 minutes and one tutorial. Like, I think it'll take me years to get [music] this good. But I don't have years. I barely have a week. I'm new to gamedev. Just getting the game to open feels like a win right now. I got to work on that. But I also need to know if I'm even capable of making pixel art for my pixel art game. So, I'm giving myself to the end of the work week and it's already Tuesday. For some reason, I genuinely think I can pull it off. I don't know where that confidence is coming from. Certainly not my previous work. So, I'll start simple and make something harder each day. The problem is when it's your own art, it's kind of impossible to tell whether [music] it's good or not. So, luckily my wife is an art teacher, so she's agreed to grade each piece brutally. Just like school, I'm aiming for C's. A pass is a pass. But don't tell her I would really like a B or a two. So, day one. Today, I'm starting with objects because look at this. [music] It's supposed to be a bottle. I I don't know if that's obvious. It's actually really important that I can make pixel art drinks because the whole game is about running a bar for bees to get them buzzed so that they go home and make more bees, increase the bee population and such. It's called Population Buzzed. There will potentially be hundreds of drinks in this [music] game. So, if I can't get good at making pixel art drinks, I don't really have a game. So, I watched this video by what on the basics of how to use a sprite because today is the first time I've ever opened this piece of software. And I clearly [music] have no idea what I'm doing. He goes over the basic tools and gives me the confidence to get going. Uh, too much confidence because I have apparently bitten off a little more than I can chew. The blind faith I had in myself is evaporating quickly because for some unknown reason, I decided that the first drink I would attempt would be a glass of beer. Here's my reference image, and I'll let you spot the problem here because I am clearly too dumb to see it for [music] myself. Oh. Oh, I'm just going to get started with a brand new art medium with software I've never used. My art experience itself is lacking. And I figure I'll start with [music] a glass of beer. Not a bottle, not a can. A big old seethrough glass of beer. And suddenly I'm expected to have expert level knowledge on how to portray refraction through pixel art. kill. Nope. No biggie. Give me Give me 20 minutes. Using all 15 minutes of prior experience, I whip up this little number. I mean, it looks like beer, which is wild. For my first real piece of pixel art, I'm actually pretty impressed with this, but it's it's not very polished. It's [music] lacking the finesse that I would expect to see of a quality pixel art game asset and I don't feel comfortable moving on to the harder stuff. Uh, looking at this, it's just not good enough [snorts] is what I'm saying. I can't continue with this art challenge until I get better because tomorrow I'm going to create some UI elements. Thursday, I'll [music] make an actual character. And on Friday, I mean, I want to create a whole [music] izakaya bar scene. And there's no shot I can do any of that until I nail down a basic drink. No pressure. This is a really easy challenge. I wasn't going to submit this one to my wife for grading, but she saw it anyway. [music] And and actually gave it a C. She said it's all right for a quick like passing graphic, hence the C. But if it's going to be a main asset, then it [music] would need a little bit more work. Honestly, that feels generous. But hey, it's a pass. But I can do better. And to do better, I must do worse. Like, let's lower the bar a little. Let's try something that isn't as difficult. Like, like, let's do a sake ceramic bottle and cup. Something solid. So, I spent some time watching some videos by Hannah Pigs that that go over some things like basic shading and line work. I can highly recommend her videos. They built up my absolutely crushed confidence levels to something a bit more reasonable. And back to it I get. Hopefully, I can pull this off and create a sake bottle I'm happy with so that I don't have to cancel my entire game. I started off with the basic shapes, added some shading. Hated that. So, I swapped out the color palette. Didn't hate that. tried [music] out this blur effect, which was actually kind of sick, but not the vibe I'm going for. Added some rope and a logo. Then worked on a little cup and finished it all off, which took about uh 40 minutes all up. I don't really know about the outline. I kind of prefer the version without, but I guess it depends on the actual art style I end up going for in my game. But I'm going to say it, this this is kind of banging, and I know bangers. I don't mind this. It could be better, but man, it could be so much worse. Like, this is possible in my eyes. I'm okay with putting this in my game and pixel art of of this quality. I will show this one to my wife, but at this point, this is all I have. And I'm not 100% convinced that this [music] was not a fluke. So, I wanted to make one more drink. And I couldn't get the idea out of my head that I wanted to make the Gajjun Killer, a strong zero. Now, this is not the kind [music] of drink that you would actually get in a bar in Japan. It's more of a convenience store drink, but this is my game, so I make up the rules. Also, I'm just practicing art, okay? It doesn't have to be lore accurate right now. But here's the reference image. And my ego is all inflated because of how well the last one turned out. Other than the the rope part, I couldn't figure out how to make that look [music] good. It's better without the outline. But anyway, this time I started with some basic outlining to get the shape right. Realized I didn't want to use this palette because I want my can to be can colored, not blue. So, I swapped it out for a different one. And now we're talking. I don't know about you, but this shading is looking satisfying. I'm going to knock this one out of the park after I spend like half the time fixing the shape because I don't know, man. Can shapes, why are they so hard? After that, I draw the rest of the owl. I don't know. There's there's not much to say. I just redid things [music] until I was happy with them. And let me be so real with you right now. This thing slaps for me personally. And I don't care if it can be better. That will come with time and practice. But from my perspective, the progress in one day is about as much as I could possibly hope for. I'm [music] happy with this. But pairing these two together, what does my wife think? Collectively, these drinks got a B. She said they're a big step up. She said like they were done by a different person. And they weren't. They were just done by a guy with like an hour extra experience, which is a pretty huge win for day one. This whole thing might actually be possible. Day two. Okay, so I can make a pixel art drink or two. That's very important. But if you've seen my first video about this game, you would know that the only thing more important than the drinks is the bars. The progress bars, not like the bars, though. That that's also important. In fact, that's like the ultimate goal. That's what I'll be doing on Friday. But for now, progress bars, which is a whole lot more difficult than you would think be because yes, the first thing I had to do was figure out how to make wood. That was surprisingly hard. But once I stumbled upon something I was happy with, we had this little progress bar box in in no time. Sometime, not I mean, not heaps of time. A little bit of time. But I don't just want a basic gold colored bar that fills up like I showcased in my last video because my ambition knows no bounds. No, in this case I want the bar to fill up with beer. Uh so I tried out the gradient tool and made like a bunch of gradients within gradients. Ended up with this and and at this point I'm like 40 minutes in. I mean there there's more time left in my day. So, I get a little bit more ambitious. I want it to be filled up like it's being poured in with foam on top and everything. So, we're doing animation. And now I've got to figure out how to do that. And I mean, it it ended up looking incredible. One of the most satisfying things I've made. But to get there, however I figured out how to do it is probably most definitely 100% not the actual best or even good way to do it because Brother, this was tedious. I would [clears throat] duplicate the beer layer, hide [music] the original, delete the old duplicated frames leading up to the current frame, squeeze the layer down to the correct size, use the jumble tool to jumble up the layer and add a bit of motion, merge the layer down into [music] the previous animated frames I'd already done, create a new frame, and do it all over again and again and again [music] and again. Then, when I get to the end, add a little bit more jumble while fading it all out. Go back through and fill [music] in any missing pixels caused by the jumble. And then when I feel like it's finally all done, hand animate one tiny little bubble that goes up at the end, but try to wing it for some reason with no consistent motion over [music] time. So when it plays back, it just feels weird and kind of like jumps around the place a little bit. So instead of deleting it and going [music] back and just doing it properly like a good normal person, instead [music] I try to delete, redo, and add individual frames to try to fix the problem areas. and inevitably end up with something that I consider good enough. And I mean, it it looks sick as a whole. This it I'm happy with this. Uh but through this whole experience, I've realized something devastating about this piece. But um first, check it out. I mean, I'm I'm pretty happy with that. It looks so satisfying, but in the end, it it doesn't even matter because I don't think I'm actually going to use this graphic at all. and and I'll explain that, but I do want to know what my wife thinks. Okay, so she really liked this one. She said it's super satisfying and she gave it an A. What? That's crazy. I mean, she's glazing here. I think she likes me, but I'll take an A. There's just one issue. I highly doubt [music] I'm actually going to use sprites for the progress bars in my game. I'm pretty sure I want more control over them than that. Ideally, I'll work out some pixelbased physics simulation or something. so that it's actually dynamic with the input. Uh I'm not sure, but in the end, this ended up being good practice. Potentially not even usable, but still very cool. And you know, at this point, I'm definitely getting more comfortable with pixel art, and I'm starting to think that my goal to create a huge Japanese isakaya bar scene on Friday might actually be possible. Not yet, obviously. There's limitations to what I can do, but I actually am surprising myself with these little pixels that I'm putting down. There is one thing though that I've been shying away from and and I don't know if you've caught it yet, but every single [music] thing I've made so far have have just been objects or UI [music] or whatever, but I haven't made anything with a face. No character elements. And honestly, it's a huge part of why I wanted the game to be about bees rather than people. Because while my art experience is not very large, my character art experience is practically non-existent. I've got all the confidence to pull off stuff. But deep down, I know that if I want my Japanese izakaya [music] scene to really slab, it's going to need a bartender. [music] So, that means tomorrow, I'm going to have to rip that band-aid off and design [music] an actual character. Welcome to Thursday, and today is a disaster waiting to happen. I [clears throat] can already feel it. I don't have my usual shipper can do attitude that I'm known and loved for. I'm a bit worried, panicked, spooked about what I'll be able to pull off because I don't think it's going to be great. The large, menacing elephant in the room that I don't want to address, the bartender, the playable character. That's the challenge today. And I don't know if you ever heard the advice, eat the frog. It suggests doing the hardest thing first in your day so that the rest of the day comes easy. I'm not going to do that. I'm going to I'm going to start with a simple NPC. I'll be a little tiny innocent bee. Surely that will be less terrifying. I mean, bees can still be a little scary, but I'll just do what I usually do and pretend I'm completely unbothered when it's alarmingly up in my grill. I'll I'll play it cool. The bees aren't supposed to be too detailed, but they are still characters. It should be the perfect practice for designing an actual bartender, which won't be bees. The bees just visit the bar. The bartenders, though, those are the playable characters with different looks, stats, and skills and stuff. They're very important, and I'm definitely not ready for that, and I'll honestly never be ready if I can't even make an NPC B. So, [snorts] B. Okay, by now, we should be getting the hang of this. Step one, basic shapes. Step two, ignore the wings, legs, [music] arms, and get straight to coloring. This is fine. Why would this be an issue? What could possibly go [music] wrong? Everything. Everything went wrong. What is this? I don't know. I I really don't. It's a glass and it's supposed to be tiny bee lips, I guess, sipping from the glass. And here are his little bee hands holding it. I mean, this is this is a disaster. Let's pause that issue and make a new one by adding some legs. They're I mean their legs. Let's try the wings. And I'll tell you right now, I I don't know. I don't know how to wings. For whatever reason, no matter what I tried, I could not get these wings to [music] look right. And I tried a lot. In the end, I came up with these and I guess called them good enough. I tried to fix up some other monstrosities and after a long Arabic battle with tiny limbs and a stupid little glass, I fear I may have lost. In the biz, we call this giving up. Now, I don't mean that literally. I mean, I do. I don't I don't mean it permanently. It's cute, but it's just a bit of a mess. The wings are weird. But the whole hand situation with the with the drink here is just it's committing art crimes and I don't know how to fix it. I've been defeated by a bee and his tiny little arms. I don't think I'm ready for designing a bartender, let alone a whole entire scene. If I can't even make a bee I'm happy with. I mean, the whole game is bees. I collapse into the couch muttering something about being bested [music] by bees and I wallow. You know, sometimes not even creatively, just in life. [music] The right antidote is a bit of wallowing. Not too much. Don't overindulge. You know, it's it's quite rich. But a couple of hours later, I return [music] a little bee battered and not necessarily ready to conquer, but maybe just loiter for a bit. Maybe with a bit of jostling, I can fix up this bee of a thing. Man, this was hard. [music] I realized that no matter what, I cannot get these hands to look right [music] in front of the brown stripes. So, the stripes have to move. I I don't I don't know. This is what I ended up with. And at this point, I I don't even know what I'm looking at. I can't objectively tell whether this is good or the worst thing ever. It's okay, I guess. What does my wife give it? Uh, she gave it a B. A B. Oh. Oh, dang. I I have survived to live another day. The pixel art gods have gifted me with a B-grade B. It might be a good time to mention that my wife is a junior school art teacher, which may or may not be inflating the scores. But who am I to question the objective authority of a trained professional? Either way, little talent tantrum included, my day has ran out. I've been humbled by bees and my own incompetence. So, I'm just going to cross my fingers and come back tomorrow and hope that sleep gives me a little dose of that blind optimism that I came into this whole challenge with. Friday, day four. Today is the final day. Today, I see if I can actually make art. I'm staring at the potential future of my game in the face, and I'm nervous. Honestly, yesterday didn't give me all the confidence I needed, but it's probably just the nature of trying to learn something so complex so fast. You'll hit roadblocks that don't feel possible to overcome. Like tiny little bee arms. You might not run into issues with tiny little bee arms, but I'd be careful, mindful of their little arms. But you've just got to adapt, learn a new technique, or be lightly influenced into a different direction that is within your capabilities. And honestly, that's why I set out to create a game with no experience in the first place. I want to learn new stuff, and Trial by Fire is a great way to do it. Setting arbitrary time limits is just a way to control the scope of any given skill acquisition because, yeah, I will continue to learn more of pixel art skills, but I've also got a whole bunch of other stuff I've got to learn. and quick intense bouts of focus and dedication with fixed end goals may be insanely ADHD coded, but it works for me. Or at least I hope it does because today's the final exam, the submission piece. This is really what it's all been about. Whatever I make today is where I will mentally position myself in terms of pixel art skill level while I get on with the other skills I need to make this game a reality. My confidence has been knocked [music] around and yeah, I'm I'm I'm hoping for a good grade. But honestly, the most I'm really looking for is to just not disappoint myself. I just want to make something I'm proud of. So, here it goes. Okay, there's no shot I'm going to be able to pull perfection out of my butt here. So, I'm going to make a quick janky collage reference image in Photoshop. It's not supposed to look good. It's a bit of a convoluted, chaotic mess, but just like my brain, it does the job in [music] theory. like I can look at it and know what I'm supposed to do next. Uh when I'm making my pixel art making, creating. Am I allowed to say painting? Is it is it uh drawing? What's the verb? Anyway, here it is. And now I'm ready to get started with my shapes. I'll fill all those in and get a tiny bit distracted with the lantern before getting stuck into my first challenge of the day, the wood texture. Now, the good news is I have some experience with wood texture. Remember that? It's nice. The bad news is there is a lot of wood and I'm also starting to see a fundamental issue with this whole design, but it hasn't completely dawned on me yet. That will come. Uh either way, I spend entirely too long on this wood texture. I don't want it to be a repeating pattern. Got to fix that. Add some [music] variety. Cool. It's looking a little flat. I'll just use one of my new favorite things ever, the shading brush to add some level of gradient to create like a shadow effect from the bar bench. And hey, that's some pretty freaking sweet looking wood, if I say so myself. Who knows how the [music] rest of the image will turn out. I mean, I I still have to design a whole dang bartender. But for now, man, that's some nice wood. I should probably stop talking about wood. Next, I worked on this screen door thing. Good enough. I then I mean I I don't know what happened here. I gave some texture to these shelves and then just decided nah, no texture for the shelves. Don't ask. It was a vibe thing. I was starting to feel like the whole thing was just becoming one big wood texture. And I'm going to add details here to sit on the shelves anyway. So I I'll just get started with the roof. [music] And when I say I spent the next hour just fumbling, I mean, dang. How hard can a tiled roof be? Right? Apparently pretty freaking hard because I tried so many different approaches and just none of them felt right. And this is one of those perfect examples of the fact that you can only really learn so much in 4 days on a brand new skill because I just no matter what I tried did not have the ability to pull off this child roof. So I pivoted slightly to something I could actually do. This is like the equivalent of always drawing characters with their hands behind their backs because AI is not the only thing that sucks at drawing hands. And that's some more foreshadowing for you. Just just drawing hands, not not the AI. What I'm saying is I ditched the idea of the tiled roof and I just went for I don't know whatever this is. It's not as pretty or ornate, but it's a roof and that's all I needed to be. No one other than everyone now will know what it was ever supposed to be. Then I went ahead and did the counter not in the the curtainy thing. I debated putting some text on it but decided in the end to not disappoint myself any further. And instead I just moved on to the background which actually turned out kind of cool. Now my my plan is to blur it later to add some depth [music] of field and to give a bit more of a 3D pop. So So the details don't have to be perfect. There's the blur. Some of it's a bit sketchy, but I've got a plan to to cover that up [music] later. Now, I feel like it's the time to start working on the details because I'm looking at what I have and it doesn't look super impressive, but I also feel like there's not really any foundational work left. Uh, ignoring the bartender, but that means I got to get stuck into the details. I attempted to keep the cabinet details varied so it didn't look like a repeating [music] pattern. Then I started working on these hanging glasses. The shape is maybe questionable, but I'm happy enough with it, so I duplicate it and call it done. chairs. Make one. Copy paste the lantern. Yeah. So, now we get to watch my multiple attempts at the lantern. [music] And okay, I paused the recording to grab a snack and forgot to hit record again for the time lapse. Uh, you didn't miss much. There's a bit of fumbling and it's just that this this is what I ended up with. Honestly, I don't know if it's all that great, but it's possible in my eyes. So, now I should start work on the bottles and yaki to grill. And damn it, I did it again. forgot to record this part. I'm honestly surprised that didn't happen more often. I I think that's probably the last time. But we missed making a few more bottles. I also moved the bottles to the other side because the whiskey colors weren't really working against the screen door. I don't know. I'm I'm figuring all this out as we go. Okay. And unfortunately, the time has [music] come. It is now or never. Other than some finishing touches to tie this whole thing together, which is honestly the most satisfying part. So, I'll I'll do that last. I got to get to work on this bartender. It's not going to be a human. It's going to be a bird. I mean, there'll be a bunch of different bartender characters in the in the actual game, but today I picked a blue J. Not because they are ever found in Japan. In fact, they're native to North America, but instead because they look cool, very distinctive. Like I said, there'll be a bunch of different bartender characters. Not all of them will be geographically lore accurate, and maybe I should have picked one that is. for the first ever hero shot. But I'll be honest, I didn't really put any more thought into it other than it it looks cool. And holy moly, this arm, I I hate it. I can't get it to look right. And what I ended up with is probably barely possible, but I've already spent over 7 hours on this pixel art. And it's like Interstellar over here because 10 minutes of pixel art time is like an hour in real life. So, I can't afford to commit any more time to this poopy doodoo hand. So, let's finish up this blue J and get on to the finishing touches. a little toilet sign because this is a kaya just happens [music] to not yet be overrun by loud toilet needing tourists. So, we get a regular old sign instead [music] of an old lady guiding you to an unassuming back alley door and handing you the giant spoon, which is honestly part of the charm and not a euphemism if you don't know what I'm talking about. A little [music] tree over here could have been a cherry blossom in hindsight, but didn't mama ever tell you not to stare at people's hinds. Add some chips to the roof and bada bing bada boom, there we have it. my first ever full pixel art scene. And now there's only one thing left to do, which is to get my final grade. And honestly, there's mistakes here and there. There were pivots, definite areas I could improve. But I'm actually just so happy with what I managed to achieve. Like I said, I I had faith in myself. I didn't actually expect to be able to pull off something that looks like this, especially after only 3 days of practice. It might actually be too good. And and I'm not being facious. This this is impossible. I don't care how much better I get, how much faster I get at it. This will still take too long for me to realistically be able to make a bunch of them for the game. Meaning, the biggest takeaway I got from this specific art piece is to drop the resolution of the canvas size for the actual in-game assets. We don't need this much detail. And while I love how it looks, it honestly feels like overkill compared to what I had in my mind for the [music] game. But that's for the game. For the sake of the video and the challenge, I couldn't be happier. And regardless of what grade I get, it's still a win for me because more than the skill it taught me, the most valuable thing I've learned is that it's actually even possible. I've never considered myself much of a traditional artist. I've never been good at drawing or painting, and I've struggled through a little bit of digital art in university, but I think it's all too easy to selfidentify with these ideas that you have of yourself, things that you are and and things that you [music] aren't. Even now sitting in front of this artwork, I would absolutely not call myself a pixel artist because there's still so much I can't do. There's still so much for me to learn. But I've proven to myself that that doesn't mean I can't make pixel [music] art or learn to code or or maybe even develop an entire game because without ever trying, you can't ever surprise yourself. I'm surprised by what I made, which means that just given a bit of practice, I was always capable of more than I believed. [music] And that's exciting because who knows what else you're capable of. The only thing to do is try. And for what it's worth, my wife gave me an A and said, "I love it. I want to go there. [music] Let's go.

This one is HUGE
unless you're me right now in this stream no one's like expecting you to like mail at first go the basic things that you want to really call out and you're looking at your character where is the head where are the eyes like where is the character looking it's very important the next is what are the bounds of the character right like if I if I run up against a wall or something what is my character end I can probably imagine that like maybe the edge line where like I run into something is like here and like here but if I have my if I have a character with like a big long tail like this right and like arms that look like this like where is the point where the characters gonna run into a wall is it like here is it here is it here it's kind of confusing this is why I like

constraining your character within a small sprite is helpful not just for you know harkening back to that historic sort of pixel art design era it's only part of that history because of the practical side of it and so ignoring the practical side is a big mistake if you haven't made a game before so let's do it like like this let's try to brighten this up a little bit that's a bit better they were there for green for the body just for the sake of this process and I actually encourage you to use colors that aren't part of your actual intended design just for this step of the process because it will help you avoid thinking about color so we're using this like really weird green because we won't be thinking about whether or not green was the right choice because we know it's not the right choice right we're just thinking about the shapes now what's really obvious here is that we don't have a lot of space to work with right we've got so little space to work with in that we can't draw a properly anatomically correct person in this space right like a proper person their heads would be like this and then their body would be like this and then their legs would be like this and it's sort of you know it's okay I mean even this is kind of like stylized you know and you can go for that but we were talking about what was important for our character and it was being able to relate to the character that was like one of the most important things and if you can communicate how they're feeling even at this level give them in this platform of space then you're doing really well I did something just now right before we started we're fighting the cigarette the second time I zoomed out this is really really important it's very easy to zoom in and get stuck at this level I'm zooming out to give myself that perspective again traditional artists would do this with a canvas they would step back from the canvas and have a look you should do this too like I said we were talking about caricature rings so we don't have that much space to play with here so what I'm going to do is try to create something that's feminine in the most recognizable way possible and for me when I'm drawing female characters the obvious thing to do is smaller upper torso like you got that hourglass shape right and then the legs can move from there that's very feminine whereas masculine is more of a big triangle right big shoulders and then small small legs and it's okay to like get it wrong for a while before you get it right you know it's like no one's standing behind you unless you're me right now in this stream no one's like expecting you to like male at first go you know you can like muck around and like get wrong and I encourage that you know and I'm gonna introduce the second shade now just to sort of introduce the arm maybe she's gonna have her maybe she's holding it back back all the time or maybe like in this part of the game like her weapon is the pencil like maybe she has a weapon this has become a Metroidvania now whoops that's all I know how to make so like I mean I don't know I think platformers with weapons are cool you know what there's no harming that there's nothing wrong with that so kind of got lucky here with this orange time it's very pencil pencil like and I think like the other thing we would want to do is have like a uniform now in this sprite I think it might be nice to have an outline at the end of it but we'll see if that kind of goes against the shape that we were looking for am I using your mouse right now no so that's the other thing that's really important I use a tablet I you don't have to use a tablet but I highly recommend it everything that I've talked about today sort of falls outside of the how do you place the pixels it's more about where you place them and what pixels to place I'm not going to place judgment on you if you choose to do it one way or another I think I'm gonna jump back to the Insignia palette just just to help us out here a little bit because this is about character a lot about palette for this part of the episode I'd rather make this a meaningful episode I think what we we should go for here is like maybe like a schoolgirl like a Catholic schoolgirl sort of outfit like a skirt or something like that now this gets kind of gonna destroy the form here so I'll I'll do it as carefully as I can and this is something that I would spend like hours on getting this right because it's so important like it's really it's like especially with something like a skirt that takes all that shape away you got to be really careful and I'm gonna try to bring back some of the shape with shades I kind of like it and we could we could like accentuate some of this stuff as well we could give her like sort of puffy sleeves and this is kind of like what I was talking about when we mentioned at the very beginning of the series we were talking about at the pixel level so the choices that you make over this level here like single pixels will communicate details that you don't that you aren't normally able to communicate so for example let's just say I want the contrast between the shadow of her hand and the ND shirt cuff to be very obvious so I need to use a bright color just on the logic of it a bright color next to a dark color but I still want it to look like a cuff so what I can do is then go to a darker color and now you can see kind of like that cuff coming through as well as like this shape around it maybe and you can play with it it doesn't you don't have to get it right perfectly now I'm trying to I think for this character it's probably gonna be most effective for us to go for this sidon look cars at least for the head because I want to show off that ponytail because it's part of the characters design that we decided on right so we have like a ponytail and pencil and glasses you might not be able to show the glasses well we'll see how we go but let's just work our way there first we'll go with the most obvious details the sketch program is called Leonardo but I want to see if I can get this flyaway hair then I think of maybe the flyaway can be above actually that might be even better oh I think I like that I wanted her to look a little disheveled not disheveled but like like she's got kind of bad hair the one who studies too hard kind of thing you know she like woke up late because she went to bed late because she was - she was up all night studying kind of thing so we've got the low ponytail we've got the flyaway hairs all I'm doing here is just trying to pick out the things that I think are going to make the characters stand out the most on the screen again we we accentuated the head the major features of the character we've tried to really really reinforce the feminine form even though like we could make the waist even smaller to get that hourglass shape in fact I like that even more the goal always is to make the character as recognizable in one frame as possible okay and that's something that if you're animating it's gonna be a challenge for every every single frame of animation is to try to like reinforce them and you can you will as you make your game come up with like a language of how you describe that to the to the player you know like what tools you use maybe like there's a pixel here that when the characters facing this way you used to show her shoulder things like that like for example maybe like if I want to show her collar a couple pixels then reinforce that V in the neckline you know there's little things you can do and that's something that you figure out on a per sprite basis really [Music] so what was left we wanted to get some glasses in here I kind of want her to still be relatable so I still want some eyes I'm not gonna go for the bug eyes but um basically at this scale you've got limited options really you can go for the two lines you can go for some eyebrows in there as well one thing that I like to do for female characters is the lights either side because it looks like like eyeliner or eyelashes you can you can do a lot like depending on like you've got so much there it looks like there's not that many pixels to play with but you actually have a lot if we want to be more chippy you want to make the nose I'm gonna lower here which gives you more space to play with now I'm gonna be trying to use the glasses to actually kind of like reinforce her attitude so if I was to like bring these down it kind of looks a little angry right if I want her to be like a battle if it want the skin to have combat in it maybe you go for the angry but if you want to be more like innocent-looking you might use this angle to be before like sympathetic you know that's like sort of more sad even right it's very important to understand how to draw eyes now I don't have my pressure turned on at the moment this might be a bit weird but like what makes an eye and eye and like what it is that you think about or how you interpret things like for example if you draw an eye and the pupil or the iris doesn't even touch the bottoms or the top then it looks like shock right if the eye if the iris does touch one of the edges suddenly now it just looks like you're looking in a direction okay up or down if the IRS touches both then it's that's sort of more natural right and then you start referring to like okay what's happening with the eyebrows right and there's lots of different things you can do as well as people's you know like a larger pupil but as the smaller people means different things so knowing those things are even a basic level can help you choose between like where to put your one pickles that you've got if you've got four places to put your one pixel you might be able to like pick the right one hey pal thanks for watching and thanks most especially to the patrons and twitch subs who support this channel and my game dev project insignia to find out more I'll click the links in the description below and if you liked this video tell youtube by clicking the like button and then YouTube will tell me and then I'll make more videos that's nice

I grew up makinging pixel art and I got back into it in the past year and a half or so and if there is one thing I have learned that has really helped me out with my character design it is to push it there's probably a better phrase for this that's not already a song but either way this philosophy has really transform the way that I do my work it has helped me immensely so I want to help you guys out with these three easy to follow tips to help you push your art to the max I don't know why finger guns hey guys I'm Andrew Robinson of Robinson pixels the first tip is to push your character design sometimes after I've designed a character I feel like I did a great job I'm done and I'm moving on but that's not always the way to design the best character there's a couple of ways that you can push your work to be a lot better I'm going to go over two different methods now the first method is research focused so one of the videos on this channel is about shading and for that video I really wanted a character that I could practice on and I decided that that character would be a ninja turtle knockoff but not exactly a knockoff I wanted it to be its own legally distinct thing the first I knew that I wanted it to not be green because we always color Turtles to be green after a little research I found out that there's a cherryhead red-footed tortoise that lives in South America yes I know that tortoises and turtles are not exactly the same thing my tortoise would need a martial art that's not Ninjutsu so another research push told me that capaa is is really big in Brazil capaa that informed his outfit his hairstyle and even the pose he uses for the kick I would make him extra legally distinct from the Ninja Turtles by making him an actual lawyer now did I need to do all that for a video about shading I mean no not really but I did get one comment on that video that said uh I got to find it ah so he's an adult non-mutant lawyer tortoise who does capaa in his free time and yeah that is what he is I didn't even use the word capaa in the video because I didn't know the word and it's hard to say and I hope I'm saying it right right now capaa but either way the fact that somebody caught that just from this design that I crafted is really cool and it was really fulfilling for me as a person now if you want to use a push method that's not so research-based I would recommend going for the second one that is Discovery based last night I was designing an air themed Monster at first I thought maybe it could be like a cloud Cyclops but that looks like like a peanut so let's push it to a fo monster a little better but why is it standing if it's a cloud let's push it into the air but why does a cloud monster even need legs at all yeah this is a cute little monster now that can punch you around which is a thing I feel like wind does this design could be done but what if we push it just a little bit further I want it to be a storm cloud hell yeah that's rad it's not even that far off from where we started but I feel like the extra push is really made this look like its own character that could exist in something after you've pushed that character design into something you're really proud of and you're actually working on the art I highly recommend that you actually also consider pushing your contrast a lot of times when people are new to pixel art or even just art in general their pictures lack a lot of contrast this is something I'm very guilty of like with this Mario Sprite I did last year let's just look at those colors and bump up that contrast if I make the brights a little brighter in the dark's a little darker oh buddy that's better and just to push it one tiny bit further cuz I could not help myself I went ahead and introduced one extra shade for his face to bridge the gap between that lightest shade of his skin and the next shade down that helps smooth out how Jagged it looked before big improvements with just a little push fixing the contrast isn't even just necessarily A shading issue it's also a big factor even in like 8 bit style flat colors I am super proud of this power suit I just designed in a recent video but I had some comments about the color choice is not exactly working and yeah flip this thing to grayscale and you can see there's very little contrast between the yellow orange and that blue I used see I had drawn some concept art for this and if I had just Ed those exact colors it would have worked much better but I didn't because I really wanted that yellow to match the same color as samus's armor now if I really wanted to commit to that I should have used a totally different color scheme without that bright blue either NE of these options works better than what I started with once you've got your design and your colors down pat and you're actually making your character do stuff I strongly recommend that you push your poses sometimes I get stuck making my pixel art characters look a little stiff or you know rigid that can work depending on the mood of the scene sometimes it's just necessary to get things done but it's not even always the case really lots of times you want your characters to look more Dynamic I'm going to use Mega Man Sprites as examples because I cannot help myself a fre love them they're adorable and they're just so damn good okay so Mega Man's standing here even in this pose his legs are spread apart he looks like he's kind of ready for action and then to make him shoot they could have just made him hold up his gun but they didn't do that they redrew the entire Sprite with him leaning Way Forward it's so much more Dynamic and it wasn't even necessary I like to push things as hard as I can and I'm always working to try to find new ways to do that like with this alien slug that I drew last year he doesn't just scoot I pushed each frame to make the character look like he's really trying he's exhaling hardcore when he scoots forward and then opening his mouth really wide to take in another breath I think that's really funny for a monster villain you know it's kind of goofy it gives him a lot more personality now once you finished pushing your characters to the actual limit you may notice that there are some critical mistakes you've made with the actual shading process if that's the case check out my video on the capoa tortoise right over here I can't Turk a [Music]

so you can see here this part is the nose you kind of need to have a few pixels to create that distinction but once you do it's pretty uh it's pretty stable hey pals welcome to a new video today we will be covering isometric pixel art particularly isometric characters this is something that has been suggested that i make a video about pretty frequently since i started doing isometric stuff and um since i've been working with isometric characters myself making throneless and working on the ludum dare entry and experimenting with this whole thing uh i thought i would make one if you haven't been acquainted with this format in the past then you should watch some of my more recent videos about this topic because i cover some of this information in those videos and it'll just be handy if you started with a tile set before you get into characters so assuming you've done that let's get into it okay so isometric characters what are the basics what are the things that you need to know about before we get into it the thrust of it is that as we've spoken about many times this all comes from essentially one studio you know heralded this entire sort of aesthetic and that studio was quest which moved into squaresoft now square enix the team that made the game tactics ogre previously ogre battle and then final fantasy tactics final fantasy tactics advance etc so this is a character sprite from final fantasy tactics this is tactics advance you can see what's going on here we've got this very sort of like chibi format we have these nice big eyes small feet particularly smaller feet in the smaller games so this is a gameboy advanced game this is a playstation game and the format has been done a few different times and with different variations more recently because the actual historical basis for this stuff it's not super saturated there weren't that many games in this format with characters like this but it's taken more of a i guess a popular turn in recent years with you know indie developers and uh you know prominent pixel artists in the community i really love this age of empires um isometric take by ref as well as i wish i could pronounce its name cyan cyan mel uh these awesome uh knights here you can see that the style is pretty consistent so small feet we've got big heads you could explain that away with perspective so you know the characters are being viewed from the top down so the head is closer the feet are further away this helps kind of enhance that perspective i've also got games like witchbrook i wish i could source this image i don't know where it comes from but i really love this character as well there are a few things almost all of these have in common that comes down to beyond the proportions pose work so there are two major poses there's the toward and down and there is back and away now what i want you to notice here is that if we were just doing a side on character you can see all of these lines are horizontal they're parallel with the ground right the shoulders are square the eye line is square hips are square feet are square once we move into isometric because we're at a higher angle of elevation looking down all of these lines follow the ground line and because the character is still facing the ground they're still square with the ground this diagonal line appears everywhere okay between the hands the hips the shoulders the eyes everything follows that same line and there are two major poses like i said towards and away they have the same angles as well so once you have these two poses you can basically just flip them right this mirrored version here is the same sort of thing and that covers all cardinal directions so you can basically go any direction from here it is possible of course to have games where characters can move in more than these four directions if your character isn't strictly speaking bound to the grid you might also need to do top down angles as well so away toward left and right and those play a little bit closer to just like a normal top down game so so i won't be covering those angles today i'm just going to be covering these major two so let's get into it hey pals just quickly i'm here in post and i thought before we show you the actual character creation i think it's probably worth showing you my experience doing this and just kind of the things that i've learned while creating characters in this format so these are the characters that i've been creating the last little month these ones were all created in during ludum dare and this was just a quick mock-up that i was doing for the tutorials that i was creating while doing the tile sets so this is armin he is uh the main character of insignia and what i really like about this uh which came together quite well although it's hard to see here is um just how simple the actual implementation is there are really only two colors per material i think this is a actually quite a good choice uh mostly because these characters being as the size that they are if you're looking to make a game that's in this sort of tactics ratio these characters aren't really much taller than say 32 pixels tall and so i think keeping the amount of things in the character design quite simple keeps them recognizable and uncluttered so in each of these characters i think that's one thing that i'm getting quite good at um sort of just being intuitive about which is just keeping the designs to a minimum you know your characters as you create them for you know portraits or concept art or whatever they might have lots of different elements that make up the actual design of the character but at this scale these are almost more like chess pieces that's how i like to think of them and that's how they present in games like final fantasy tactics so here for example you know we have our ranger he has a hood and cape he has a black tunic and he has brown shoes obviously in the game he's also carrying a bow and each of these characters has their own weapon and that's kind of like part of the silhouette as well although here it's not part of the sprite because um that's just not how it works uh so the goal here was to keep these characters recognizable from both angles and also to use the same color for all of the characters to signify which team they're on because there was a red and a blue team so we have the ranger we have the witch now i did create these very quickly i mean this was the whole thing probably took me like two and a half hours so just note that it's not probably my finest work but i think working under those constraints is actually quite a useful thing because it teaches you what you should and shouldn't do so the first sprite that i created was the night and i didn't really have a lot of opportunity to do much research on on which elements i should you know you know should stand out and be the most salient features i kind of just was in such a rush that i put together okay he's got a breastplate he's got some pauldrons he's got some you know gauntlets and he's got boots and that's it you know like that's as much as i could do throw a cape on there put a feather on his cap and away we go but when i was creating this you can see how many more colors i'm using and i think you know especially here at the back of this helmet it really serves to just muddy the sprite it doesn't really help and it's more work it's more pixels that you have to create and be mindful of the you know the placing of so the goal really here is to say as much with as little as possible that's really how you should do this the priestess was i think one of the last that i created and i'm quite proud of her i think she has a few elements that really stand out you know she has her her apron and the um you know the dress that she's wearing she's got these gold little i guess clips in her hair and that's really like enough you know there aren't any more than three colors on any of these materials and it makes her look quite clean compared to say the night tonight looks good when it's simple but you can see in certain areas you know it's just noisy right so my tip to you is to start simple try to stay simple and only add when you absolutely have to so that's my experience enjoy watching me create a new character from scratch okay so get yourself over to your favorite pixel art editor mine's a sprite a-s-e-p-r-i-t-e for those of you who are asking and uh basically what i want you to do is create a tile so start from the bottom corner drag up so you've got a nice diamond shape here with two by one angles and we can come down like this okay there's your uh there's your tile once you have this tile it becomes a little clearer about where your character is going to stand and how they're going to stand okay so what we'll do is we'll just take another color here on a new layer and i just want to fill in the silhouette so let's get the shape right first okay so once we have our base silhouette we can start filling in a little bit more shading so i want to really emphasize what's on top here because that's kind of the most visible thing it's very easy to pivot into just doing a side on character where the eyes look like this and suddenly it's not isometric anymore so just uh paint the top face as if it's being lit from above and i usually do light from above anyway so and if you're stuck basically you just want to stick to that rule of following this line here so anywhere that you want to draw if you can follow this line across this down to here that means the feet are square okay this down to here that means the hands are square same thing with the eyes that's the angle the eyes are going to be at so once you've got that basic shape worked out i want you to refer to your character design so don't create a character here from scratch unless you know what you're doing basically you should have some kind of sketch what your character looks like you know who they are in the game so i'm going to take a character concept from insignia and i'm going to see if i can create her in isometric i might just give her her feet here so we can see that since it's a female we might want to change some of these proportions a little bit give it some hips okay so now i'm just going to start replacing some colors so this here is going to be this color in fact most of this is going to be red so i'm just trying to bring in those features of hers trying to make sure that everything's in there as it's supposed to be now with these kinds of characters the eyes in like the final fantasy characters what they tend to do is be a little bit cross-eyed like that kind of like the um characters in minecraft so if we have a look here you can see they've got the irises and pupils dead center and then either side of that we have the whites of the eyes here with this three quarter view you can see we don't even have the second pixel if we go to tactics advance they do have the second pixel on the isometric so you can kind of play with it until it looks right to you i like this idea of having a third pixel up because it just kind of makes it look a little less derpy and then you can even do a little bit to darken this off so you really want the darks of the eyes to be pretty dark so let's try this looking a little better now i like to create shapes out of shadows with the face so we're going to try to make this a little bit bigger and then what i'll do is this is the nose here so you can see here this part is the nose and you kind of need to have a few pixels to create that distinction but once you do it's pretty uh it's pretty stable cool nice okay so see how we stuck with that original silhouette and because we established that first it was pretty easy to morph it into the character that we wanted and what i was really doing as i'm working through here is i'm picking out the bits and pieces from the character that i think are really important that make them you know stand out and unique so in this case the proportions of the character you know being a feminine character means the shoulders aren't quite as broad we've got uh the hips here are a bit wider but the waist is a little bit more narrow so trying to take that and you know convert it over without breaking too much that's basically the way to go about it and i'm just gonna see how much i can add before i need to start taking stuff away again so can we get some of that color in to create some depth in the pants and what i like to do because a sprite's uh background here tends to be a little bit tricky sometimes like you know this white square going into this gray square right at this junction where the shoulder is could be a little distracting so what i tend to do is on a new layer just you know darken it off and see what it looks like under a more flat condition you can even go straight to black if you want and just test that so here she looks kind of square i might just take away these pixels and that helps us out a little bit nice and because this is like a metal i might go out of my way to make this a little brighter and we could do a little comparison here i'm not trying to do you know a one-to-one copy but it's nice to just see the differences so i've gone a little bit taller i quite like that to be honest um what else has changed i think we did a pretty good job i quite like what we've come up with here it's a bit cleaner it's not as grainy we could maybe extend this foot out if we want something a little more heroic if we want to be a bit of a wider stance a bit stronger and of course the outline will always bring it back into that final fantasy tactics look so if you really want the outline it's there for you and if you do use the shift o version of the outline it's always nice to just come back and have a look to see if there's anything you can touch up or that looks a little bit better because it's obviously an automated process and you are you know you're going through and creating this by hand so you might as well just take a critical look at it and just make sure you didn't you know lose anything although there's no opportunities that you're going to miss out on so for example this line here you know maybe we actually want this to extend up to further you know differentiate the head from the shoulders you know if that's what we want and just working on a little bit more effort here to separate the head from the rest of the body you really don't want to lose those major components okay so like the head the torso the legs the arms those need to feel separate you know you can even work on adding those internal sort of like outlines to yeah further enhance that separation and it becomes a little bit easier to see now and i always like to experiment i mean these are single pixels here so we can always play and just see what comes out so when we added this black line what it did was it made the feet relative to the waist a little bit further out so if i bring this back in like this now it's a little more square and maybe we'll keep the shoulder pads in as well okay i really like that i think it's cool and we can add a little bit more character to the silhouette i think just one of those extra little touches awesome okay so that's the first sprite the next one we want to do is obviously facing the other way so let's do that next so the way i've tended to do this is to just take what i made and make it totally symmetrical this is going to feel a little bit a bit mean to just erase major parts of the character but honestly like because this is the same character this pose and this pose are so similar it's just that the the aspects are flipped the stuff that was behind is now in front and everyone everything's turned around so uh basically you can just take this and flip the um some of the posture aspects so you can take this and move it a bit closer in you can take this and move it a bit further away and then you really want to just erase all of the details you know you can even fill the head with hair and that starts to already look like it's from behind right so toward and away and you just want to bring those major shapes back in so the more confident you can get with you know treating pixels almost like it's clay i think the faster you'll become cool so you'll notice that a lot of this silhouette from one to the other is pretty similar right especially the arms and this is something that i don't think is necessarily a problem um particularly you know for games where this is the case where you've got characters that can move away from and toward the the player and particularly when there are a lot of characters on screen you do want the each individual character to have a pretty clear consistent silhouette you want the colors to be consistent you want the shapes to be consistent because it allows the player to very easily recognize who they're looking at so this is a pretty quick exercise to be honest making a character that's this small shouldn't take you very long and uh you can sit here playing for a long time when i was working on these characters for throneless i found myself trying to limit the sprite creation process to about like 10 minutes per sprite so i i dedicated you know for four or five characters i really only spent like two hours working on all of them that was like towards away kneeling sprites so again because these characters are so small you know it is something where you kind of can create new poses by taking a lot of the information and just copying it for you know the next frame i would just advise that it's probably worth mocking up the character with like a silhouette first just the blob maybe like a shaded blobbed character just so that you know like where the pieces are going before you start copying and pasting elements around okay so as you're working with this make sure that you don't lose those angles right the ones we were talking about before so you can just always double check if you're a little bit stuck especially on the hips here i found it quite difficult to stay and keep this line but you know from the shoulders everywhere you can basically draw that line and you should be able to draw that and have the bottom of each of those elements line up and if you're struggling with one area just do that check to make sure that that's the problem because it could be that you know the actual pose is a bit off-kilter for other reasons so there you have it an isometric pixel art character both towards and away you can flip these for the different directions we had our reference here and we came up with this lovely sprite from here of course we could animate these i won't cover that in this video but maybe in a future video we'll do attack animations and walking i'm currently still knee-deep in development on throne list obviously it's only been a week since i started the project i've made some really good progress and i'll be moving on to character animations soon so if you are a little bit patient and you wait a couple of weeks maybe i'll have a video coming up soon with animations for this kind of thing based on my experiences animating my characters for that game so there you have it thanks for watching catch you in the next one hey pal thanks for watching and thanks most especially to the patrons and twitch subs who support this channel and my gamedev project insignia to find out more click the links in the description below and if you like this video tell youtube by clicking the like button and then youtube will tell me and then i'll make more videos that's nice thanks again and until next time you

pixel art is kind of deceptive in how simple it appears to be like you'll go into it knowing that a pixel is one square but exactly how many of these does it take to make the kind of pictures that you see online or in games well hello my name is Brandon and I make pictures out of tiny squares and post them online for fun and in this video I just wanted to give you some points of reference if you're starting pixel art how to choose a canvas size or a character size so let's just roll the credits and get right into it [Music] [Music] when you're selecting the sides to make pixel art one of the things that I find to be a useful point of reference is just to consider the resolutions of classic consoles often there's a look from a certain game or maybe just a single component like a character sprite or a portrait style that provides some sort of clue for what size or style you want to attempt in your own work so rather than rattling off all the resolutions here I think it's best just to go through each with some examples to see these canvas sizes in action okay first steps the Nintendo Game Boy and the Sega Game Gear which I've grouped together because both had the same resolution of 160 by 144 pixels in the case of the Game Boy they only had four shades of color to work with but they're still able to pull off iconic looking games like the Pokemon series I think just look at the computer that's on the desk there it's only 16 pixels tall but it's got a monitor with text there's a keyboard with like a dozen keys there's a tower and there's a mouse that has a left and a right click on it that's the kind of simplicity that's important to learn when you're starting pixel art you kind of want to program your brain into learning how to scale down these kinds of details and small canvas size is a key part of that I've actually been using this exact size and some of my own recent work and having a blast because it's the low resolution that brings a challenge that's kind of fun to figure out and work with in moving up from there's the gameboy advance at a resolution of 240 by 160 pixels which again is another versatile size for stylized pixel art and if we look at this screenshot from Star Wars Episode three in particular we can see lots of great examples of portrait sizing within this limited space there's the main player portrait some extra length indicators and of course the head of the player sprite itself there's even high-resolution portraits that pop up on screen for dialogue and I just love the fact that they were able to effectively create different scaled versions for each purpose rather than just reusing assets and it shows that they really embrace the resolution of the Game Boy Advance another great thing going on here is how they use geometry in lower details towards the background to create a sense of depth Chancellor Palpatine aka the Senate sitting in the distance has a smaller sprite and simpler coloring to convey his distance from characters in the foreground in fact his heads only seven pixels tall but it works within the context of the scene the other thing that helps is not a flat tile game there's sort of this vanishing point angled look that drives the perspective of this shot in particular yeah sorry how the hell of looking at screenshots from Star Wars Episode three this is what's about that you've got to find stuff that inspires you and this one's definitely one of mine so look for screenshots you know stuff that grabs you and and study it moving up from here is actually gonna be a two-for-one it's the original Nintendo Entertainment System and the Super Nintendo which have very similar resolutions both have a width of 256 pixels but the height of the NES is 240 compared to 224 of the Super Nintendo that's right the Super Nintendo is actually a slightly smaller resolution compared to the NES any graphical superiority it has is not based on resolution it's based on color count whereas the NES can display up to about 25 colors at one time from a palette of about 60 colors the Super Nintendo is able to display up to 256 colors at once and that's from a theoretical palette of over 32,000 colors that's not to say that more colors will always make your art better though and there is a charm to the central look that is certainly worth exploring I guess the main takeaway here is just that a resolution in the 200s is a solid pixel art look that allows for a lot of freedom of style and level of polish that you want to give it [Music] lastly we've got the Sega Genesis at a size of 320 by 224 pixels there's not much to say here because it's pretty much just a wider version of the Super Nintendo resolution already looked at but even in the screenshot from restart or ristar I don't know how it's pronounced actually let's go with ristar he's a star he rises I guess ristar so here there's great examples of environmental details and background landscapes at this resolution also I just love the color palette for this game it's vibrant greens and purple shadows of the main elements to the pastel II pinks and blues the background if you've already spotted detail on the bottom left corner you know I have to mention that fantastic little 16 pixel tall icon of the character it's such a nice reduction from his main sprite and this brings up another good topic which is sizing and pixel are character sprites in general how large should you make them and what level of detail can you achieve based on character size so let's take a look at some examples so here I pulled a bunch of examples that we might be familiar with just to look at what kinds of pixel Heights are typically used for character sprites I like to think about this lineup in three distinct groups and these aren't any kind of official designations just my own interpretation so there's a small sprites from about 0 to 30 pixels well okay I guess not zero cuz I'd be nothing but at least 1 to 30 pixels so even one pixel is too low I guess that'd be I don't know the ball from pong or something can we include him sure either way we've got pac-man in there he's 13 pixels tall he's an iconic and simple design at a low resolution there's also the guy from Pokemon I guess ash or whatever name you gave to him is his name the Polka man he's only 16 pixels tall which is the same size as the NES Mario sprite actually then there's also linked NES Mega Man and Super Mario from Super Mario World you'll notice that the common thing shared by these small sprites is that they maybe take two or three character elements to give focus to when you're working with low resolutions it's best to communicate a smaller number of details really well than to try to pack in too many details for example the sprite of link from Legend of Zelda linked to the past has well-defined elf ears a hat and then largely uses color to define the character look as a whole the NES Mario sprite has a hat a mustache and overalls with buttons small sprites rely on eliminating unnecessary details as they compete for valuable real estate and be confusing to look at you wouldn't want to give them a one pixel watch shoelaces a graphic t-shirt and glasses etc because it becomes too much of a mosaic be like color in your character with a literal checkerboard in general less is more here and this range of sprite sizes the perfect place to play around with learning reduction in details if you just started with pixel art next up are the medium sprites from about 30 to 65 pixels now this one's my favorite group to work within because it's where the restriction that we saw from the smaller sprites opens up to allow for a lot more freedom of style to come through while still maintaining some kind of obvious pixel art look to it for example the sprite of Samus from Super Metroid is able to achieve a lot of great mechanical detail and highlighting through the use of gradient colors without being too distracting and you can still feel the contribution that each individual pixel is making to the overall sprite another great thing within this range of sprites is that it's where character proportions starts to play a major role if we look at the sprites from Megaman X and Lupin from Lupin the third we notice that despite being from completely different games they're actually very similar stylistically the jump in sprite size from 34 to 62 pixels results mostly due to lupins lankier proportions requiring a taller sprite to maintain this particular style of art and shading and finally of the large sprites from about 70 pixels and up sprites at this size have a great sort of illustrative quality to them like the sprite a cyclops from x-men mutant apocalypse it's a faithful rendering to Jim Lee's iconic x-men artwork so playing a game with that kind of art is like playing the comic book of course the downside to sprites this large is that they require a lot more time effort and skill to pull off certainly they're worth aspiring to though but if you're just beginning with pixel art again I'd recommend to get a handle on the basics by playing around within a small or medium sprite range first generally when I go to design a character I'll open up a campus size large enough to give me a lot of breathing room let's say about 180 by 180 then if I've decided an approximate sprite size based on personal preference I'll rough out the character as a stick figure before rendering further line work detail and color I like to copy paste often and keep each iteration of the process so I can look back and perhaps bring in ideas from earlier that I might have strayed away from it's also kind of cool just to see the process along with the final product so that's all I got for now thank you for watching I know that that was a bit more of an information blast in the tutorial necessarily but I hope it gave you some things to think about especially if you're starting pixel art and just getting some ideas of where to begin so thanks again for watching take care and keep its where has that a cheesy tagline [Music]

[Music] foreign hello there my name is Brandon and I make pictures out of tiny squares and I've been doing that for a while now in fact it's been eight years since I first started making pixel art with about the last three and a half years of that making videos here on YouTube I'm often asked about the best way to start pixel art and I often give the same advice which is I say to start by using a low resolution and to restrict yourself to only a few colors and that's good advice I do absolutely stand by that advice but it's the kind of advice that somebody gives when they've been making pixel art for eight years and have in some ways forgotten what it's even like to not know what to do with those parameters having a small canvas size and handful of colors is great but it doesn't really do anything for you when you're not sure what pixel art is supposed to even look like or how it's constructed so today I wanted to retread my first year or so of making pixel art and share not only the way that I got started but also what I think the most important factors were in those early days what you may have already guessed based on what you're watching here is that the way I got started with pixel art was to make Sprites in the style of the NES Mega Man and this footage by the way this is an Unearthed archival footage from back then I filmed this this week for this video and I started these Rush Street Fighter paint overs just so that I had some relevant b-roll for the video my actual first attempt at pixel art was that I made a Sprite of the Hulk in this style and you can see that I knew he needed to be larger than Mega Man so I just kind of kept extending the legs outward at 45 degrees to make him taller I've since remade this Sprite back in a video for my five year pixel anniversary and I'll probably have to do another one at 10 years but as far as getting started it was really helpful to have that Mega Man Sprite to use as the basis for the kind of style and level of constraint that can go into a pixel art character and for me I was particularly interested in emulating that sort of retro look so this seemed like a perfect place to start now I've since come to realize that making Sprite edits and particularly NES Mega Man Sprite edits is quite a common place to start with pixel art to the point that I've almost sensed it to be perceived as kind of cliche to a degree and indeed if you do think of a character you can probably Google them with Mega Man pixel art and there's a good chance that you'll find some interpretation that already exists what's Curious to me though is why this is so common why are there so many Mega Man style Sprites out there 8-bit characters like Mario and Simon Belmont from

Castlevania are also iconic representations of that era and certainly you can find similar character edits for them as well but I believe it's because Mega Man has two things that those other guys don't first and foremost his Sprite is essentially a blank character template that shows you where everything goes the entire body is laid out so clearly and contained within this bold Line work where the color divisions of the gloves and the boots make it easy to pinpoint how different kinds of hands and feet could be placed over the Sprite and second of all he's got those eyes I think we're really drawn to the cartoony eyes here because they allow for expression in a character by comparison you don't really get that from the Mario Sprite and even less so from Belmonts plus they each only show one side of their face the Mega Man Sprite presents you with a really nice solution to characterize which is this little 2x2 unit surrounded in white you leave a little space so that it's readable and then there's a 1x2 for the other eye in order to show that it has some perspective and direction to it so with these basic principles I set off to create my own designs but when following this template in reimagining other characters I found that I wouldn't get that far just by following the existing Line work example it becomes necessary to add features that don't have any basis of reference in Mega Man's costume and it's at this point when coming up with new shapes that I started to really learn Pixel Logic for example with the Street Fighter characters here there's things like you know ryu's got a headband what would that look like in this style and guile has that incredible flat top how can I make some angles that capture that same Majesty right and that was the kind of thing that I did for about the first year that I was making pixel art and I was having fun reimagining a bunch of characters as if they were the cast of their own retro games every character face and every little accessory or augmenting piece I made for a Sprite like this was a small lesson in translating shapes into a pixel art style and once I'd created enough characters there was this cumulative effect that was almost like forming this mental library of design solutions for different features and shapes things that could go on to inform not only future characters but also small objects for environment work too I also want to be careful and not give the wrong idea here I'm not really suggesting that you build an entire portfolio exclusively based on tracing and edits of existing work I say humbly while drawing over a bunch of Mega Men it's just that as a beginner you're dissecting and taking in these Styles multiple Styles ideally and I think it's only natural that a lot of earlier work tends to be more strongly derivative even without intending it to be at times these things are sort of like training wheels and eventually we'll find our own intentional stylistic choices to bring to the table for me that jumping off point came when I felt like the Mega Man style Sprites were too restricting too tiny to fit in the kinds of details that I wanted to they have such tiny bodies that wouldn't it be nicer if the bodies were more proportionate and I could fit in more detail and create better poses some of my earlier attempts at this were the Star Wars characters and it was really refreshing to have that extra room in the Sprite sizing especially after having laid the foundation with the restrictive smaller size of the Mega Man style Sprites you can even almost still see the underlying DNA of the Mega Man style in their faces a little bit one of the major pieces for me around this time were these Sprites are made of the faction leaders from Titanfall 2. I remember being up till three in the morning or something just obsessed with making these and how amazing it felt while making them to actually be able to do it and to be proud of it too it was as if I was able to create something that truly felt like it had my own stylistic influence for the first time regardless of the Sprite size or the style that you start with another important thing is just to experiment with the design and see where that can take you and this helps you develop that personal influence that you can have on your work going back to the street fighter Sprites I kept these ones really tightly adhered to that Mega Man template just to simulate the kind of outcome I might have had as a beginner but we can pick one of these out now and try to play around with it like maybe let's start with Blanco because he's probably the one that's got the most room for improvement in his original design he's got a little sharp tooth poking up that I like but that doesn't really exactly fit into my interpretation here because the eyes in the mouth are butted up right against each other so I might just select the top part of the head and bump it up to give myself an entire row of new pixels to work with so the tooth fits fine now but now it's weird that the white of the tooth is blending with the eyeball so let's just get that eye shape out of there all together maybe and come to think of it this version of blanca is depicted with entirely Whited out eyes anyway so maybe there's a different way to arrange the black and the white here to give it more of that look the other thing of course is that he could be larger than this so again I'll grab certain parts and drag them out just to create more space then I'll do my best to fill in the gaps by connecting the existing colors and shapes the hair was another thing that's quite rough so I'm noticing now this flow of how it rounds quite closely over the back so I'm gonna try to fill that out a bit more and find that kind of shaping to it I've also added an extra Corner pixel in some spots of the black line work because I'm finding that that makes it look a bit more spiky and that really fits for him the final thing then is to maybe find a bit more of an accurate posture for him kind of hunching over rather than standing straight up for this I started by turning the hands forward more like the reference pose and then selected the head and dropped it one more pixel over the body from here I'm continuing to play around with some of the other finishing details like maybe he could look bulkier maybe the feet should be larger and more monster-like and slowly what comes out of this is actually quite a departure from where it began I found this little thing with the foot connecting up to the knee there so he's got sort of this squat Crouch kind of stance and it's interesting because it's not much taller than the original Sprite but he feels so much larger because he's wider and he's almost having to crouch into that same height all right well let's try another one I think I want to try guile because I think it'd be nice to capture his fighting stance and that's going to require putting those fists up and having one of them partially cover the body and this is something I used to avoid in my earlier work to the point where it kind of elongate Limbs just so they wouldn't have to get in the way but this is all right what we can do is work out the arm coming from the front and then just copy paste this over the Sprite and nudge it into somewhere that looks appropriate and doesn't cover up the face and once again I think we can try something different with the eyes the big cartoony look really does take up quite a lot of room so I'm just gonna try sizing them down and creating something a bit more focused for Kyle I also wanted to see if I can make the flat top more Dynamic by just tipping It Forward at a slight angle I've also added this motif of using a dotted line to help separate the hair from the head without being the full weight of a solid line and it also gives it a bit of texture in a way too I've used that again along the feet except in that case it's used to convey the grip on the combat boots and that sort of stuff where single pixels are so effective and multi-purpose like that that's one of the reasons I recommend starting with small size pixel work as a beginner at this resolution it really doesn't take much of an adjustment to see quite a dramatic effect like that in an angle or a small detail on a Sprite so it's really worth playing around with the contouring here and there and just seeing what it looks like eventually you get more of a feel for it and have a better idea of what to expect or what kind of control you're looking for but even a bit of trial and error in the meantime is a good way to find those Solutions so you can see with the whole crew now just how much more the Persona we're able to capture by thinking about those ways to improve by deviating from those starting points and just like how there was a lot to be learned about translating shapes to even make those starting point Sprites there's an entire other education in taking those through this Improvement phase now you can imagine that an alternative way to reach the same result could be to plot out a circle with some eyes on it add some shapes for the hands and feet and that might offer just as much as the Mega Man template did the benefit of that template though was that it was providing some built-in guidance on the character size and the way that color is applied but we can learn how to select those sorts of parameters as well if you recall my second piece of advice for beginners was to try using a limited set of colors and I just want to show you an example of what that looks like I'm giving myself the task of recoloring these Sprites Now using a maximum of only three colors each for Sprites that use Line work quite prominently like this this usually comes down to using black white and then some accent color and you can almost pick any color that you like and it inherently looks interesting when it's the focus against the dark and a light color like this I also have a lot of fun just actually being constrained and deciding on the exact color themes to represent each character's costume and compare with those Originals the color limits here have their own kind of charm and they also force you into just reevaluating how to be resourceful and repurpose the colors and creating all the detailing you can see actually that it wasn't until doing this that I got the idea to add in some highlights for the hair and this is mostly because I was just looking for a way to create more Dimension from this limited palette if you want the real Pro tip though instead of using a pure white as the lightest tone try using something that has a slight Hue to it in this case I've got this yellowy color that's used for some of the skin tones here and this is a great way to squeeze in another color into there and it's really pleasing to look at too because it's a softer contrast than would be the case when using white all right well I think that's about as well as I can cover my early experiences in pixel art and the approaches forward from there like anything there's probably no one size fits-all solution and this was what happened to work for me and set me on my path I wanted to pay it forward for those of you interested in getting started with pixel art and I prepared this empty character template using some of the principles discussed in this video and there's also a handful of colors there to get started that are adapted from the NES style ones that I used today so there's gonna be a link down below to download this image at its original pixel resolution and I hope it can be helpful to some of you out there just as the Mega Man Sprite was to me all those years ago so thank you for watching and take care and keep it square [Music] foreign [Music]

[Music] hello there my name's Brandon and I make pictures out of tiny squares and in this video we're going to attempt to make Sprites in the style of Chrono Trigger which to me is one of the best and best looking 16-bit RPGs that there is I've got a few characters that I'd like to recreate in this style and you can probably already guess like who these are going to be but before that we should just try to figure out what the main ingredients are that go into the look of a Chrono Trigger Sprite so let's pull up the Sprites from the main cast here just in their neutral standing poses we can see that they're all around 32 pixels tall um frog is only 24 pixels obviously distinctly meant to be a shorter character but I guess the takeaway here is just that the uh 32 pixel mark That's sort of the average or the default to start from when you're just thinking about you know your basic like nonf frog human design now the overall height is only one piece of the equation though because you could decide on a certain height for your character like 32 pixels but that's not necessarily indicative of any particular style on its own and one thing that really affects the style is the way that that character is proportioned into that space to demonstrate what I mean we can break down these characters into a few basic shapes and we can do that by tracing out uh circles over the head and the hands and the feet and just trying to ignore things like the hairstyles and the costuming for now and we'll end up with these sort of blank skeletons that are kind of underneath all of that decoration so if there's anything characteristic about the Chrono Trigger Sprites as opposed to the other styles here one of the things that stands out to me is how the Chronos Sprite has a smaller head to body ratio than a lot of the other Sprite Styles here it's more toward a realistically proportioned human and that's not to say that it's it's accurate in that sense it's it's very much still in a kind of hyper stylized proportion it's just that it's like less chibi or less deformed compared to some of these other Sprites here probably the strongest example of that is uh if you look at the Sprite of omori you know you've got a big head big like bubly head that actually kind of takes up more space than the body does whereas if we look at Chrono you know smaller head more room accounted for the body so the trade-off is that there's less space to work with for things like facial features I guess but more space allocated to uh costuming details and and maybe posing of the limbs and that sort of thing what's cool is that if you continue breaking down the other Chrono Trigger Sprites in the same way you can find essentially this same skeleton repeating itself across the roster just in in different poses well again you know for the basic human designs at least uh but it is also fun to see like the proportioning on other types of builds like with Robo so I think what we can gain from this is that when we go to build our own Chrono Trigger style Sprites we can use these sort of formulaic building blocks and pieces these uh sort of stick figure mannequins or whatever to to pose around and we'll end up with something that has approximately the right size and proportion to it now the other component of studying this style is going to be the use of color from a design standpoint each character really has their own strong color things going on uh you know kronos's got like the teal and orange thing going on Frog's got green and blue is is his main colors so as a cast each one fills a nice little pocket of color selection what I also noticed when analyzing these Sprites um from like the pixel art perspective is that most of them here have a palette of exactly 12 colors that are used um all of them except for Aya who has 13 and frog who's got 11 apparently there's a limit of 16 colors per Sprite on the Super Nintendo so these are just under that limit which is totally fine it's just interesting to see that I guess from an artistic standpoint there is this consistency of about a 12color palette that's going on here so let's try to make our own Chrono Trigger characters and I'm going to be remaking some Final Fantasy 7 characters in The Chrono Trigger style starting with cloud and I've just got his character art there off to the side uh to use as a reference for what we'll be translating obviously we'll grab that SK SK framework that was derived from the Chronos Sprite and start connecting it up to give us a basic outline for the body starting with the arms and sort of taking the outer track around those plotted points of the shoulder and elbow and having that meet up with the hand for the boots I notice that Chrono Trigger Sprites often use a prominent cuff so we'll put in some Line work to shape that cuff around the foot uh in between the linework I've used this structure where it's like three pixels in a row and then the next one is like a single Pixel up from that which I think helps to show it sort of rounding around the leg right and with that in place this only leaves a bit of space to connect up the rest of the legs and in doing that I'm trying to capture the sort of poofiness or those the like angled silhouette of the pants and to help that I'm going to use some of the black line work as a shadow on the lower part of the leg and that should help just show a bit of that volume the final prominent features here on the body are the pauldron and the harness that's over the chest and those should be easier Define once we get to the coloring stage but for now I'm just using the line work and the space available to kind of block those in now moving on to the hair I'm going to try to do what looks to be what was the approach for the Chronos Sprite which is basically to assemble the hair out of a sort of Mosaic of smaller shapes uh you know these are both your typical spiky-haired protagonists so uh what I'm trying to do here is use mostly like rigid triangle shapes and fit them together into a similar you know geometry or whatever as the reference art has uh this is also where it's nice to have started from that blank head underneath everything because you're really able to visualize like where the skull is underneath this and kind of build this framework of hair on top of it and giving it that appropriate volume that sits away from the head all right let's start adding some color here and we can do a first pass of just dropping solid colors inside the lines and this first wave of colors is going to be kind of the midtone shades which means it's just like the base color for that part of the costume and we'll come back around with a second wave of colors to add either a shadow or a highlight or both on top of this and get more of that detailed and Polished look so because we're going to be working with only 12 colors we do have a fair bit of freedom to a degree but there's always opportunities to try and be economical with the exact color choices and overlaps uh that being said in in our sampling of Sprites from Chrono Trigger I I found it was generally the case that the skin used two colors there's sort of the base tone and then a shade tone for it and for the most part these two skin Shades were only used for the skin like they didn't really show up that much uh on other parts of the costuming that weren't meant to be skin and for the hair it's pretty common to step through three or four colors depending on how you want to count what belongs to the hair uh for cloud I've stepped through yellow and orange and then there's like a dark red or brown that's used uh to color the line work within the hair and I've even used that same outline color to around the face uh as some colored Line work that's like at the edge of the skin there by the time we're done rendering the approach is going to be that we'll have colored over a fair bit of that original black line work using these different darker tones that suit that particular bit of costuming and I think this is part of the formula as to why these Sprites look so rich and colorful is that instead of black line work you've got this subtle uh colored Line work that's going throughout the entire

Sprite and it just punches up the vibrancy in a nice way this also means though that it it's something to budget out when thinking ahead to the set of 12 colors that are going to be used for that palette as a final touch I'm going to add Cloud's Buster sword onto his back I feel like that'll really properly make this feel like it's Cloud uh even within this Chrono Trigger look so I'm sketching this out on a separate layer just to know um what kind of area it's going to take up and then just popping that back in behind the Sprite for the finalized Sprite work I made a slight adjustment to the colors here that I'd been working with the original palette were all colors that I had sourced from various Chrono Trigger Sprites and I took those and just kind of dialed them into my own selection so I feel like that came out looking pretty good uh I mean he's essentially a rehash of the way that that original Chronos Sprite looks but it was a nice safe one to start with and and just kind of get our bearings on on what the style entails uh next up though I want to recreate Tifa in this pose that she's in from the artwork so I'm going to start by getting those building blocks into place again um approximately in a way that looks like it's this pose for this Sprite I'm also going to do it where uh one foot is more forward than the other we see this on a few of the other Sprites so it would be nice to kind of break things up and try this posture this time around now with Tifa the focus for the line workk is going to be to find Clarity in the arms and the legs and the way that I'm going to do that is by making sure to leave at least a one pixel Channel between the line workk that we can go in later and drop color into we want to have the ability to follow that line of pixels that creates the limbs so that this pose can read properly one of the speed bumps that I'm running into here though is is that there's not really a lot of space left over for the legs after carving out space for the tank top and the skirt so I'm going to raise the entire Sprite up by one pixel which sounds like such a minor thing but it actually gives like the exact amount of breathing room that you need to kind of make those adjustments sometimes with the coloring one of the things I'm prioritizing here is putting in these bright highlights for the gloves and the boots uh her attacks are mostly martial arts focused so it makes sense to draw attention to these spots on the Sprite and using such a strong color ramp like that to finalize the Sprite I'm going through to add the colored Line work and tia has sort of a simpler color scheme than Cloud did so this is a bit more relaxed just finding that space for everything within the pallet of 12 I wanted to dig in on a few of the choices that were made here that I think help add to the uh like capturing the rendering style of Chrono Trigger firstly I I mentioned the white highlights in sort of the highly reflective or areas of interest uh bright highlights like this sometimes also appear on the skin as well so for Tifa I've done a single dot at the shoulder and another at the knee another thing that's going on here is like fitting in smaller details like I've used the Gray Shades at her elbow and wrist to suggest the pads and sleeves that she's got and normally when a detail is this this small or there's not really a lot of room to account for it I might consider losing that altogether but it feels like with the Chrono Trigger Sprites they they really go for this sort of thing like having all those little details of uh wristbands hair ties any kind of belts or straps uh even if it's only like two or three pixels and it's the difference of only a subtle shade of color uh maybe just try to make it work and go for it and then you end up with this interesting level of detail going on within the Sprite and there's some actual intention there because we know how it relates back to the character design all right the final character is going to be Barrett uh given the size difference here compared to the other characters I think he's going to make a good one for testing out this workflow under kind of extreme conditions so I'm starting with those same uh idea of the building blocks just like before but this time we'll make the circles a little bit larger just to account for his exaggerated build in connecting up the line work and and starting to find the structure here I'm mainly focusing on adapting that core area of like the uh The Ripped uh jacket you know vest and trying to give Clarity to the musculature but also having room to create a few of the tatters on the vest there's also a fair amount of layering or costume blocking to think about here for Barrett like he's got uh the the metal bands over his stomach there's also a dog tag and chain there if we want to try and catch that detail so just to give even more breathing room for all of that that's going on I'm going to push the size a little bit taller here the original intent for this was to maybe keep him at 32 pixels but pushing him up to 36 should actually just give an appropriate height to have him like towering above the other characters anyway with the color selection uh I'm very quickly running up to that Max of 12 just because there's a few distinct gradients to try to capture here you know we've got the skin tones uh there's a selection of greens for the pants the colors on the vest and then there's the blue Grays of all the gunmetal Parts Plus the line work that's going to be ass with those so I'm looking for a few efficiencies here things that can uh overlap in their color use so in this case I've ended up reusing um some of the darker Browns throughout the skin and the boots and that brown or one of them also gets used as the colored Line work within the vest I also kind of went all out with the highlights here hopefully you've got an eye to spot those now but I I've dropped in some white pixels along the tatters of the vest just to give it more dimensionality and then there's of course some metallic shine on the gun hand but let's look at that full crew together and we'll of course bring in the rest of that Chrono Trigger cast just to see how everybody fits all together and you know for first attempt I I feel like these came out looking decently authentic I really tried to to figure out you know what the formula was here and I think I could at least imagine these Final Fantasy 7 characters in like the Future Part of Chrono Trigger but we'll close up with some CRT time to look at these in their natural environment so thank you for watching and take care Cub Square [Music] [Music] [Music] oh

[Music] hello there my name is Brandon and I make pictures out of tiny squares and in a lot of my videos where I create character sprites I've shown how I like to build them up from a kind of stick figure construction I've had several questions about this process so I want to share a sprite time-lapse and take it slower than usual because there's a lot to discuss in how to arrange a character pose and also how to go from the initial stick figure to the final sprite as always this is just me sharing my own preferred workflow and I'm not saying that this is how you have to make character sprites but it's what works for me so if there's even a single piece of information here that helps you as well then that's awesome the character I'm gonna be making sprites for is actually our old pal captain Jace who I created in my perspective video a few weeks ago and I didn't really imagine him having much life beyond that piece but he'll serve the purpose of having some simple human character to pose and draw the first thing I normally do is to find the approximate sprite height I kind of know by now that my personal preference is generally around 35 to 50 pixels tall but if you're still new to pixel art and you're unsure about what height to use then I recommend starting small maybe no larger than 25 to 30 pixels or studying the sizing of existing work if you're looking to achieve something similar to that next I lay down some initial frame work for a neutral standing sprite pose and the main focus for now is on establishing the character proportions so I begin by positioning the head size the torso height and the leg height I'm always trying to improve with Anatomy and gesture but for relatively straight legs I tend to use straight lines if you wind the leg head on and sort of an s-curve when viewing it more from the side will be any definition to the legs later but this helps indicate the overall flow right now for the torso rather than using just a line I find it more intuitive to use a wedge or some kind of block shape where it's easier to visualize how it may look when rotated to different angles in a 3d space in this case character is facing slightly to the right so I shade out the side of the torso and I don't really need to position the gesture the arms quite yet so I leave a circle to indicate where the arm might connect in the meantime I began detailing the face and hair just to make sure I could achieve the kind of look I pictured within this space sometimes it can be a hassle to resize and redraw character once they've been made so it's important to have some rough idea of the desired care size and style early on in the process as far as proportions go an accurate human character will be somewhere around seven and a half heads tall on average but more stylized or cartoony proportions will run smaller than that and I wanted this character to be a little cartoony looking so I've used a head size around 12 pixels tall within an overall height of 50 pixels give or take a cue for the hairstyle next I start adding pixels to thicken up the lines and create a basic silhouette for the lakes when moving from a stick figure to a silhouette it's important to try and preserve the angles or curves you've worked out in a stick figure stage while also building up the muscle structure such as the curvature of the thighs and the calf muscles this can be a delicate balance because sometimes even the difference of one pixel in width can make your character go from looking skinny to beefy but I usually try to err on the side of having a more interesting silhouette rather than one that's too flat and simple but of course this hugely depends on the art style and the character design anyway for drawing in the arms I like to start by placing down a circle for the hand and then work backwards towards the elbow and the shoulder to connect everything together for a more dynamic pose it might also be useful to draw a path for the arms kind of like I had done for the legs but in this resting pose I just wanted the hands to be placed where they wouldn't interfere too much with the visibility of the character I create the arm silhouette with the fill color but then I also leave blank areas that define the costume elements like his gauntlets and shoulder armor if the entire thing were created as a completely dark silhouette I might lose track of that placement so I like to leave these empty areas for those costume elements that'll be colored in later and that kind of leads to the final step before I move on to coloring which is refinement of this line work what I'm looking for at this point is to have a readable character design with clear separation in these costume details just from this single color alone so it's a combination of refining the silhouette but also erasing certain areas to indicate those different costume details or just things that I want to pop when looking at the character I mean there's stuff you could write off like oh I'll just fix that when I add some color but you know if you're able to make a readable sprite out of the line work alone it should translate into the final product being that much better by the time you do add color to it in a way this is essentially like doing one bit work where you're just trying to get some strong readability out of only two colors and I think it's important to explore simple styles or restrictions and pixel art like one bit because the limits imposed by that type of prints direction can really play nicely into learning how to create readable line work so here's a look at the finalized line worked with a few of the previous check points at this point we've already got some clear separation of the different armor pieces on the upper body and readable silhouette for the legs that preserve the initial construction along with a few of those cutouts to indicate additional detail like a knee pad and some anklets on the other leg just throw in some asymmetry and interest my next step is to start shading the line work and when I'm creating colors from scratch I like to begin by working in greyscale I gave myself an evenly spaced grayscale palette of brightness values from 10 up to 90 and I don't think I use them all but it just gives a wide selection to work with the easiest place to start with this is to first fill in all the open spaces I left in my line work just deciding singular gray values to assign within each component this really focuses the design choices because we don't need to worry about color influencing the design quite yet we only need to decide how bright or dark each piece of the costume is compared to another from here I continue layering on different gray scale tones one at the time to create more complex shading I start with a single shade tone and highlight tone for each costume elements and then add extra as necessary if I want to create more steps of shading to smooth things together the other thing I'm doing is lightening some of the line work just to soften it up a little bit and then it also gives the opportunity to make the line work of one area be distinguishable from that of an adjacent area whereas if the line work was left entirely as black it would all just blend together I guess I should mention as well this is obviously a sprite without lines as opposed to one that is created without strong line work to border and break everything up I find that outline look to be more cartoony and that's kind of my favorite look for pixel art I also just really like the way you can use an outline to your advantage such as creating sharper corners in a design or breaking up that line a bit to give more control over the shape and angles of certain objects I feel like it also helps readability when placing the sprite on various backgrounds once we've got the greyscale shading in a good spot it's time to add color and to achieve this I create a new layer and set the blending mode to the color option then when I paint on this new layer the color will overlay itself on top of that greyscale contrast that was already worked out in the previous step if you've watched a lot of my previous time lapses you've already heard me talk about the color blending mode a lot and if you haven't I'd recommend checking out my video about creating four color palettes if you'd like this process explained in even more detail and I also show it within a sprite in that video as well if that happens to be a program of choice it can be important to learn how to develop your own color palettes from scratch because then you can really fully customize your art to match whatever vision you might have for it and this method of starting from greyscale kind of systematizes that process and it's really the best way that I found a great color so far the only side effect of using the color overlay like this is that applying the same color over various gray scale tones looks kind of bland because there's no change in hue that's built into it for the Photoshop people out there sometimes I do use a gradient map instead to get around this but otherwise what I do from here is copied a sprite over and then color pick the different shades and shift their hues apart a little bit using the color slider so instead of the hair being the same hue of blue for the main shades and the outline color the outline gets pushed towards purple and then the highlight gets pushed more towards cyan for example it just breeds more life into the colors and I think makes the sprite look more interesting and appealing for the metallic pieces of armor also building a slight hue shift as well it's kind of a range of really desaturated purple colors it could always be left is a completely d saturated gray but I like playing around with adding a touch of color to it something like cyan blue or purple works really well for adding a cool color temperature to the metallic texture and I mean cool isn't cold rather than like what a rad color temperature and then the final step for this sprite is going to be refinement of the color palette and to do that I arranged all the colors as a palette so I can easily look over the full set and make any tweaks to the hue saturation or brightness based on what I feel is necessary it's also a good opportunity to spot any colors that might be close enough to each other that they could share the same identical color for this set I ended up making the shade color of the pants and the colored outline of the armor into the same color the other thing I noticed at this point is that I wanted to introduce more color steps into my gradient of the gray tones just to get more options for smoothing the shading together and so this is the final I sprite after the refinement another thing I should mention is that I always find it best to design sprites on some kind of a neutral background that doesn't influence or interfere with the design too much meaning no color to it and not too bright or too dark that way the final sprite can hopefully suit a variety of lighting and backgrounds all right so this is great we've got a single sprite but if we want them to be doing more than just standing there we need to explore a few other poses to redraw the same character in a different pose I like to kind of extract the stick figure skeleton from the existing sprite and then pose that again I just use a combination of simple shapes and lines that way we can find really basic construction and proportions that make up this finalized sprite and then it's much easier to play around with a simple stick figure skeleton I wanted to try a completely different and sort of challenging pose and I sketched out this one of him sitting cross-legged eating a bowl of ramen to get going I took that initial stick figure skeleton and started creating the pose by moving each piece into position one at a time again at this stage we're just looking for the pose to be recognizable through use of the stick figure one of the things that happens when you go to make a pose is you think like oh yeah I know what running looks like or I know what sitting looks like and then you go to actually create it and it's like wait how do lakes work again another way through this is to get in that pose yourself or to look up reference images online just to study the angles and the orientation of the limbs this is why the stick figure skeleton is a really nice place to start because it's a simplified form where you can study these poses and then place the limbs and joints based on what you studied for this stick figure pose I also added in a small shape to represent the bowl of ramen because I want it to work out how the arm and the hand would need to be placed to be holding the bowl and also making sure that they're all positioned above the legs so it doesn't really block the recognizability of that pose and then similarly I worked out the positioning of the hand holding the chopsticks at this point I didn't really know how I'd render the chopsticks in the final appearance but having them as part of stick-figure early on will make sure I keep track of their placement throughout the process in order to build up the sprite this is again a process of working with the character silhouette and the line work and just trying to find the cleanest representation of that line work before moving forward for this sprite there's the added importance of having clearly visible accessory components like the bowl and the noodles as well I love how seriously I take this but it's true though if you're making a sprite that's trying to show or communicate something in particular you need to focus the attention to that thing so in this case it makes sense to prioritize the bowl and the noodles within the line work the other part of course is that now we're looking at the character head on so requires reframing some of the features like the headpiece and the hair and I also wanted the face to be a little bit more exaggerated or comical here or you know it just looked like he's really enjoying the meal so here's a look at the completed line work we've got the bowl in the noodles and the stuffiness face expression and then I've also used some heavier shading here to create a bit of dimension in the hairstyle on the left leg there's a little bit of a dashed line just to remember the angle of that front leg and not have the whole thing become a sea of dark silhouette moving along to color what's nice here is that we've got the palette established from earlier and I'm already familiar with the way that it was applied across the design so I can start by filling the line work with a few solid base colors I also already had the bright yellow from his anklet and I reused that as the noodle color and then for the bowl I started with the lightest gray tone just so it's bright and draw some attention that way with the rest of the armor kind of starting at a mid-level gray I continue along applying the shading here in a similar manner to how it appears in the first bright by coloring the outlines and the shading of the hair and giving that slight amount of specular highlighting to his armor pieces I keep the first bright on screen just so I have it to compare back to and to make sure I'm achieving a reasonable level of consistency between them because the lighter tones used in his armor I ended up introducing an even lighter shade to use for the bowl it's almost kind of a bright white just to make that piece super noticeable there's also a highlight of light gray on the tip of his front foot again just to help it pop a little bit more to kind of draw attention and anchor the cross legged sitting pose alright that was a pretty specific pose so let's do one last one with just a bit more action to it for this one I'm gonna do a kick so I begin from that neutral stick figure and start by moving the arms out of the way then bringing the foot up to kind of find the kick height and finally work back from there to lay down an even angle for the leg from here I move the shoulders and ribcage piece around just to give them a more dynamic angle and also to work in a bit of a twisting motion around the torso continuing on it's kind of the same old story of filling out the silhouette and refining the line work in sharing this process I feel like it's also important to mention the different ways that you could actually make pixel art I actually just use the trackpad on my laptop so that's why I respond to having a process that lends itself to basically clicking a single pixel at a time or just using very short motions like this if you're the kind of person who prefers using a drawing tablet you'd likely take a different approach especially if you prefer to draw a character forms immediately rather than building up this kind of framework first a question that comes up a lot with route to pixel art is like do you need to know how to draw to make pixel art and the short answer is probably no not really at least not when you're first starting out but at some point if you want to continue improving or developing your own style I think it's really important to learn like some drawing fundamentals outside the typical scope of pixel art and so a workflow like what I've shown here is just kind of an attempt towards that sort of idea and it's what helped me have more understanding and control over what I was making alright we've been through the coloring twice now so I'm just gonna jump ahead here to the finalized kicks bright with a few of the process check points you'll notice I chose this foot with the anklets for this kick and I was kind of messing around with the idea that they'd provide some sort of extra power for that so I also tried experimenting with a version that had some like kick II swoosh lines to bring in that ability idea and here's the full set again altogether just to wrap this up so this is pretty much the process I follow to make most of my character sprite work in some cases a few of these steps might not be necessary like in cases where the character design is quite simpler than this or if the art style is something entirely different but for the most part this is a pretty good representation of my typical workflow I hope you found it interesting and if there's anything that helps you improve your own work in here then that's even better so thank you for watching and take care and keep it square you g'day pals welcome to a new video today i wanted to talk about top-down pixel art following on the previous video on pixel art styles i want to talk about styles in this perspective lay down some principles for top down pixel art and then maybe do a bit of a tutorial on starting your own okay you ready let's get into it all right top down pixel art this is an area that i don't have like a whole lot of experience in i haven't made many games with this uh format and um yeah it's not it's not something i spent a lot of time doing but it's been really fun doing a bit of research and breaking down what i've noticed and what i can share with you today so let's do uh do it like we did last time and look at a nice big grid of games that have this this top down feature so what is top down it's essentially a perspective where the camera is above the action looking down in a kind of orthographic way where everything is laid out as if it were on a grid right so you can see the perspective of things like uh square plots of land appear rectangular to us they don't appear to fade into the background i'm essentially going to be protracting out from games that inspired the ones that we'll be talking about today here we've got the original final fantasy this is uh that's on the nes this is uh we're looking at secret of mana and then uh chrono trigger this example here of chrono trigger is kind of like the one outlier where you will ever see a sky and an ocean in a top-down setting like this this is a very unique perspective trick i think it works okay in the snes but it's a very seldomly used technique and it can go wrong very easily so use it at your own discretion working back on handheld we've also got the pokemon series of course and on the snes uh the legend of zelda are linked to the past this is i think probably like the granddaddy of most of what we think of as like um kind of like in vogue or fashionable in a top-down game i think a link to the parts really set the scene for a lot of that stuff so pulling out we've got a lot of more modern kind of indie games that use this top-down format obviously stardew valley is one that is very popular that people make games in the image of we have games like cadence of hyrule which are almost like a sequel to or like an indie take on a sequel to a link to the past you know in the same world same sort of palette this is moonlighter more stylized flatter this is garden story a much more stylized and flat example as well and then moving sort of in this direction we have games like eastwood like we saw last time this is battle axe really really interesting very detailed art style here and eldest souls which i think is also very gorgeous too i've tried to stick mostly to similar environments here so that we can make good comparisons so we're looking mostly today at outside environments grass forests that kind of thing one thing that stands out to me as i look at all these screenshots beyond the fact that they all share the same perspective is the way they manage contrast and that's what we're going to be talking about today for the most part unlike 2d side scrolling games where the character sits at the junction of the ground and the sky and the sky is quite washed out and can allow the character to stand out against it top-down games have all of those layers compressed and you're looking straight down at them so you know you don't see the sky you see the character the ground grass you know out of bounds areas versus areas that you can walk on those are all mingling and sharing the same space and so the way that these things are presented visually is really important because it's much more important to explain to our eyes visually you know what the different parts of the scene are we can't just expect certain regions of the scene to mean certain things like we can inside scrolling we have to be able to interpret that by looking at them as we scan around the screen so the most obvious point here has to do with grass right grass is present in almost all of these this is really obvious in games like a link to the past where the grass is essentially one flat color that's uh sort of spiced up with a few different sprites here and there in games like this you know you can imagine if we tried to draw every blade of grass here even with the the colors that we that we've chosen here which are quite similar to each other if we were to fill this entire space with those grass blades it would be harder to tell link apart from the ground it would create what we call visual noise so areas that we walk on are really obviously areas that we want to be kept clear of focus the character of course needs to stand out and in all of these games um using more higher contrast colors is an easy way to actually have the character stand out things like outlines tend to be used the same way they do in other games to have the character stand out and i would say probably even more so having that black outline is used very commonly in this perspective having there be more dense uh details and more saturated colors is also common what's interesting in top down games is because of the way the light is often coming from the same place the player is looking from it's harder to use light and shadow to make a distinction between things that the player can walk on and things that they can't in a platforming game that's really obvious because we can just light the top edges of all the things that we want to be able to step on but here you know from the top looking down you know we should be seeing a lot of stuff be lit up and not a lot of stuff uh be hidden from the light so these games tend to use noise as a way of modulating the difference between things that you can and can't step on and this is part of contrast as well so looking at eastwood here you can tell that we can't walk here because of just how high contrast and noisy it is right it's kind of like a way of passively telling our brain hey this isn't somewhere you can go i think this is something that kind of comes back to nature in a way i mean like it's almost the way you think of forests right versus clear paths we want to walk in bright open areas we don't really want to walk in noisy areas where it's hard to see predators so just looking at these i mean i i think i make the point pretty clearly but if that wasn't enough let's look at an example where i would say in battle axes case i think the art here is is incredibly high quality it's obvious how well these details are constructed and that the colors are beautiful nice to look at but i would say it's quite difficult looking at this frame to actually tell the difference between what's environment what's characters what's enemies what you can pick up and where you can walk these bounds these sort of variances in contrast and in the difference between signal and noise aren't really here because everything's detailed and everything is quite saturated so what are these principles let's take a look at breaking them down okay so one of the most important principles is modulating contrast and you can see here for things that are supposed to be obvious to us like the player objects you can pick up npcs these are all important to us right we need them to be signals essentially that we can see on screen at a glance and we compare that to things that are of less significance so out of bounds areas and terrain high contrast can be used when thinking about these things and for the most part all of these things should be able to be found on the terrain so the terrain being below this line has low contrast and left of this line being insignificant that's kind of what we want to use for our terrain because all of this stuff sits on top of it so it needs to be different in contrast on the flip side out of bounds areas they can be high contrast because they don't overlap with the terrain and actually the difference in contrast between out of bounds areas and the terrain creates the separation in a lot of these cases so a little design principle that i came up with here was the idea of separating clear paths versus dense forests those are the two kinds of domains that our eyes track against and those serve as the canvas on which our npc's objects players sit on i want to take a really quick second to stress the difference here between noise and detail signal and contrast okay so noise and signal are kind of two lines on a scale that relate to how eye-catching something is okay so both this and this are not very eye-catching even if this is more detailed right the the the non-walkable terrain is detailed there are details there but their signal to noise ratio is low because of the amount of contrast used the distance between the darkest colors and the brightest colors is very low it's a small distance so our eyes are comfortable just glossing over this area even though it's got detail on the flip side looking at this bush here we can actually see quite a lot of contrast and our eyes are drawn to it mostly because of the separation between those black outlines and the green leaves the way that our eyes work essentially our eyes are really great at shape detection and so whenever shapes are closed like this that draws our eyes to those areas and especially a cluster of closed shapes is very very uh eye-catching to us it's just what our eyes are like designed to stare at so if we zoom out here and look at the entire canvas you can kind of see the different parts where your eyes get caught on areas where the signal-to-noise ratio changes right so this scarecrow standing in the sort of blank field there's a lot of detail in these textures right this is it's quite uh noisy this texture but because it's noisy and not very high signal it allows something that does have a lot of signal to kind of like stand out to us right we see the scarecrow the closed shapes the black outlines and immediately our eyes get drawn to it so now that we've established the role of contrast in making a distinction between signal and noise how do we explain the difference between these screenshots in terms of their actual aesthetic interest you know how do we make things look good or interesting when we have low contrast so preserving interest when it comes to preserving interest and actually having lots of detail in your image without drawing too much attention this comes to the question of how do we have differences in color without differences in contrast okay so light can be broken down a few different ways oftentimes we talk about rgb for red green and blue those are the channels that we use to actually store the data that describes light but in a lot of uh image editing software you'll also come across things like brightness hue and saturation to describe colors okay and what we can do is we can take one of these axes and we can change them slightly just one of them and then detail our image with just that slightly different color and what you'll end up with is something that is low contrast but high detail so here we've taken this green sort of area of grass and if we just move up right into the into like a slightly different brightness we can draw you know a second layer of grass it's still very low contrast but it's high detail likewise you know we could instead change the hue we've got yellow tones and blue tones here instead again high detail still low contrast right this won't attract your attention quite as much as if there was a black shape standing you know in the middle of this and finally we could change the saturation so you know we've got more saturated or less saturated greens more grays more vibrant colors and uh that's like a very easy way that a lot of these games create interesting terrains that need to be low contrast but give them lots of nice detail so yeah changing only one or two of these channels very slightly can be enough to create that diversity to uh to really fill out your image even if you've established low contrast in an area now i can hear you saying but adam if we just change these colors like this what happens to our palette this is a really good question i know that i've stressed in the past the importance of a palette and with a lot of my games most of my games i do use quite a constrained palette so the danger of actually betraying that palette and just having colors go all over the place becomes a question and and the threat there is that we lose our sense of style right the consistency goes out the window if we just have colors everywhere so how do we preserve a style something that feels cohesive when we don't have a palette because we're changing these colors so subtly that's what i want to talk about next so a quick kind of like statement that i want to make not necessarily a definition but just a property of style is that style's really about consistency in decisions right when we say something is consistent what we're really saying is that all of the different decisions that go into determining what that thing is follows some sort of pattern that we can follow right that we can we can identify and that the times that that pattern is broken are either none or very few times okay so it's about consistency in the decisions even if hard limitations like palettes aren't present so let's look at stardew valley for a second stardew valley has a lot of colors here on screen and a lot of different colors right so even in the green space we've got a lot of different greens being used here now you might say oh you know they're breaking palette in all these areas but you'd be lying if you said it didn't look good because it does look really nice this is partly because it models reality right if you go to like a forest like this you see a lot of lush and varied colors of green even inside of just like looking at the grass you know different trees different bushes different species of plant all have different hues and and even in a single plant you can have lots of different vibrant shades of yellow and green so this lush rich use of color is really important in actually bringing out what makes stardew valley so so good that aside there are areas i would say in this frame where we could criticize and we could say okay where are areas where decisions are inconsistent so the first i would say is how brightness is modulated looking at the tower that asset from the bottom of the tower to the top of the tower all of the bricks seem to be following quite a flat view of color they all have you know light edges on the top and darker parts in the creases between the bricks but looking at the top brick and the bottom brick they all have the same sort of base gray but if we look at this sort of rock face here the colors at the bottom of the rocks are much much darker in high contrast than those at the top so the way that the brightness and contrast are being modulated is different in different parts of the same image in ways that should be the same right these are both vertical areas that are being elevated and sun would essentially because they're all above ground the sun should essentially be hitting them pretty much the same way so i would say this is an area if we go back to the to the original this is an area that to me sticks out a little bit a second way looking at the same image would be tone okay so this same rock face here if we look at this yellow it's almost the same as this kind of clay dirt that's uh just by the edge of the water here they're essentially the same because they're all just the dirt that's you know the grass is growing out of but if we look down just before we hit the surface of the water we have quite a shift to into like warm tones obviously the water makes it cool once we get past that but just here we can see there's like a drift into like a warmer orange yet over here as we go down into a more darker space instead it becomes a lot cooler and a lot greener sort of like a greeny brown versus a warm brown so whether or not either of these are realistic i mean you could make arguments for the warm tone because it's kind of establishing this sort of very summery uh nice light effect in the same shot having this kind of like very high contrast gray green brown situation is a little bit uh conflicting right like why would these colors go in different directions under the same circumstances so again not necessary for us to have a palette but it is necessary for us to make similar decisions across the scope of an image now let's compare that to eastwood eastwood is in my opinion this is like a masterpiece when it comes to just the visual composition i think this is like the best example of the set that we looked at because it's so easy to tell the difference between uh where you can walk and where you can't but there's such a consistency in the rules that describe these things okay so let's take a look for a second all of these colors here are in the same sort of color space right it's very low contrast overall and that makes it seem very stylized because there aren't any places in the image that break this there are no parts that are suddenly very bright versus very dark they're all in the same color space all of these colors all of the brighter colors are slightly more saturated and slightly warmer and across the image contrast and brightness are actually varied based on pathing which is really really interesting what's really brilliant about this is the way contrast is being used in a very very subtle elegant algorithmic way to actually tell you where you can and can't walk right so navigable areas are very soft very similar you can easily just wash your eyes over this and not really see anything but as soon as we get to this edge you can see the contrast ratio is a lot higher and the differences in color are a lot higher just at that edge right like this yellow and this green next to this darker color here those are the most different colors we've seen as we've moved across here so far but then as we work back we can see the brightness starts dropping down and the contrast starts cutting away so looking at it here it's so impressive how consistent and smooth that transition is but how clear it is as a rule right we can see where the rule is introduced and we can see it over the course of these sprites fade into something that is less eye-catching but definitely less navigable there's almost like a sense of like visual gravity that's kind of like pulling you out of the dark areas and towards the softer lighter areas and you know this is something that again it's kind of modeling reality in a way but it's also very very clear visual communication so shall we have a try putting some of this together let's do it all right so here we are in a sprite and i've just got a reasonably large canvas open this one is a new canvas it's 640 by 360 set to indexed the reason why i'm using indexed here is so that we can shift colors around as we go because we're going to be building out our own palette here all right so i'm just going to like wipe my entire palette i'm going to start with a new index and just like come over here to the spectrum and pick a color in this case we don't have any reference so i'm definitely going to be changing this but i'm just going to start with something so let's delete that and that's a decent green it's not a bad minty kind of green if we want to go for something like more washed out or pastel i'm not really sure what i want to do with this yet so we're just going to start copying this color out and then maybe like shifting it right and i'm going to i'm just going to do what i usually do which is shift down and towards a little more blue as we go down and a little less saturated but i'm going to be very very subtle with this so you can see it's almost impossible to tell the difference between those two colors but this is going to be walkable terrain so we're just gonna create like a nice maybe like a kind of like a t intersection of uh walkable terrain so i'm gonna make some colors now that match this palette uh and i'm gonna build them around this minty green so maybe we'll take another color here and we're going to take the same thing but we're going to shift it kind of like up a little maybe this color can be like the dirt underneath nice cool so here we can do some you know patches where the grass didn't quite make it and our goal is really to to minimize the amount of draw that this has on the eye we're really just having very very minimal impact on on the vision here we want it to be you know different enough that we can see some difference but uh not so different that it catches our eye now i'm going to take this green and create a bit of a like a highlight color maybe one that's got some more yellow in it and it's a bit stronger we'll add that to the palette as well i'm going to use this as our uh like highlight on any blades of grass that we have sticking up i think we probably want to go a bit more green here and then we can underscore those with some shadow and that creates a nice very very soft bit of detail there cool so we've already started with very very simple very soft yet different details just by changing saturation a little bit a little bit of hue slight amounts of brightness here and there and already we've got you know three or four different colors that we can use to create some unique shapes and interesting details that don't pull focus um you know we'll still be able to look at this and know where the player is supposed to be walking now i want to take this dirt collar and i want to do like a shadow version of that so that we can create some depth between the grass and the dirt so i'm going to lock this off again go down slightly from this color here just drift down and then add that to the palette so i'm going to take this color here that's like slightly darker and maybe we can start using the shade brush to create a bit of a distinction between these two so i'm going to take my two colors go into this press shading and then start drawing underneath here right on the top edge because that's where the shadow is going to be visible now immediately i know this isn't going to be quite right and part of this is the depth part of it is the color but if we unlock this and then just start shifting down again we can actually start working in where we want to go and i think yes desaturating much much more helps us out quite a lot there actually okay so we can take these two colors and just wear them back now we want to be very subtle maybe even more subtle than this maybe we could go even closer it's just enough to draw some distinction and then we can do things like add little cracks and extra details here if we want and just finding shapes here that work really i'm not really trying to force too much just doing a little bit of exploring it's nice to fill in the edges usually rather than the center the center keeping it nice and open gives us a bit more space now for this to really make sense um to actually explain the differences between these two colors here we want some objects above the uh above the terrain maybe some trees stuff like that and i'm going to lock this off gonna take this color here and then shift it way down let's create like a lot of contrast and add that color and then see if we can just draw some some shapes here that really create that distinction now i want to play with the idea of there being shapes first and then we can work in the details of what those shapes are but really i'm just talking about like okay how do i want this you know this edge to look this is obviously not you know game assets this is just like a picture it's just a mock-up so we can really do whatever we want here and again my goal at the moment is just to just to find the right colors so i think i want it to go a lot darker maybe oh that sort of blue is nice yes i like that quite a lot let's lock it off again find an even darker tint and then we can paint that in as well so i'm going to start just filtering in the idea of there being sort of like trees and i'm sort of trying to create you know i'm thinking in my mind that the trees are kind of like this right maybe even a little more top down like this so that we can get more of a perspective on them yeah maybe maybe closer to this and that's just what i'm doing as i'm filling in these shapes here you can see this if you complete that shape you know that's kind of where that's going essentially but i'm using a big brush here because i don't really know yet what i want to do with this so there's no point going in here with a single pixel brush and defining details when we don't actually know what the overall shape is going to be and that's kind of the the motto of doing this kind of mock-up work you're not creating assets yet i mean you might like some of these things might become assets and if i'm going to do some sort of tiled trees like set up here you know i might take one of these zones and turn it into an asset but right now all i'm trying to do is get the feel right so i don't really know exactly like what i want out of all of this so there's no point going for quality when i can go for quantity right it's really about the volume of the space then the exact details of what's going on at least at this stage later on we'll add some details and that's kind of what i'm doing here really i'm just thinking about okay what do these edge shapes look like as we work our way over here and then now we can start coming back down here and following this this line a little bit more and now that we've got some sense of shadow something that's actually casting this darker shade now it makes a bit more sense what it's there for so at the moment i'm just sort of tearing in a little bit into this space just to create more of those shapes just to feel it out a little bit and what i'm really looking at is the contrast between this space and this space and where i want this to be pulling focus and where i want it to not be pulling focus so most of the contrast when this is all done will be here at this line and then i want to wear that back so there's much less at the end so i'm sort of thinking about this as a gradient of contrast and detail so there's going to be more noise you know more shapes here at the edge of the tree where the light meets the shadow right and it doesn't really matter at this stage what those shapes are just sort of blotting them down there's going to be light and darkness all over the place but you can kind of see what i'm going for here right where they're just becoming more sparse and less obvious less eye-catching as we work back maybe this is an opportunity for a slightly different tone so i'm really liking where this is going um if this is looking a little strange right it's because we need to work in these edge details so if you take the grass shapes and start plotting them up through this you'll start having it feel more like it's a it's a change in the grass rather than just some random you know brown splotch so like this you can even maybe work it back a little bit and you could start also fanning it out by uh having like the different shades sort of cross over in little bits here and there and it might be nice to just work in some of these extra details just little highlight bits and darker bits just little bits of variance having the colors be shared like that helps it be more consistent so so even though we're picking these colors kind of at will we are still up obeying the kind of logic of a palette and we're trying to reuse colors where we can so we could probably take some of these details and just import them over here maybe we'll just be a bit more careful with that you can start seeing now the traces of like an asset being created right like these details being repeated this really does feel quite quite close now to like an actual game screenshot i think we could go a little bluer with these colors i'm just going to shift them ever so slightly over and it might be a cool idea to put a building here i'm thinking maybe like a house of some sort and i want to try to start with a color that i've already used and i still don't really know what the shape of the house is yet or anything like that so i'm going to try and just create something and not get again not be too picky about any of these decisions that we're making get some perspective on here i can kind of think of this as i really like kind of tudor style houses i do a lot of fantasy stuff so i'm going to see if i can create something that's kind of fantasy so let's go for maybe like another few other colors here i'm just going to start picking them up a bit and then placing them down and then seeing if i can work them to where i need them to be that's not bad see if we can find the back edge as well some level of depth here on the front and then instead of this color i think i want to go for something a lot more generally just eggshell and i'm actually kind of feeling this could be a bit bigger this building maybe we double it have the same detail on the back side this kind of reminds me of the like vermillion city gym or viridian viridian city jimin pokemon blue it kind of reminds me of giovanni's gym cool now as i look at this this feels much much more saturated than this so i might take like all of these colors and up the saturation so i'll take all of them like this just grabbing all of them gonna go edit adjustments hue saturation and just see if we can play with the saturation just bump it up until it fits i think even that is a lot better if we go all the way it's too much but that's where we were that's feeling a lot better it doesn't need to be much brighter or darker i think that the brightness is fine but i think bumping the saturation up is the way to go there yeah that's pretty good and i'd like to add a color as well to this that's kind of like a tile highlight you know just like the top edge of some of these roof tiles just to give it that specular look right where the light is catching it and these details help to really tell the difference between like something that's a solid that we can't walk on and something that we can walk on right all of the saturation and the contrast between these colors here is very close together these colors very sharp very rigid so that we know because of the difference between these two colors there's something happening here and it's not this very smooth transition it's not somewhere we can just walk freely so then i would just consider like what these roof tiles look like this might be a little too noisy at this point maybe we're pulling too much focus that looks like it's going to make a nice texture there as well too having these run along and i probably wouldn't go crazy i wouldn't like put the same shape all the way up but just hinting at it here kind of gives us a nice uh a nice start and if we can kind of work it into some of these shapes here if we can you know indicate that there is a texture in the points where elements meet then we don't have to tile the entire space now i want to make quite a distinction here between rooms that we can enter and ones that we can't or at least like places that are doors and doors and top-down games tend to be very dark right very dark shadows so that's what we're going for and i just want to go a little darker right here on the doorstep and in fact maybe all the way around the building if it's casting a bit of a shadow so i'm just going to go to the layer below take this color and then go to the next darker shade and put that all the way around and this might be time for another color i really like the idea of almost outlining it with the really hard shadow but i just i think we need to be a bit more definite as well with our shapes and what we're doing with them and honestly i don't really know if you would see this shadow like this but it's kind of interesting to look at so i'll keep it there it feels it feels kind of right now of course again because it's a shadow it's going to have these grass shapes blending the parts where the shadows meet because underlying through all of this is grass right so even here you would see grass kind of adding some noise to these shadows it's a bit tedious but you get the you get the idea i think i would probably add another shade to this just to separate this a bit more i feel like the the shade we've got these two colors are very close together we could just go a bit darker now what i'd like to do is see if i could bring this set of trees over into this space right now they're very it's very freestanding it feels like it's its own thing but if we can bring the trees up into it then i think it'll be perfect so let's put it on top of the trees just to create some shapes for context shadows are an excellent way of grounding your objects into the scene once you cast a shadow from something suddenly it feels like it's there otherwise it kind of floats it's a bit bit strange to look at now of course i would never make a game like this i wouldn't just sit here and especially an rpg and just draw shapes for every screen in the game you would want to create assets out of this stuff so you know rest assured after doing a mock-up like this you could take some of these elements and turn them into assets you break them down into pieces and start making tiles out of them stuff like that what other things can we do to kind of create some more interest in this scene we could probably add something like a road in here you know it wouldn't be out of the question to have like a footpath coming from these buildings like this that's actually not so bad you know what i'm going to do that and we'll see if we can just make this without ruining any of the detail we created nice okay i'm really happy with where this is right now i think from here the only thing i really want to show you as far as like sending you on your way on this intro is just getting some detail on some of these trees i won't do the whole thing because it's going to take ages but just to show you what that might look like essentially it is a texture so it's going to be trying to create some repeating shapes that indicate you know the clusters of of leaves that the trees are made up of and we want to use some nice interesting contrast with the colors and we want to be able to repeat those textures while changing the colors into something that's lower contrast and more homogeneous as we move to the to the edge we've done trees like this before in the alpine parallax tutorial that i gave so you should be familiar if you follow my tutorials i'm a little concerned here about the fact that these are so similar to the color next to them but we might be able to to do something here we'll see what happens i want to really shade the tops of these trees more than the bottoms i think we want to go into a different color space i think it's too similar i would like to go towards something more blue and just to break down my thought process on this essentially i've got like let's just say we've got our shape that looks like this right this is like the the overall shape that we've got for the for the top and then every shape beneath that is kind of like a flick like this duplicated a few times okay and that's like a layer and these trees have a few layers okay so let's just pretend these are some details in our tree at the top i want to see these shapes inside of this so i can take this and start working them in and then if i want to go even further these shapes would probably have little kind of like details at the very edge where they're catching the light a bit more so i could do the same thing internally inside of this but one step up so we could call this the body right of the of like a frond so the body color here is the same as this and then the highlight color is one step up but in this one the body color is the stepped up version and the highlight is even higher okay and that lets you layer them see and now we've actually created one two three layers of the same pattern but with three different color steps right we've descended down we've descended down the space over the course of those three steps so that's kind of what i'm going for here is applying the same logic but stepping it down as i go and maybe for the sake of argument we'll just see if we can do like the bottom of these trees i'd like to just touch on that so i'm just going to take this timber color and see if we can work it in at the bottom of these trees and i could see us going a lot darker with this honestly i could probably take this color and just drop it down another notch maybe shift it towards blue oh that purple is kind of nice that's not so bad though that's okay cool not a bad start this definitely needs like a ton more detail you know across all of the areas you know these trees are nothing yet they're all just a big big mess but you can see kind of what i'm going for here i hope much better so i think this would be the shape that i would sort of work with from here just take this and see if i can like copy it and paste it and move it around and replicate it around the map you sort of have to go from the top down if you're just going to be copying and pasting so for the sake of this lovely tutorial i'm just going to be copying and pasting this shape just to give you an idea of what it might look like if we're closer to being finished uh i want to replace all the colors down essentially and just knock out all the highlights so they're flatter right there's less range now because we've just taken away all the highlights which means now this front edge of trees is a lot more standout than any one that we put behind it so i think we'll leave it there otherwise we'll be here all day and tomorrow as well i hope that you learned something and i hope that you have a bit more confidence now moving into the start of your own top-down journey i will probably elaborate on this in the future doing things like uh character animations or objects things like that and uh yeah that's about it for this tutorial if you enjoyed what you saw today please consider subscribing and uh until next time hey pal thanks for watching and thanks most especially to the patrons and twitch subs who support this channel and my gamedev project insignia to find out more click the links in the description below and uh if you like this video tell youtube by clicking the like button and then youtube will tell me and then i'll make more videos that's nice thanks again and until next time

it's not necessarily a question of having to study history it's just knowing that doing so gives you an opportunity to channel your creativity [Music] i've been getting a lot of feedback about my pixel art tutorials and one of the ones that i get a lot is to go into a little bit more detail about character sprites we're going to go for a little walk through history and examine a few different areas where we see some really original sprites and character designs we live in a world where you can't just release a game in isolation and expect people to come and find it you really do need to be out there sharing your work and getting people excited about it and people do really judge books by their covers so having really interesting sprites really interesting characters and screenshots is a pretty much direct path to getting people interested in what you're working on okay so i want my game to stand out i want my screenshots to look good how do i do that what are the elements of that let's start with the fundamentals silhouettes are basically the most fundamental thing if you reduce everything out of a character design what's left is just the outline it's the first thing our brains do which is define outlines identify objects from other objects everybody knows what this is it's so clear based on those simple fundamental shapes that make up the outline of the character this is obviously mickey mouse mickey mouse has giant round ears big shoes those elements those shapes that make up the overall shape are so clear that you can't mistake them even if all the color gets taken out all the shading all of the interior details we still know who this is this is the like primary uh most important thing when it comes to designing your character anthropomorphizing animals making animals look more like people is gonna come with some proportion changes but what if you're just making a person right a lot of your characters are probably gonna be people because a lot of our stories come down to people there are still things you can do to create really interesting changes and really stand out silhouettes that have to do with yeah shifting those proportions right to create a style that's unique this game is called swords and sorcery it is on mobile and tablet and i think it's a really excellent example it was very popular when it came out i think in 2010 from having this really really unique style likewise games like fez you know this shape here this sort of round pill shape uh very clear eyes the two dots with the flat mouth very very standout and the proportions that aren't necessarily human but for a bipedal character you know that's something that that makes a lot of sense for us and this goes beyond individual character designs into entire ips so final fantasy um you may if you're younger member of the audience you may not remember this as final fantasy but prior to final fantasy vii almost all the sprites for the games um specifically in battles and stuff looked like this this look is wholly uh owned pretty much by final fantasy characters and there are elements of not just the silhouette not just the proportions but the design of these characters that make them really obviously final fantasy characters specifically the eyes so we've talked about silhouette we've talked about the proportions of a character that can make up that silhouette but what about if we want something that's more natural how do we then still further differentiate the character and the next level is posing this one i mean mega man's jump sprite is i would say probably one of the most iconic sprites in gaming in gaming in the anthology of video games it's not a particularly natural pose but it stands out so clearly this silhouette is unmistakably mega man and it went on to inspire other games right games don't just exist in isolation this idea of like having the legs sort of be splayed a little bit apart the arm out as the character jumps it's persists even to today this is bionic commando on nes capcom fans really really understand this sprite uh to the point where i think in marvel vs capcom 3 uh this character is like uh the way he jumps in that game is just as lanky as this uh as like an homage to the original so we've talked about again silhouettes proportions we've talked about poses now let's talk about inside of the silhouette in classic games like games on the nes uh the amount of color work and even the pixels was super limited games like ducktales here had to go a long way very very careful pixel placement to not only create uh this character sprite which we're all supposed to know who this is already i mean in ducktales it's an existing property before the game so they had to do a lot of work to obviously isolate this as strictly scrooge mcduck right it's not just a duck it's this character that you know from tv rendered in you know you know 32 by 32 square at best the things that make this character stand out are the isolated elements that we've come to recognize from this property so there are also obviously original characters um this is shovel knight you know him from the game shovel knight i have a video series on this channel where i spoke with nick wozniak about this topic uh and about the design of shoulder as a character and he spoke to silhouettes as well and their importance and his his uh focal point here was really about this uh t-shaped opening in the helmet something about this shape being the darkest color this black shape accompanied by obviously the shovel uh head and the the horns coming out of the helmet i mean it's so it's so iconic that games since then have sort of gone on to to be inspired by it is very boba fett from star wars also things like like uh props right so we've got the original prince of persia here that game is known for this framing this two characters facing each other in a duel with single pixel swords uh stretched across you know like that's so ingrained into the history of games and in pixel art games specifically that you see it echoing later in games like nidhogg it's not only communicating at this point what this game is about it's an opportunity if you are going to look through the history books and reference something else do it in a way that's at the very least conscious of where those details come from and what those details are going to lead the player to evoke i wanted to talk about multiple characters in the same game that have unique elements but still feel like they're from the same universe i would say the original final fantasy vii uh which was remade recently but the original version does an excellent job of this probably accidentally almost to an extent this is one studio doing their first 3d game and the technology really limited them in how they could actually create these characters very very limited polygons very very limited motion not a lot of opportunity for getting really creative with intricate details so in a way it's very very similar to pixel art as such the actual elements themselves become really really identifiable this choice of making the hands just these sort of like cowbell squares it's hokey but it's so recognizable every character has hands like this we can look at the silhouettes and see huge differences between them right look at barrett barrett is really triangular this comes from the character design if you look at the concept art for barrett that translated back down because they have so few polygons to work with it's really important to use those elements that they have to actually differentiate the characters cloud obviously with a giant sword spiky hair it's clear who he is sephiroth with the ridiculously long sword i mean part of these are anime tropes but they're also video game tropes and they stem from places like this so we've talked about silhouette we've talked about proportions talked about poses props and design elements of the character colors and how those things are presented so now you know the elements that go down to creating iconic sprites let's talk about the process of how we arrive there off the bat obviously if you're doing any kind of design for anything you should probably start somewhere malleable and if you're not familiar with pixel art if you're just learning you're probably more familiar with pencil and paper so for sure like sit down and design your characters on pencil and paper right do what i'm about to show you but do it in an environment you're comfortable with that allows you to make lots of changes because just from a purely design process standpoint this is something that you should be doing anyway right no matter what you're doing whether it's a story characters mechanics you should be plotting them out in ways that are malleable so you can make decisions about those choices uh make different decisions if you want different choices and not feel like you've lost time but for the interest of time we're going to jump straight into the sprite editor and i'm going to lay down some sprites some shapes i'm going to walk you through this process pick an arbitrarily large canvas don't limit yourself to the eventual size that you know that you want to use you may have technical limitations you may say well i know because i've decided earlier that my game needs its characters to be within this 48 by 48 grid that's fine but for this exercise if we're concepting a character it's okay to start ignorant of those bounds likewise with the brush size right and the the actual colors that i choose i'm choosing a size that doesn't matter if i was trying to go for a color that i thought did matter like a skin tone i'm already making an assumption about something that might be something that i would change later and if i've got some emotional attachment to that choice it's going to be harder to change it so if i choose red it's like well i know this isn't the skin color that i'm going to go for for this character so it doesn't really doesn't really matter so the first thing i'm going to do is walk you down this process of silhouettes we're going to start with those fundamentals and work our way down if i was going for something like our man barrett from final fantasy vii here the brief on that would probably be something like uh they're a strong character you know they're much taller than the group they play the the role of tank from a gameplay standpoint maybe we've already got gameplay notes on how this character behaves and we're going to then extrapolate that out into a sprite that befits that character so we're going to go for the tank let's go for a an animal rather than a human character we're going to go for an animal and we're going to go for something that's close to what i've been doing recently which is animals in a fantasy world so maybe this character this tank character is going to be like an elephant right and maybe this elephant he's going to walk on on two legs uh because we want to anthro morphiz him but we're going to place him in a fantasy context as well i've started with a square because i kind of knew that i was going for a tank here uh but let's think about how we could create a shape that's unique obviously if we want this to look like an elephant then we have to start with the characteristics of an element so we we think about ears about the trunk the square body or the large body maybe the fact that the eyes are quite far away from the middle of the head they're quite further on the outside maybe a small tail but probably not that it's probably not that important to go for the tail and those big round hands which maybe for whatever reason is i was drawn to elephant because they're similar to barrett's hands so what i'll do is i'll just start blocking things out and i want to see i want to see these things in the silhouette i want to see the trunk if i drew the character front on right maybe the trunk would find its way in the front like this but from a silhouette perspective i can't see that right that's gone in a silhouette so we want to orient the character if we can in such a way that we can see some of these elements and maybe maybe we pose the trunk in such a way that it represents the character so maybe something's like more square and downward with maybe a little kink on the end or give us some sort of a you know almost like a bicep or something like that we'll work with that maybe the the hands will be quite low down uh but i want the feet to feel strong too so just looking at this it's close but i want to see some ears here so maybe i'll bring the head down a little bit bring the ears out this way if you uh like this content or this kind of thing nick was who i was talking about was the artist behind shovel knight he actually streams this kind of content what i'm doing right now almost all the time on his um on his channel it's nick nick was w0z i want this to feel quite strong so maybe i'll bring the arm out and i'll create some space here so that i can differentiate between the arms and the actual body you know creating gaps in the silhouette help us to isolate those elements a little more clearly right if mickey mouse if mickey mouse's ears were so big that they overlapped it would be harder to understand that this was mickey mouse right so uh this is kind of interesting let's see if we can compress him down and then the next thing i'll do is i usually start creating a second tone this is a shading exercise but it's also going to help you find your shapes internally so we've basically drawn the character you could think of this as like a complete sprite right but it's complete with one color so now let's make it as complete as we can with just two colors and we'll only add colors once we don't think we can get any further with the amount that we currently have so this is about as far as i think i can get with just one color let's now introduce a second color and we'll see how far we can go and this is obviously going to help us introduce some depth into the character physical depth and another thing to think about would be okay like we've got a shape of an elephant here but what what would we do even at this stage to give this element this element this elephant something to uh define how he achieves what he achieves so let's just say this is an rpg it's term based what is his power like what's his what's his attack right does he like headbutt things does he like carry something in his trunk yeah grapples with his trunk as well so then maybe the trunk itself has some sort of well keep in mind maybe it has some sort of like a bandage or some sort of something that would help him grip better tusks as well obviously yeah if this is a male elephant so this is just obviously like a complete like idle sort of t-pose but if we wanted to focus on that grappler aspect then i think what i would do is probably change the pose entirely once we get past this stage and give him more of our arms outstretched sort of ready to grapple something that's probably the time where we look at some references of elephants because i'm starting to my knowledge of what elephants look like is starting to dwindle so maybe there are some things that we can borrow just shapes that then make their way in so i think this v shape that comes down from the ears into the trunk is actually really interesting and maybe we should capture that i think that helps and we could add a third color in now just to see how how we go and once we've got this color we can start thinking about okay maybe we want this to be maybe we've got some more opportunity to sort of break this down a little bit more we can play with the contrast of these shapes so this dark color next to this bright color those are sort of we find these little opportunities as we add these shades and we look at how they interplay and they then contribute to the overall shape that we end up with you know maybe we start thinking about the way that the the trunks have these wrinkles we think we want to okay maybe we want to make the tusks a little more stand out so how do we create shapes that work there how do we you know shift things around shuffle them a bit maybe we go for an even brighter color for the tusks so i think this is a kind of interesting start we're definitely not there yet but we've got something right we've got something we've got something we can play with and work with and we can edit this again because we didn't spend that much time on it we can change it to meet our needs right so right now this is just an elephant who's standing on two legs and happens to be in this sort of square chibi style but maybe we could try another version with more emphasis on the body less emphasis on the head maybe we could uh keep the head and put it on a body that's a little different right we were talking about him being a grappler so maybe we'll give him a little bit of a different pose maybe he's more hunched over and maybe he is more squat and maybe you know maybe the feet in this game could be actually more like really really square square like this and elephants actually do have kind of long legs so maybe it's not so bad that we give him longer arms we have to be mindful of the trunk because we don't want to lose that shape in the silhouette right if we lose this and then bring this back down to a silhouette form it's not recognizable as an elephant anymore so we want to make sure that we keep this somehow maybe we could bring the trunk back up and that gives us room for this hand to come out and we could even bring like some perspective into this we could break make this hand come out like this maybe up here but it would be really important that we get it right so i i'm conscious here when i'm drawing this to think a little bit about lines and i can start to see something forming here with this line and this line and this line there's something going on there so i might actually seek to accentuate that because the forward lean emphasizes this aggression that we're trying to get in the pose so you know maybe i would go further let's make this even bigger let's try to get these proportions really wild so this is good but let's go crazier still we're not like it's important to sort of play with the sliders a little bit right so maybe we get these to be really big to the point where they're like tree trunks and maybe we can do something i'm not really satisfied with this space down here just kind of lacking a little bit turkish oil wrestling maybe or like specifically hakan this something more like this you know i like this idea of having the legs be kind of more squat i really like this process i find this fun doing it in a video makes me feel like i need to be a little more like a little quicker just to finish the video up notice what i'm doing i'm taking the elements i'm copying and pasting so that i don't lose what i've done but i can still see it in the same context to influence what i'm doing next and if there's something that i don't like i can always look at something else that i've done that was it you know at a different state and say oh i preferred the way that this was and so now i'll move back towards that for example if we think about uh this being fantasy uh you know we've got war elephants being something that existed in real life thailand vietnam indian subcontinent so straight out the gate you know there are elements here that we could use i really like this sort of idea of like there being some sort of helmet these tassels are kind of interesting and the chains something about there being chains involved is kind of interesting it makes him seem a little more monstrous i can imagine this character even as we're thinking about this pounding the ground with his big big feet so definitely here we'd see like probably we'd have to come in and start giving some more shape here it might make it a bit difficult i like the idea of maybe like a like a wrestler's belt in the form of a chain you know keeping him all together there's something in that maybe we're adding things but we don't want to take anything away that's the sort of puzzle right we don't want to remove from that silhouette what are some changes we can make that help enhance the silhouette this is weird this is a weird character i'll say that much i've never done a character like this before it's a fun exercise right locks it on i knew there was an x in there this is what i was saying earlier right so we've talked about silhouette we've started creating these elements that help with the pose character design elements that help orient the character we've got some proportions that help accentuate what we're trying to do right now we're down at this point here where we're thinking about props different other um elements that then help the player ground this character to tell a bit more about the story of the character and so understanding how these elements have been used in past media that will help communicate to the player more clearly because they'll have these references you don't always want to harmonize with those references but something like an elephant having a helmet this is inspired by history so it's not a bad thing to pick and choose reinforcing ideas that have been used in the past whether it's history or other fiction and then departing from those things at other points in time i don't mind this idea of wrist guards i mean these these guys obviously have hands do good artists have to study history too it's not necessarily a question of having to study history it's just knowing that doing so gives you an opportunity to channel your creativity so i think that this wrist guard idea actually would be a good fit for our elephant and so let's help do that as well and it could be something that we just used to separate the uh the base the palm of his feet and the actual rest of his arm kind of has that effect of like almost looks like training weights or something what would be interesting is going more for muscle than fat so if we if we bring down some of the belly we bring up the pecs that might swap out the comedic aspect we're talking about getting some of that dignity back if he's got huge pecs it makes him being shirtless less funny so how do i now that we've got our shapes how do i make them more expressive how do i make them clearer you know they're rough shapes now but when we're done they'll be clearer shapes i think we can bring this up a little bit so we're going to do this guy and then we're going to do another character in the same universe maybe a different class okay so now i'm going to transition into the real colors so here's a here's an issue right the chains are the same they're going to be the same color as the skin so now we're like huh i think i'm going for oh yeah copper could work like a bronze kind of look although with this brown we might end up closer to uh like leather if we go in that direction that would then come down to shading really so high contrast will make it look less like leather well he looks expensive he looks like it looks almost like golden and then i'm noticing there were just some issues here with some of these colors where we kind of got a little bit away in terms of the um uh got away from the color space that we were in so like this lightest color here we should see that here right so then this should be somewhere around here maybe even on the sides here as well so i'm gonna go for three toes now that i know they have giant toes i think this actually could be good for helping us orient the uh you know where the hands are facing but we're going to be really careful that we're using the right shades to not lose too much so just trying to be careful here you see adding in these details like they've sort of in my opinion taken a little bit away from the hands i don't hate them but they're definitely making it a little more difficult to read just trying to get them in there let's do a little bit more shading work just to further emphasize some of these elements oh earring could be cool yeah for sure now you got to be a little careful because the more the further down you go for this stuff the more you don't have to animate that right so you can't go too crazy unless you have a huge animation budget but you know we could do something like this for sure in this work where do you picture the light source so i almost always frame the light source from above almost always humans are pretty much programmed to read things based on a an upward a skyward light source so i i just for simplicity's sake that's usually what i do and you can see here right like there's light coming down downwards here downwards there downwards there it's almost always above and he's leaning over his chest so this is obscured there's less light coming through here people always say like the eyes the window to the soul that almost always comes down to like reflections in the eye from what i understand if i give him this now he has a personality just showing the reflections of the light like the light in his eyes it's like oh that's a character now he talks you know just adding those two pixels now it's an enemy weird how that happens but that's kind of the way that it is all right cool maybe we could swap this out for something else i feel like there's a lot of opportunity here maybe uh like a ring of some sort might be a bit boring oh this kind of looks like it could be tassels i like him i like him a lot i think it's great i i would be interested in seeing what happens if we throw something out the back maybe a feather something like that booyah elephant war lord grappler tank guy there he is and remember like we started with this this is where we started right we started with something that was like just the idea of the trunk and it being square right we made him smaller more squat added some tones tried to accentuate tried to give him a pose that worked remember we talked about this silhouette pose we started decorating him with more interesting elements decorations that then tell us a bit more about the story of the character that also help to explain uh the silhouette you know these trunk elements here this feather will help us know which way he's pointed it all is supposed to serve this purpose of having him stand out i think we did a good job hey pal thanks for watching and thanks most especially to the patrons and twitch subs who support this channel and my gamedev project insignia to find out more click the links in the description below and if you like this video tell youtube by clicking the like button and then youtube will tell me and then i'll make more videos that's nice

[Music] Hello there. My name is Brandon and I make pictures out of Tiny Squares. And today we're going to be looking at the wonderful world of Pokémon trainer sprites. These are the character sprites that show up when you first pop into a battle in Pokémon. And they lend so much flavor to the tone of the battle because you get to see so many different personalities and attitudes and aesthetics of these different opponents throughout the game. So, I really want to unpack what's going on here and understand this particular character sprite style. Given the long history of Pokémon games, there's obviously many incarnations of trainer sprite pixel art styles from over the years. We're going to focus on the Gen 1 designs today, partly just because it's an obsession of mine, you know, kind of from a nostalgic point of view. But regardless, a lot of the techniques that we're going to be discussing can also be applied to other pixel art styles. So, let me know your favorite game or generation, and maybe we'll do a deep dive into another Pokemon game down the line. So, let's start with the highle specifications here. Uh, these sprites are all designed within a canvas size of 56x 56 pixels. This is kind of that pocket of space that's allotted on the battle screen. And this is also that same area that's reserved for the actual Pokémon creature designs, too. And for the most part, each character is scaled to about 56 pixels tall. They're they're kind of scaled to maximize that 56 pixel height with some exceptions being for characters that are intended to appear a bit shorter like with the youngster and other characters like that. One that amazes me the most is the biker sprites. The pose, the perspective, all all these details on the motorcycle as well. It's just so well executed and really makes amazing use of this space. The fact that you could actually fit an incredibly readable character and their vehicle and have this much energy to it. I I think this one might be one of my favorite sprites from the whole game. But even with things like uh Beauty where she's contorted into this kind of interesting flow of the arms and the hair and you got that Pokeball being tossed out the side, you know, you you could have just had her standing in a relatively upright position, maybe kind of like how Lass is standing. But again, this more ambitious posing is making great use of that allowable footprint. And it's also exuding a lot of personality that in this case is really like fitting and really memorable for this character. Now, with these sprites being from the original Game Boy, everything is constructed with that fourcolor palette. I'm showing them here in a super game boy appearance that uses this combo of the sort of orange and blue look, but really you can just think of these as being composed of four steps of brightness. You know, you have the black, you have the white, and then there's two in between shades. So whether that's the Super Game Boy coloring like I'm showing here, uh the Game Boy green tones, or just a grayscale appearance of the original source sprites, it's all just that same logic of there being four steps of colors to work with. One of the things that stands out to me the most about the rendering style here, the way that they actually use the colors, is how bright the highlights can be. It's often the case that we see a character's skin is colored using the lighter of the midtones, uh, in this case, the orange, and then that allows the white to be painted over that as this sort of blown out highlight. The lighting on Brock, for instance, is so strong that it looks like he's holding a flashlight telling a ghost story around a campfire. Like, just with the directionality to it as well. It's this really dramatic look just from that contrast, which is even just two adjacent tones. The strongest example of this kind of thing, though, might actually be with the engineer sprite where he's got like the bright white highlights that step through the orange and then finally down into the blue. And that's not just for his hard hat, which you might expect to be kind of specular or or reflective. Uh but it's also on his clothing. It it's this very dramatic look and it's also conveying a very specific and a very strong light source. This isn't the only way to shade stuff, though. I want to pull up beauty again because they've done something interesting here where instead of using that approach of the orange skin tone with the blown out directional white highlights. Uh what's going on here is that her skin primarily is just in the white color uh save for a little bit of orange toward the edges and some of the parts that are overlapping uh to give it some dimension. And my interpretation here is that her skin tone is reading a bit more pale than the other characters. And I just find this interesting for the idea that we're really just still talking about the same two colors, the orange and the white. And it really comes down to the way that they're being applied and kind of the proportion of that usage that's resulting in this different interpretation of skin tone. Another great way to extend the reach of this limited color palette is to use a technique called dithering, which most often is applied as a checkerboard pattern of two colors. And so just in a broad sense, it kind of averages out to being like an in between shade of those two colors. And we see this used a lot on these trainer sprites. And it's used in really clever ways, too, because the good thing about dithering is that it can lend a kind of physical texture, kind of a roughness to what it's being applied to. So for instance, uh the hiker's beard I I feel has kind of a bushy feel to it with that dithering. And on Lieutenant Serge, the dithering does a good job at conveying the fact that he's got this haircut where the sides of his head are shaped. For the bug catcher, there's dithering to create the feel of the mesh for the net and even a different kind of dither pattern for his straw hat. Uh, this one's more of a sparse dither pattern, which I think is a really smart choice just so that it's different from the net, which is actually really close by the hat, and it still conveys that feel of being a straw hat. So, in general, dithering is a great way to expand the appearance of tones within a character and works particularly well for conveying physical textures like fabric and hair. The last little trick we can do with a limited color palette comes down to how we play with the line work. You'll notice how even though these sprites are very prominent in their use of line work, it's not always a continuous solid black that's used throughout those lines. often there's a touch of the darker midtone uh in this case the blue that kind of interrupts it every so often or in rare cases the black line work is is broken up and drops entirely. These sorts of applications almost give the effect of line weight variation or tapering which helps really soften the line work and can sometimes help indicate dimensionality. And often for hairstyles, you'll see some of those color tones actually placed outside the outline entirely, which has a really pleasant softness about it. It can kind of round things if you needed to. In some cases, this gets kind of extreme, like for the Team Rocket dude, whose outfit is entirely black. This steps through both midtones at at the edges of that silhouette, and it's almost giving this like out of focus soft touch to it, I feel. But it's also kind of cool in a way that's like he's being backlit, I suppose. So, I can kind of see where they're coming from with that one. I think the best applications of dark clothing, though, are from sprites like Giovani, who uses just the blue shade to soften a couple of the inflection points on his silhouette, and the gentleman, who uh does that same thing, but there's kind of a small touch of the orange in there as well. Additionally, he's got some internal line work that uh is using the blue as the line work over the black clothing. And so it's kind of like this inverted linework approach and that works really well just for getting this read on the different pieces of his dark clothing. So when you combine all these parameters and techniques together, you get the basis of the rendering style for these Pokémon trainer sprites. But of course, this discussion isn't complete without mentioning the notso secret ingredient in all of this, and probably the most important ingredient in all of this, which is that the art style itself is adapted from the concept art of Ken Sugamorei, who's the primary character designer and art director, notably in a lot of this foundational work and throughout many generations of the franchise. And you can find his character concept sketches online, uh, like these ones that became the game sprites. And at first glance, I see things like this and I think, "Oh, okay. So, they must have scanned in the drawings and traced that and then just kind of rendered it up into the final pixel art appearance." And certainly, these do look incredibly close to the sketches, but they're not an exact one for one match to that line work. Uh, the thing with pixel art, especially low res pixel art like this, is that you want to prioritize the space for a character's head or the thing that's going to be the most recognizable for that character, let's say. So sometimes this means distorting the character's proportions just to make sure that you can allocate that room to those character elements that are going to be the most recognizable. In terms of how these were handled, I I think what's most likely the case uh whether or not they were traced is that the pixel artist would have to make these kind of adjustments on the fly to find out what works best in the pixel art appearance while still maintaining that flavor of the original concept art. So, just as an exercise, I want to try making my own trainer sprite and kind of playing in this weird hypothetical workflow of adapting Ken Sugamorei's character art. I want to take this concept art that he's done of Officer Jenny and turn it into a trainer sprite in the Gen One style. If we look at this, I think we can already imagine that scaling this exact art in this proportion down to 56 pixels tall is not going to really leave a lot of resolution for detailing the face and the character expression. So ahead of time, let's just try to adjust the proportions by increasing the space allocated to the head and then kind of squishing the area allocated to the body to make up for it. So we're just kind of redistributing the artwork. And listen, I know we're getting into weird and uncanny and perhaps sacriiggious territory for this moment. I agree with you. But just stay with me here. This is going to be our stepping stone for the purpose of this exercise. So, we've got our distorted officer Jenny. And if we resize that down into the 56 pixel canvas size, uh, the image quality obviously takes quite a big hit. But this is going to be our proportion guide, our trace guide for adapting this character art. So, I'm going to go through and get an initial trace of line work going. And I've placed a layer of translucent white to separate the original artwork from the layer that I'm drawing on. And this just helps me see the lines that I'm making, not get entirely lost in all the noise here. Now, one of the issues I'm having trouble with here is that her pose with this Pokeball pistol, which by the way is such a cool accessory for this kind of character, but her holding this, you know, in this kind of perspective over her body, while it looks cool in the original artwork, it's giving a really difficult read for the pixel art interpretation. So, I'm going to make one of those on the-fly calls to reposition the hand away from her body so that it can kind of occupy its own space and have its own silhouette. A bonus here is that, you know, we do have a lot of room to kind of spread out here and make use of this 56 pixel width that's available to us also. And on that note, I I think what's going to help overall is actually to widen the character a bit. I know we already distorted the original source artwork, but I can see the potential benefit of readjusting things just to have a bit more room for detailing the face and resolving that whole area with the hat and the hairstyle as well. So, here's the finalized line work from that round. And I think it's looking decent so far in its approximation of the original artwork. So, we'll move on to the color and shading by first dropping in some solid tones that are going to be the dominant color of each area of the sprite. The blue tone here works well. It's a sort of a perfect fit for the outfit. And I'm going to try that approach of starting from the orange as the skin tone and then applying those white highlights over top of it. And we can bring that same logic down onto the leg as well, of course, and kind of make sure things are consistent just in the way of her being lit from the left side of the screen. I want to try a bit of dithering on the skirt because I feel like that's an aesthetic choice that's kind of in line with the set of sprites from the original game. And then we'll also take some of that blue and use it as a shade tone on some of the edges of the skin and also use it to break up some of the line work on the legs as well. In finalizing the appearance here, I'm mainly targeting a lot of the line work areas. Uh I'm finding that the best approach here is really just to experiment with the placement of these edge pixels. like try to have some reason or or or some kind of like intent in the in the shaping or the softness that you're trying to achieve. But even with that, you can get pretty loose with it and end up with some pretty cool combinations just by kind of experimenting your way into it, if that makes sense. You know, just sort of placing the pixels in the outline and seeing what looks good. And if it doesn't look good, you can just kind of walk it back if you need to. Um, some of the more straightforward ones for me for this design is just to tuck some of that shading into areas where there's like angled line work. And you can just kind of like neatly place one of those shade pixels into those corners. And that provides a nice bit of anti-aliasing in those spots. And then there's also just like the idea of targeting areas of the costume that could use a little bit more of a pop of color that's kind of infused into them. [Music] So after those rounds of detailing, here's the final result. And I got to say that I'm pretty happy with this one. I think she fits in relatively well with a lot of characters from the actual game. Um, you know, maybe one of the things we could debate is is whether or not they would actually have maintained her having this Pokeball gun. Uh, but when you think about it, you know, one of the fun character details are all these ones that have these little accessories or there's almost this like implication of physical threat sometimes from these uh trainer sprites, even though we know this is just going to result in a creature battle. Um, but regardless, this was a cool exercise to try. I think if I were to do it again, I might I might try just adapting by eye instead of the whole like wonky sort of proportion trace and that sort of stuff. But it was kind of fascinating just to see like that angle from it, too. So, this was a great learning experience. And if you're looking for more videos about Pokémon artwork, I'd highly encourage you to check out this render breakdown video by Tom Mccclac, which I'm going to link down below. Um he shows how to capture the pixel art style actually across different generations of Pokemon games. Uh he does really cool stuff over on that channel. So go check him out. Um but over here we're going to close out with some CRT time. So thank you for watching and take care and keep it square. [Music] [Music]

[Music] when drawing pixel art or digital art of any kind we're presented with millions of colors to choose from it can be quite challenging to select a handful that work together for your project a really great way to make your artwork more cohesive is to use a color palette hopefully this video will help and by the end we'll be able to select some nice colors welcome to my call tutorial let's get started breaking down the color settings one of the best ways to select colors is hsb or hue saturation and brightness hue is the actual color with nothing else applied to it i like to think of it as a rainbow where you pick a color based on its name like red blue or orange saturation is how intense or strong the color is and you can visualize it as a scale from gray to full color finally brightness refers to how light or dark your color is so how do these settings relate to pixel art and game design well cooler hues such as blue and purple can indicate calmness green can indicate something natural and the warmer tones can suggest more dramatic things of course this is an oversimplified approach to hue and no specific hue has to stick with the classifications listed saturation can indicate how fun or playful something is with more saturated assets being used for cartoon comical art styles and less saturation for realistic mature artworks bridal colors can be used for bright happy things and dark colors usually mean more dramatic dangerous themes in this sense brightness is usually paired with saturation but not always now we know what the color settings do let's start picking some colors designing a palette a common mistake a lot of people make when choosing shades for their colors is to only adjust the brightness just using the brightness can make the darker colors look dirty or muddy to change this we can saturate the colors as a decrease in brightness and desaturate them as they increase in brightness but the most important change we can make is hue shifting shoe shifting is changing the hue of the color as the brightness changes let's take this red palette for example because in the real world highlights are more yellow and shadows are more blue when hue shifting we can apply the same idea going up from the red we go through orange and into yellow going down from red it starts to become more purple by hue shifting your selected colors will look more vibrant as they get darker and in a game or scene having all the shades colorful usually looks a lot nicer so generally speaking try to shift your lighter shades towards yellow and your darker shades towards blue here are a few more examples your color palette could consist of a couple sets one for each of the main colors you want to use now we have the color palette we can refine it to more we want to have as few unique colors as necessary to minimize double ups a good way to retain cohesiveness and refine your color palette is to do something called color bridging when you bridge colors you make some of the colors on the highest and lowest ends of the spectrum the same and this helps tie the sets together here's a drawing that uses a more beginner's palette and here's a drawing with a few of the color rules applied okay so we understand the basic rules and principles but how can we use these to give our art a specific feel setting a mood i think one of the biggest factors in setting a mood isn't the colors but the lighting lighting can play a huge role in the scene and shouldn't be overlooked low dim lighting can mean your scene feels scary and dangerous like the character is fighting a boss or a villain if your scene is well lit then it can have a more pleasant feel take these two drawings one is dark one is bright even without changing the content we've made a huge difference with lighting another thing to keep in mind when setting a mood is temperature naturally warmer colors can feel more inviting and cooler colors less this is because we associate color with temperature in these two images the left one looks much more inviting just through the use of warmer shades online resources when going about a drawing or a game i think it's important to finalize a color scheme and mood beforehand there are a couple of free websites that can be quite useful for making color palettes the first is coolest.co a free website for generating and finding color palettes by hitting spacebar on the generate tab coolers will generate a unique color palette it won't take long for you to find a color palette you like and when you do you can export to save or edit it on the explore tab you can discover some of the best palettes made by other people the next is lowspec.com a pixel art centered website with hundreds of pixel orientated palettes to choose from if you're not comfortable making your own palette just yet i highly recommend this site both are linked in the description thanks for watching this video i really hope you could find something useful from it thanks to soggyhum for suggesting the idea if you have a video idea you want turned into reality comment below and i might get to it this was more of a theory video so if you enjoyed it and want to see more then i've got a similar one on the left if you prefer a more step-by-step approach the video on the right might be more suitable thanks you

hello there in this video you'll learn everything a pixel art beginner should know when practice this video is part two of pixel art short scores and it's a compilation of shorts on the topic of pixelr Basics check out the video description for other parts of this free course as well as timestamps for specific topics so let's jump into it you have to avoid this rookie mistake if you want clean lines the only thing that matters is consistency in thickness one pixel thick lines are used the most so how do we ensure that consistency a single Pixel has eight possible neighbors out of those he should only have one or two neighbors if a pixel has more like three neighbors the line becomes thicker in that area and people call these unnecessary pixels doubles aprite also has a great feature that will help you out by checking Pixel Perfect property you will create clean lines without worries of course you can use the line tool but that's not good enough if you want to become a true Pixel Art Expert you have to understand the essence of straight lines first a diagonal straight line consists of sections like one or two pixels Etc you have to keep these sections as consistent as possible both in length and in the ratio of appearance in cases when you have to mix length of these sections the maximum difference is one pixel so if your line has section that is 2 pixels longer than any other section you've messed up and your line won't be straight won't approve here's a test you can do duplicate and connect your line to itself then zoom out and you will easily notice the difference so if you only use 90 and 45° angles rethink your decision and you might ask yourself why not just use the line tool don't trust the tools blindly yes aprite has wonderful tools but like all other software it was coded by people and thus is prone to human error allow me to show you this on a simple Circle example this one was created using aprit ellipse tool it is 13 pixels in diameter and the other one was custom made it is same in size but it's a bit different personally I think the custom one looks a bit more rounder which one lookss better to you using outlines is not only an artistic choice but also a game design Choice here we have the same rock with three different choices on the left when we don't use outline it Blends quite nicely with the grass which is useful for non-interactable objects in the middle we have a full outline which covers the entire Rock making it more visible it pops from the background and that's quite useful for interactable objects on the right we have a partial outline where the object is still quite visible but because we erase the outline on the bottom Edge The Rock feels more connected to the ground which is quite useful for non-interactable object that could be used as a landmark in the game but we still have one important choice that we need to make regarding the outlines most people only use black outlines however you can also use white outlines you can use colored outlines and so on and you can also use shaded outlines meaning where the object is hitting the light the outline would be of the lighter tone and where the object is in the shadow the outline will be darker mixing pixel size is considered in pixel art one of the greatest scenes and these are called Mixels it's when you want to have different pixel sizes in your image for example you can see that even though this three is now larger it has different pixel size this is a standard pixel size however here on the side each pixel so to speak is actually 2 x two in other words four pixels beginners often do this when they want to have a larger object and they just don't feel like droing it from the scratch then they just take the object resize it to the size they want to have and call it a day and this results in Mixel as you can see this looks very unprofessional and if you are doing it just for fun it's all great but if you're trying to sell a game if you're trying to create a product that is visually appealing you definitely want to have your pixel sizes consistent all throughout of course sometimes in games and UI this cannot be avoided but in general you want to have your pixel size consistent it doesn't matter what you're trying to draw everything can be broken down into these five basic shapes Cube sphere cylinder cone and pyramid and if you learn how to shade these five basic shapes everything will fall into place naturally for example a pencil is a simple combination of a cone at the top and a couple of cylinders a truck is basically just a modified cube with a couple of cylinders for the wheels but that's not all even the human body can be broken down into simple shapes this is why it's so important to understand how to shade these five basic shapes because once you understand that you can draw pretty much anything there are a few simple rules you can follow to ensure good shading for Cube It's one side one color why is that because every side of the cube is flat and the only thing that matters is the angle of this side to the light source in this case it's on the top left side so the top side will be in the light tone the left side we can leave in the mid tone and the far side should be in the shadow tone if you want to increase the contrast of your lights and shadows you can certainly do so but the basic premise stays the same one color per one side a lot of beginners will try to create something like this thinking it's correct because the light source is coming from the top left and naturally the light should be strongest near the source of the light and then fade out slowly this isn't how the light works you really need to have only one side in one color for the cubes because remember all that matters is the angle of the side of the cube to the light source that's all that matters in pixel art you'll be often shading as spherical objects meaning around arounded object the first thing you need to decide is where the light source is in this case it's going to be on the top left side then you need to define the light area and even this by itself should be enough if you really want to add more tones you can certainly do so but be careful not to overdo it if you overdo it and introduce too many tones even though you can do it I don't think you really should because this is no longer pixel art in pixel art we are trying to simplify everything and that includes the color palette shading a cylinder should be straightforward if you know just two simple rules and because cylinders are used a lot of times when creating organic things like humans for example it's really important to learn these two rules so first first you need to understand the structure of the cylinder we have two flat sides on the top and on the bottom which is invisible and they are connected with this curvature in the middle the cubes since they are flat will only use one color per side and sphere since it's rounded will use multiple colors but in the curvature of the direction from the light source and since cylinders are a combination of both meaning they have a flat and rounded surface we will use only one color to represent the flat side and we will use multiple colors to indicate the roundness of the cylinder starting from the light source and then curving away into the shadows and that's the most important principle dividing into two rules if a surface is flat use one color if a surface is rounded use multiple colors in the direction of the curvature so if you know this that the sphere is rounded in all directions which is why you get millions of colors on a sphere if you overdo it and this is why it's important to simplify things in pixel art but it's rounded in all directions so starting from the light it will fade towards the shadow in all directions but in cylinder is a bit different it's not rounded in all directions it's rounded only in one so it's curving from left to right meaning horizontally which is why it's shifting colors horizontally and only on the rounded portion but since it's flat from top to bottom specifically for this middle section it will use only one color vertically and this is why we get these Stripes of color so if you understand this basic principle flat surface one color rounded surface multiple colors you'll be good to go okay now that you know the basic principle of shading which is one color for flat sides and multiple colors for a curved side what about cones a cone is nothing more than a cylinder which is tapering on one side towards the point it still has one flat side which is not visible in this specific case but it's here on the bottom and if it was seen we would shade it in one color just like a cylinder but since we can only see this rounded portion which I want to showcase since this entire section is curved we want to use multiple colors starting from the light and going towards the Shadows but notice the big difference between these two since the cylinder has two flat sides on the top and on the bottom we have these Stripes of color but since cone is angled towards the point the shadows will also be angled towards the point so we are still honoring the basic principle multiple colors in the direction of the curvature but it's only going to be one color if the surface is straight or flat in a single Direction meaning in this case from the bottom to the top and finally the last geometric shape to learn is Pyramid It's a combination between a cube and a cone meaning all the sides are flat just like on a cube but it tapers towards the point on one side just like a cone shading is very simple since sides are flat we use only one color per side taking into consideration where exactly is the light source and the common mistake I see from the beginners is that if the light source is from above for example they will do something like this create a gradient of light and Shadow why is this wrong it's because the only thing that that matters for flat surfaces is the angle at which they are looking at the light source or if you want to put it in the other way the angle at which the light source is hitting them since the light source is directly above the pyramid both of these surfaces have the same angle towards the light source meaning they will be both in the same tone pillow shadding is one of the most common beginner mistakes in pixel art the term comes from shading pillow specifically in case where the light source is directly coming from the camera meaning you are the light source not only is this a bit un natural because we are not the light source unless you are in the dark having a flashlight but also the lighting is very boring and while this could work on the pillows in this specific case it doesn't work on other shapes and it is characterized by having a light tone in the middle and bands of Shadows towards the edges let me show you two examples first you have this hexagonal shape with some light in the middle but in fact this should be a cube but because the shadding is so poor it loses its readability it loses its shape and three-dimensionality this is why we have those basic rules for shading the other example is where every side of the cube is shaded like a pillow you have light in the middle and then shadows as it goes away from the light so please always stick to the basic principles of shedding that I've mentioned in the previous videos now that we have covered the basic principles of shedding let's talk about highlights and these are very specific in pixel art highlights are a reflection of the light source so if you have an object with sharp edges you can highlight it the edge to emphasize it beat on a cube like this a cylinder like this you can even extend the highlight a little bit into the midtones or even in the shadows which you'll see a bit later or a pyramid but you can also use highlights on rounded objects as well but for this sphere notice how the Highlight is not directly in the middle of this Center light this is because a highlight is a reflection of light towards you so whenever a light rate finds that exact nice angle at which it can directly reflect into your eyes that's where the Highlight will be and lastly you have a cone in this case I'll just place a highlight directly in the middle of the center light now let's look at a very familiar scene from the Pokémon Red notice where the Highlight is here on the edge of the table even though this is not realistic lighting and where the Highlight would be it still works and since human face is very complex you'll find lots of different shapes and places where you can find highlights here we have a couple of portraits from Chrono Trigger I decided to use this to honor Akira tuyama and here we can find highlights here on the nose if you know that dot on the nose that you'll always find on portraits but you can also find it here on the cheeks on the foreheads and and so on so the main takeaway from highlights in pixel specifically is that you can use them to push your shading a bit more towards the realistic side but you can also use them to define a shape more clearly especially if your color palette count is very low regardless of your choice keep in mind that as long as it looks good it's all fine now that we have learned how to shade basic geometric shapes now we have to expand upon it first thing you should be doing is trying to combine shapes here we have a couple of cylinders one for the rubber gum one for the wooden part of the pencil and yes I know there's a graphite in the middle but still though for the basic understanding of shapes this is what you should be thinking about and lastly we have a cone for the tip of the pencil after you've created some objects by combining shapes then you need to learn how to start modifying the shapes to fit whatever you need for example we have this front part of the truck I can start by simplifying it into a cube but then I will start cutting some parts away specifically this top portion to get the shape that I want then on top of it I can introduce some details and the same thing applies for the wheels or tires I have just created using this basic cylinder I have narrowed it down quite a bit and then I just punched a hole in it so look up reference images for the things you love drawing the most and then simplify them in the basic shapes modify the shapes if needed and then shade them here's how to calculate cut Shadows quickly and properly first we need to define the position of light source and we do this by creating two points one very exactly the light source is and the second point is on the ground which is directly below the light source next will we find the bottom edges on the cube and mark them in the same color as the point at the ground then we'll cast Rays from the point at the ground through each of these points on the cube next we cast light rays coming from the light source but this time directly through the top edges and the phenomena will happen for this side of the cube notice how two light Rayes connect under this specific point right here and we'll mark that specific point with another color then you can clean up the rest so it doesn't confuse you and you will do the same thing for the other edges as well now we can clean up everything now connect this point to its corresponding Edge do this for other points as well then connect the points themselves to each other and finally fill in the area that's all this is how you get a properly calculated cut shadow in relation to the light source diering is a very specific shading technique in pixel art its purpose is to create the illusion of more colors on the screen without actually increasing the color pallet size in the past it was invented only because of technological restrictions but nowadays it's just just a stylistic choice the most common form you will see is this checkerboard pattern where you take two colors and just create a checkerboard pattern with it creating this illusion of midone you can use it on both flat and rounded objects and if I use it here on the cylinder you will notice how the transition between midtones light and shadows becomes a bit smoother and if you're still skeptical about its use I will select this checkerboard pattern and use the blue tool to blure out these colors in order for them to blend then I will take this color with Color Picker and you will see it's pretty much the same color as up here in the midtone within the gradient tool you have two options no diing and diing and within diing you have bare matrics each of them are higher and higher but what do these numbers mean luckily it's quite simple these numbers represent a size of a tile used for a specific pattern for example 2x2 bar Matrix means that we are using tiles which are 2x2 pixels in size starting from the middle we have this checker board pattern which I'm sure you can recognize right here in the middle then if we go to the left towards the black color we have 100% black on the left side but in between we have 75% black color and 25% white and vice versa for the white side now if I use this bare Matrix to go over this entire screen and I go to view grid and grid settings and place it width and height 2x two and I activate the grid you will notice these patterns 100% black 75% black 50/50 checkable pattern 75% white and 100% white there's an easy way how you can create your own detering pattern first decide on a bare Matrix meaning how large your tile will be for creating the detering pattern the larger the tile the more complex pattern you can create in this case I'll use 4x4 start in the middle and create a perfect 50/50 blend between two colors in this case it's a checkerboard pattern then if I go to the left I will just keep removing one pixel of the opposite color in this case I'm going towards a pure white color so on each tile I'm just removing one extra black pixel and just one by one I keep erasing them until I get pure white color and I can do the same thing for the opposite side as well but you don't have to use checkerboard pattern there are plenty of ways how you can start in the middle for example you can see one of them is here you can see another one with vertical lines horizontal lines and so on so please do experiment an easy way for a beginner to start diing is to just choose a shape apply one Shadow tone so you have two tones one for the light and one for the Shadows then create a checkerboard pattern with these two colors create a third color for a midtone and Shady the object with the said midtone wherever it would go remember the area of that midtone select the checkerboard pattern Press contrl B to create this brush tool then press G to select the paint bucket tool and click on this midtone area and you will have your detering effect this this is how you can easily practice where exactly your diing pattern should go this also applies to flat surfaces like the cubes and rer surfaces like spheres in fact if I zoom out of this sphere notice what will happen in a Sprite it will automatically create a midtone even though I'm zoomed out I'm still on the same frame so if I zoom in you will see the ding pattern if I zoom out it will blend it into a single tone this is how you know the placement of your detering pattern is good unfortunately this is not enough shading just conveys the information about the shape of the object and of course its relation to the light source before we can start creating objects we also need a texture a material and we'll get to study plenty of these when I start covering materials very soon a basic practice is to create a texture and then wrap it around an object in this case we have a simple sphere first create a mid tone for the wood then add some STS which are going to be the planks since the light source is usually coming above an object we'll Place some highlight tones on the top Edge and then finally we can indicate some wood grains you don't have to be too precise with this just relax and have a little bit of fun with it experiment and that's all this is how we get a very basic wooden texture but now I have to wrap it around an object so I can take a sphere it can be a box it can be cylinder anything then we take this basic concept of having few planks and very dark streaks which separate the planks and try to replicate that texture around this sphere okay before we start picking colors and creating our own color palettes let's just talk about what color actually is so it's a combination a formula if you will of three different things saturation brightness and Hue here on this rainbow strip you can see all of the colors this is Hue a red Hue meaning red color blue hue meaning blue color and so on but that's not enough if I go to the left side I will start to desaturate the color because saturation is the intensity the vibrancy of the color so if we remove the saturation it becomes gray and finally if we go up and down we will influence how much white or black color there is meaning how bright our color is so for the example let's take a brown color so you can see that the Hue is within the orange Hue slightly towards the reddish tone and it's desaturated and darkened this is how we get a brown color listen carefully because this is important if you plan on printing your pixel art the RGB color model uses a red green and blue as primary colors and it's called an additive model because we are combining rays of light into a white color color your screen right now has red green and blue emitters of light and if you turn them all on you will get a white color so we are combining the colors this is in case if we are emitting light and this is great if you only plan on having your art in digital format however if you plan on printing it you will need CMYK model here the primary colors are sign magenta and yellow letter K stands for key or black color this is called subtractive model because when a white light hits an object that object will absorb all light except one so if you have a Apple once a white light hits that Apple it will absorb all colors except red red will be reflected into your eyes and this is why we see the apple as red and if you overlay too many colors that object will absorb all of the light spectrum meaning we will have only black left and this is how we usually get black color in traditional paintings the colors value is all about how we humans perceive that color some colors we perceive brighter than the others for example we perceive yellow quite a bit brighter than the blue color and the way we check the colors value is by transforming it into a gray scale and how do we do that whatever your image is create a new layer and fill your entire canvas with black color then double click on that layer be sure it's on top of everything and change its mode to color and then your entire canvas will be in Gray scale and then you can check the values how something bright or dark is and this is especially useful when you are checking the depth of your scene specifically the difference between the foreground and the background understanding values is very important especially for color blind people and if I if I turn on this color value Checker notice what will happen they're all the same they see no difference all of these colors have the same value but they are quite different aren't they the most common mistake I see pixelr students making when creating color pallets is not Hue shifting as the name suggest when you're are creating your color palette you must Hue shift between midtones so for this orange color I will Hue shift towards the light sources color meaning yellow in this case because of the Sun so for the shadow tones I will just go in the opposite direction from the yellow in this case towards the blue color or even closer looking towards the red and the same goes for the darkest Shadow as well and I have here another example of this same color palette but without Hue shifting so observe how this dot on the Hue wheel doesn't change at all and now look at its prite before and after this is without hu shifting and this is with Hue shifting hu shifting brings a lot more life and a tiny bit of realism as well but because there's a higher contrast in the colors your Sprite will also look more interesting without hu shifting your colors will look very flat and a bit washed out so be sure to H shift when you are creating your color palettes let's talk about creating color ramps the whole point of them is to create a cohesive color palette where we have some multi-purpose colors for example this white color is a highlight tone not only for this blue color but for this yellow color as well likewise this purple color is a shadow tone not only for these purple and pink colors but also for the blue color as well and even though this blue color is a midtone between these two colors it's also a shadow tone for this green color as well and funnily enough even this mid tone yellow color is a highlight tone for this green color so when you are creating color palettes try to find their purpose we want to create a cohesive color palette but also to optimize it and make it as small as possible you can consider these multi-purpose colors as glue that holds the palette together you can start with something as simple as black color being the common Shadow tone the darkest Shadow tone for all the colors in the palette and likewise the same for the white being the brightest in case you don't want to waste time creating your own color palettes and just want to experiment with other people's palettes you can go to lp.com palet list it's right here and you can filter bunch of different palettes depending on tags and number of colors and so on likewise if you're go into a Sprite there's a bunch of different preset colors that you can use if you just want to jump straight into drawing pixel art want to save a lot of time in game development better learn how to create tiles a tile is a rectangular Sprite that can be reused as much as needed in order to quickly create Landscapes and it also uses less resources here's a Pokémon Gold stage and if You observe closely you will notice patterns that repeat these are all tiles and this entire scene is made of them and if I turn on the grid it will be even easier to see and recognize some of them are used more often and some of them are used less of course but these are all tiles have you ever noticed these in your favorite games a common problem with tiles is that you can e easily see their pattern the most common mistake is having seams seams occur in a tile when you are creating a texture in the middle of a tile and because all of the texture is in the middle of a tile you will get these seams these streaks which are very obvious and you can notice them at a glance the way to fix this and create seamless Styles is to overlap this texture near the edges and with a little bit of work those Styles become seamless yes you can still notice patterns if you pay close tenture but it's very difficult to see a tile set is a collection of tiles that you can use to create your game on the screen you can see tile sets from Pokemon Gold one of my favorite games that I used to play and it uses styles of 8 by8 pixels in size as you can see once you bring a lot of different tiles together you can create a lot of different combinations and you can create entire Maps just made out of these tiles when you are creating tile sets for your own game you have to think about their purpose what do you need and what you don't need so you don't waste unnecessary time for example if you have a ground and water as your base Styles will these two connect and if the answer is yes you need to think about transitioning tiles between these two bomes so to speak because when you make these transitions your world looks fuller it looks bigger it looks more fulfilled even here in Pokémon go you can see even though they do have a specific tile for water and grass due to the perspective they still have these transitioning tiles that creates this illusion of depth there's a difference between a tile set and a tile map a tile set is a set of unique tiles and a tile map is a map made out of tiles so this would be a tile set as you can see every tile is different and this is a tile map where you can see a lot of copies of the same tile sometimes you just want to add some variety into your tile set if your tile set looks like this and it's very plain that's totally fun and if you want to create more details there are two ways to go about it this is the first one keep the tile set the same but on a layer above it add some details as a separate object that that way you can move them around and you have more flexibility another way would be to create variation tiles this is essentially just taking a tile duplicating it and changing something in the middle it can be something as obvious as these gems but it can also be something less obvious for example these styles are a all of faces and they can be combined between each other as I wish for example these two top tiles are the same but the tiles below are different and you can even notice the same face right here which is a little bit different on the left side but nonetheless the edges of a TI are same so the way they connect the way you create seamless Styles is that you overlap the details on the edges and you only change the middle this is why it's important to know how seamless tiles are created because then you can create variation tiles on top of them antialiasing is an advanced technique in pixel art that we are using to blend the transitions between shapes and lines when used on a straight line it creates the illusion of the line becoming thinner on the edges un curved lines it creates the illusion of the curvature being a bit more smooth and it's always good to keep keep the antialising on the inside of an object because if you change the background color that way the antialising is still there and the color values of the anti-as pixels should always be between the two colors which we are anti-aliasing the best way to start practicing is to create some clouds because they are unpredictable use two colors one for light and Shadow and the third color will be used for anti-aliasing and then try to recognize either the straight lines or the cures and use ntls to blend them for example if I recognize this specific shape as a straight line I can ntls one portion of it likewise if I recognize this as a curvature I can antias the middle of it and then you just keep on going until your entire Sprite is finished

I've gotten a lot of comments as I've started my full time indie game development journey asking me how I got so good at pixel art, which is a great compliment because I actually don't consider myself to be that great of a pixel artist. However, I did go from basically having no art skills to being able to produce decent and presentable pixel art. And I did that by learning a couple of very important rules and guidelines. And so in this video, I want to show you some of those rules and guidelines that I learned that can take you from having absolutely no art skill to being able to produce pixel art for your game that looks good. So let's get right into that. The program that I'm using here is Aseprite, which I think is one of the best programs you can use to create pixel art. It's what I use to create pixel art for all of my games. However, the tips that I'm going to give you in this video are not Aseprite specific, they're going to work for any program that you use to do your pixel art. Okay, so tip one is start small. I'd recommend starting with a small canvas size when you're first starting to draw pixel art, something like 16 by 16 or 32 by 32 is what I would recommend. There are some pixel artists that would say that smaller art is actually harder. And it can be in certain circumstances, but the point of starting small is to give yourself a big constraint. And the overarching theme of a lot of these tips is going to be that constraints breed creativity, the fewer options you have in terms of how to produce your art, the more you can focus in on the things that actually matter in terms of making the art look good. So when you start with a small canvas size, you're essentially going to be forced to work within the limited confines to produce good art that conveys the message or conveys the elements of the artwork that you're trying to convey in a proper manner. So instead of having a huge canvas where you can place as many pixels as you want and kind of get lost in the details, starting small allows you to think about each individual pixel and how it makes an impact on your overall artwork. So here in Aseprite, I've actually got this canvas size is 48 by 48, but I've subdivided it with these blue squares into 16 by 16 squares. And so essentially, I'm going to be using the 16 by 16 canvas for this video. The next tip is to use a limited color palette, I would recommend starting with a one bit color palette. So over here on the left hand side, I've got black and white as my colors. If you start with this again, constraints breed creativity. If you have only two colors with which you can make your art, you're basically going to have to focus on how to do the most with your limited tools. And this is actually going to draw out what's most important about the art and how you convey it to the viewer, it's going to force you to focus on the form, you're not going to have millions of colors to choose from, you're not going to have highlights and shadows to worry about, it's just going to be about can you convey what you're trying to convey using only two colors, if you focus on that, and you practice that, that is going to give you a very solid foundation on top of which you can build more complex things like shadows, highlights, colors, that sort of thing. So since it's almost Halloween, I'm going to go ahead and draw a jack-o'-lantern. I'm going to show you how I would approach that using a one bit color palette. Before we get into the next tip, I want to tell you about this video's sponsor, ClickUp. ClickUp is an app that gives you all the tools you need to manage your game development projects. You can create lists, which are essentially Kanban boards to track and manage the tasks you need to do for your games. What makes ClickUp stand out from other task management software is that it has a high degree of customizability. You can add any columns you want to your lists, and you can even add custom fields to each task in the task list. Besides lists, ClickUp has a number of other useful features for game development. It offers a documentation feature, which allows you to write rich text documents, which is perfect for preparing Steam, store page copy, writing game design documents, or even documenting your code. Additionally, ClickUp has a really useful forms feature, which is perfect for collecting feedback from your players. You can customize a form through the ClickUp interface to specify the data you want to collect. Then you can send this form link to your players. Every time one of your players submits the form, the form data gets logged into a ticket in the associated list, making it super easy to triage the feedback you get. When working on development and ensuring you deliver a polished game, being organized is not just a nice to have, it's essential. Game development cycles can be long and complex, often spanning months or even years. You've got multiple moving parts like asset creation, coding, QA testing, localization, marketing plans, and platform specific requirements. Missing one detail can delay your launch or even lead to costly errors during submission to platforms. That's why tracking every task and staying on top of deadlines is key to ensuring nothing falls through the cracks. With ClickUp, you can set up productivity boards and organize everything with calendar views all in one place. And by the way, ClickUp is free, so go to tryclickup.co/firebelley to get started. ClickUp gives you all the tools you need to effectively manage your game projects. Thanks to ClickUp for sponsoring this video. Okay, after just a few minutes, I've got what I think is a perfectly decent jack-o'-lantern. Now, again, I'm not an artist. I'm showing you how you can become decent at pixel art with no art skills or no art background. And so me being a non-artist, I think I've produced a pretty passable jack-o'-lantern. I'm just going to mention a couple of the things here, the ways in which I've used the one bit color palette that you can pay attention to when you're attempting to draw your own things. So first, you cannot make sections of the black collide with each other. So I have to make sure that the mouth, for instance, is separate from the eyes because if any of these black pixels touch like so, I get a very muddy image that doesn't really convey the meaning that I'm trying to convey. And so you have to be very careful about where you put your black pixels. And you'll notice too, that I can use the black to represent multiple things. So in the case of the eyes and the mouth, the black represents a lack of a pumpkin, right? It's the holes that have been carved into it. However, this line on the left hand side here, this black here, that's not taken to mean a hole that's been carved in the pumpkin. I think that most people will look at this and understand that that's meant to indicate curvature. That's meant to indicate that there's a ridge or ridges, plural, along the pumpkin. Something as simple as that can make or break the impression that your viewer gets. If I were to fill that in, you know, this still looks like a pumpkin and this would definitely work. But that little extra detail, I think, just kind of puts it over the edge. And then for the stem, you can see how I've made the stem basically its own little floating island here. It's not connected to the base of the pumpkin. It's separated by this moat of black here. And I think that just adds a ton of depth. It looks in the small preview like the stem is going into the dip of the pumpkin. Right. It looks like the pumpkin dips in and the stem goes into it. Now, again, you can definitely connect it. And that wouldn't necessarily be a problem. But these are just things that you can keep in mind. Like you can experiment with it. There's multiple ways of doing this. And I would encourage you to experiment. The whole point of starting small and the whole point of starting with a limited color palette is to experiment with how to make the limited colors and the limited amount of pixels work in your favor. Once you've gotten comfortable with a one bit color palette, you can move on to a limited color palette that actually contains more than two colors. So for instance, you can go to this website Lospec and browse any palettes you want here. I would recommend going from one bit to something like 8 or 16 colors. You still don't want to introduce too many colors at once. Again, we want to keep the constraints relatively tight so that you're learning within those confines. You're learning the fundamentals because what those restrictions are going to do is they're really going to bring out the necessity of the fundamentals of drawing the forms well and making the most of your colors. You're not necessarily going to produce better art by having access to all of the millions of RGB colors. So what I'm going to do is I'm just going to choose exact colors of eight. And I'm going to go ahead and find one that I like. I actually really like this funky future eight one. I've used this in one of my previous games Ace of Sphynx. And I think it's a really great example of how a limited color palette can go a long way. You can see even in this example image here, you can see that on this island, the goats are blue, right? And I want to mention that because we can make colors work in a lot more ways than we think we can. When you think goat, you don't think blue, you think white. And so if you were to draw on your own, you would probably immediately go to some kind of white color. But the interesting thing about this funky future eight is that they're blue goats. And you can look at that and say, well, I know that they're goats. They look like goats, but they're blue. And so you can make colors that are usually nonsensical work when you're using a limited color palette for all kinds of different purposes. Okay, so I've loaded this funky future eight into Aseprite as a palette on the left hand side here. And now I'm going to show you how I can color up the pumpkin using these limited colors. Okay, so using the funky future eight palette, I've colored this up, and I'm going to talk a little bit about the choices I've made. So you'll notice that there's no green color in this funky future eight palette. Now, if I was choosing my own colors, I'd be tempted to choose a green for the stem of the pumpkin. But because I don't have a green, I used this light blue instead. And what's interesting is that it doesn't look out of place at all. It's a blue stem. It's not something that you're ever going to really see in nature. And yet viewers are going to understand exactly what that is, that's the pumpkin stem. And then for the highlight, I've just gone ahead and used a yellow color. And for the shadows, I've used this pink. Now, the shadow is a little bit too light. But that actually leads into my next tip, which is to use appropriate contrast. One thing that a lot of new pixel artists do is that they will not have appropriate contrast between adjacent pixels. And so I think that right here, this orange and this pink, those contrast values are not high enough, such that as you were to zoom out on the preview, you actually kind of lose definition, the pink and the orange become not super distinguishable from each other. Now, because this is a well designed color palette, I think you can definitely use orange and pink next to each other. But I'm just going to, for example, purposes, choose a lighter pink here. So I've made it a little bit lighter. And you can definitely see now that as I zoom out, it all kind of blends into one color. If I zoom out enough, it actually just looks like orange. Now you can easily tell the difference when you're super zoomed in, but players are most likely going to be seeing your art in this kind of size. And the problem is that it's either going to lose the detail that you think it has, or it's going to become muddy. And so what we can do about that is choose higher contrast colors. So I've put that pink back as it should be. But for the shadow, I think I'm going to choose the darker color here, and I'm just going to go ahead and draw that in. Now this is going to look really harsh close up. But as I zoom out, you can see that that actually doesn't look terrible. And if I wanted to, I could reduce some of the shadow on the bottom there, and I could even fade it out to a lighter color to make that transition a little bit less harsh there. And what we can see now is I've kept the lighter pink arches here to indicate the curvature of the pumpkin. But then I've made the part of the pumpkin that's in shadow this darker pink or purple color here. And this may not look entirely great when you're super zoomed in. But again, as you zoom out, that looks great. Actually, you can see a very clear contrast between what's in highlight, what's in shadow, and then the smaller details, the lighter details are represented by a lighter color, which still has enough contrast to be noticeable without interfering with the other elements that you're trying to emphasize, like the eyes. If I were to make this dark color everywhere where the pink is, you can see that we kind of get, again, we get that muddiness because now there isn't enough contrast between the black and the purple, and then it kind of creates just a mess when you view it from a distance. So manage your contrast well. Pixels that are adjacent to each other should have plenty of contrast so that they stand out. Tip number four is hue shifting of shadows and highlights. So this is going to apply whether you're using a color palette or whether you're trying to choose your own colors. So I want to point out something that maybe you didn't really notice, which is that why are we using a purple color for a shadow when the pumpkin is orange? And furthermore, why are we using a yellow color for the highlight when the pumpkin is orange? What a lot of new pixel artists will do when they're trying to do highlights and shadows is that they will try to just choose a darker version of the base color and then a lighter version of the base color. So for instance, if I were to select this orange here, what that might look like is I might go into the HSV slider here and just turn down the value and then draw that as the shadow. For the highlight, I would do the same thing. Select the base color and then go ahead and down the saturation if I'm already at max value and just paint like that. So you end up with something like this. This is what a lot of new pixel artists do and I really honestly don't think that this works. This makes your art look very bland. Now compare that to this where the shadows are not a darker version of the orange. They're actually a different color entirely. And so how do you choose the colors for your shadows and highlights? Well, if you're manually choosing your own colors, what you can do is hue shift. So we're going to select this base orange here and I'm going to open up the HSV hue saturation value slider and for the shadow what I'm going to do is I'm going to shift the hue toward purple. So I'm going to shift my hue to the left toward purple. I'm going to go even a little bit farther and then I'm going to turn the value down. Darken that and then if I go ahead and paint this now I've got a pink now instead of an orange but that looks better immediately. It looks so much richer. The art is popping much better and likewise for the highlight what I can do is select again my base color here but instead of hue shifting towards purple I want a hue shift towards yellow now. So I'm going to shift towards yellow and you can play with the saturation and value sliders here but I think just a hue shift works here and now we get to yellow. And what you'll notice is that I'm actually closely matching the colors that were already available in my color palette. So just those minor changes to how you do your shadow and highlight colors makes a huge difference in how much your art pops. If you're not picking your own colors and you're sticking to a limited color palette you can apply the same principle. Just look for a color in your palette that is shifted more towards purple for your shadow and a color that has shifted more towards yellow for your highlights. In other words go for cooler colors for shadows and go for warmer colors for highlights. All right now I've got some tips for outlines which are to avoid jaggies and to make your outlines pointy to emphasize pointiness. Now what do I mean by that? Outlines are generally pretty straightforward. You take your dark black color and you can also use a non-black color and you simply just draw around. So here I am taking the pumpkin that we've drawn a couple of times now and I'm just outlining it in black here and we end up with something like that. But there are a couple of things that you want to keep in mind. First avoid jaggies when you're drawing any sort of line but especially outlines. And what a jaggy is is it's essentially this kind of line where you've got a line like this and then you've got a line like that and then you've got a line like that. Do you see how there are doubled up pixels here? That just kind of looks awkward. Now you can definitely make a style work with this. However you want to avoid that. You want to make it a nice smooth step up without any pixels doubled up. So that's the first thing you should know about outlines. The second thing is for something like this stem I may want to emphasize that this stem is pointy but the way that the outline currently is is that it looks a little bit rounded. Now watch this. If I place a single pixel now the stem looks pointy. So I've just placed a pixel up here in the top corner of the stem and now the stem looks a lot sharper. Now this is especially useful when you're drawing things like swords and whatnot because depending on how your outline is drawn it can actually make the sword look blunt or rounded. Whereas if you just add that extra pixel to emphasize pointiness that's actually going to make the sword look sharper. So this is kind of a violation of the jaggies rule because we're kind of doubling up pixels instead of having a nice uniform outline but that is a trick that you can use to emphasize to the viewer that this is pointy. And then another thing about outlines is that you can also feel free to remove the bottom edge of the outline for things that are supposed to be sitting on the floor because then that makes it look like the object is connected to the floor and then the outline takes over to emphasize that it is elevated or it has some separation from the floor. So this is actually something that I'm doing in gunforged for any entity that is designed to be attached to the ground. I actually don't have a bottom outline. I just remove it like so. Okay so to wrap up this video I just want to recap the tips that I gave you. So tip number one you want to start small. You want to use a 16 by 16 canvas or a 32 by 32 canvas and the reason that you want to choose these sizes is because this is going to really force you to focus in on how to represent what you're trying to represent very clearly. This is going to allow you to focus on forms, the individual impact of a single pixel. You're not going to have as much room to go deep into the weeds on detail. You're really going to be, I think, focusing on the high level forms. Like how do I make the viewer think I'm looking at a pumpkin with a very small canvas size and it's going to take you a little bit of practice at first to figure out how to work with such a small canvas size. But Google is your friend. Go to Google images or any other image board and just look at what other pixel artists are doing and learn from them. Study how they're placing their pixels, how they're making the most out of a 16 by 16 canvas size, and try to borrow that and use some of those learnings when you're practicing on your own. And again, you want to use limited color palettes. One bit to start, I would recommend, because then you can draw things like this jack-o'-lantern here and just keep drawing things of all different types. Draw flower pots, draw trees, draw swords, whatever you want to draw, try to draw it using a one bit color palette on a 16 by 16 canvas. And it's going to take practice, but you're going to get the hang of it because the constraints are going to breed creativity. There's only a limited number of ways to make something look good in a 16 by 16 canvas with two colors. So keep trying until you get something that looks good. Okay. Again, contrast, make sure that you've got enough contrast between your pixels when you start using colors. So when you go into using limited color palettes of eight and 16 and 32 colors, make sure that when you're putting pixels adjacent to each other, that there's enough contrast between them. And then if you're going to do shadows and highlights, hue shift the shadows on highlights. Highlights need to be shifted towards yellow and shadows need to be shifted towards purple. And of course, with outlines, avoid jaggies. You can make an element of your art look pointier by creating a hard right angle like so, and you can remove the outline from the bottom of the entity if it's going to be connected to the floor. So again, these are some of the tips that I've picked up as a non-artist to become decent at pixel art. I think you'd agree that these are not the best pixel art jack-o'-lanterns that you've ever seen, but I think you'd also agree that each of these jack-o'-lanterns would not look out of place in a game with a similar theme. So I really hope that helps. I've got actually many more pixel art tips that I've learned, but this video has already gone long enough. So if you'd like to see a part two with some more pixel art tips, let me know and I will go ahead and produce that video. Otherwise, I hope you found it enjoyable. I hope you found it useful and I will see you in the next video. Thanks for watching this video. If you'd like to support my work, you can purchase one of my courses, you can wishlist my upcoming game Gunforged on Steam, and you can sign up for my email list at firebelley.com. The links for all of those things are in the description below.

welcome dummies I'll be your host for this evening and today we will be doing pixel art for the people who don't know how well first of all what is pixel art pixel art is a digital art form that most people use for video games and those pegboard bead things anyway it's an art style that was heavily used in the early days of video games as they were very limited in their color palettes and their file sizes however nowadays has become its own respected medium that a lot of people actually admire and it is also pretty easy to get into because of the size restriction and how many squares you have to work with so it is definitely a go-to style for non artists so you might be asking yourself where can I actually do pixel art well in my opinion the personal best place to get started with pixel art has to be a spray you can use other tools like Photoshop if you have it or even things like if you're on a budget of free but you should see your software as an investment in making your life easier for the long run so you might as well pick something up that is built for pixel art from the get-go all right let's talk sizing if you want to get into pixel art and have no idea where to begin start with an 8 by 8 or 16 by 16 canvas eventually you will be able to do art in a 32 by 32 or 64 by 64 canvas but for now I would honestly say that 16 by 16 is the best resolution to get used to as you can still get in a lot of detail at this level without the headache of getting in over your head and having too much blank space to work with being on the topic of blank space let's move on to a tip I will give you that you need to really soak in to take your art to the next level you've probably heard this a thousand times but this is a tutorial for dummies so I'm approaching this as if you don't know crap hey guys we are having a brief intermission so just stay tuned for the tips but I need to talk about a few things before we move forward so if you guys have been watching my channel then you prob remember this guy if we can smash a hundred likes by the first week of this video coming out yeah well you guys ended up completely killing that goal and getting a huge number of likes on that video so today I'm here to ask you guys to help me out even more I've been putting an insane amount of effort into this YouTube channel lately and I'm not showing any signs of slowing down but it would mean the world to me if everybody right now went down and liked this video because it's gonna help me get this video out to more people so if you guys can drop 300 likes on this video I'll know that you guys really appreciate the content and I will be dropping even more heat and just some insane videos that I thought would never be possible and lastly before we go to the video I just wanted to say that I am having a Q&A video happening right now so if you want to be a part of that go into the description and the first link should be my discord channel so go into the discord Channel and drop your questions in the Q&A Channel I'm answering two questions per person so if you're watching this when this video comes out I'll be collecting submissions until next Tuesday having the video drop a couple days after that that's all for me so back to the video anyway something that is very valuable to understand about pixel art is that you need to make sure that every pixel in your art serves a purpose what do I mean by this well you have an insanely limited amount of pixels to manipulate on your image make sure to really use each and every one to its fullest potential I'll talk throughout the video on how you can actually put this information into practice but for now just remember that you should have little to no pixels in your drawing that are randomly placed so now that we have a basic understanding of what pixel art is let's talk structure when creating an object you want to start by creating a silhouette of your image first block out what the general shape is for your object then work on adding color details and shading afterwards as far as colors go you have quite a lot of rules and things to follow so I'll keep things brief and stick to the important stuff so usually you will have three to four colors permits here so let's break that down you're always going to have a base color that you use to color an image in then usually you'll have a shadow of this color in a highlight of this color sometimes you find images with two base colors a shadow and highlight but it's usually up to the artist and what they find fits their needs to create shadows you will almost always want to take your base color decrease the value decrease the saturation and shift the hue closer to blue the reason behind the hue shift is to bring out the coolness of the shadows color aka making it blue highlights are a bit more difficult but you should get the hang of it with a little bit of practice for creating lighter colors there are two main ways of approaching this first you can do the complete opposite of what you did to make the shadow take your base color bring the value up increase the saturation and shift the hue closer to yellow or instead of increasing the saturation you can also decrease it you might be wondering when to use each method of creating lighter colors well to keep things simple increase the saturation on your highlights when you are creating a lighter color on a solid object this would be for things like clothes and usually things in nature and decrease the saturation when you're working with reflective surfaces so things like shines on metal glass and even things like skin an alternative to this is going to a website called low spec it is a very handy tool to get started in pixel art and they actually already have hundreds of color palettes that you can easily refer to for your own art if you don't understand colors right away a tip I can give you for adding shading is to focus on one main light source for your image most lighting will usually come from either the top left corner or the top right try to be consistent with this and change the lighting source as your image rotates around to keep the lighting consistent in your piece next let's discuss ordering your image when creating outlines you have three main ways of going about it you've got the no outline where your image has no outline the hard outline where your drawing is outlined in one solid color and you have well the overly complex and very time-consuming outline probably not the best name for it but it is by far the hardest outline to pull off basically you're going to take the colors of your image make them darker and add them as a border for your image it might not sound like a ton of work but when you realize that you have to do this for every single image in your game yeah it can get pretty time-consuming last but certainly not least is probably the most useful advice I have for you today because pixel art has very few squares you can actually use it is very easy to see each and every last pixel another artist has used to make up their own art now when I say this I am NOT saying to just grab someone else's art copy it and move on but here's the thing because you have to manually input each pixel on screen it means that no one really has an upper hand on another person whether you use a five thousand dollar drawing tablet or your trackpad every 4x4 cube will look the exact same look at references when you're drawing to see how other people have solved a challenge you are currently working on eventually yes you will be able to make your own art and have your own style but when you're first starting out look at references see how the best sword was made are the most perfect cardboard box start with good habits rather than spending three years just playing around with your art and being proud that you are 100% self-taught through your own imagination that's honestly complete bullsh and there are way too many resources out there now for you to be thinking like this in today's world analyze other people's art and breakdown each pixel placement study what they did right how they went about creating a smooth curved edge where you should set your lighting how something should be positioned for a more interesting picture also look at what they could have improved on take someone else's drawing and fix it you don't call me a wise guy for nothing learning is a great way to learn obviously but teaching is just as important to really drill in those concepts and make them a part of your brains comprehension well that just about wraps up the basics of pixel art and how you the dummy can get started right away [Music] [Applause] I'm going to be putting some insanely useful material in the description of resources that help me shape my art to where it is today and what I continue to use moving forward on my journey I'm by no means an expert in the fields but because of that I find it easier to relate to beginners and help them start their journey in the next episode on tutorials for dummies we will be taking a look into how you can create games quickly and easily from scratch again obviously no experience is necessary so please make sure to attend the next video make sure to subscribe for more content like this soon thank you for all of your continued support here are some comments from the last video that gave me a laugh so make sure and drop some heat in the comments below to possibly get featured in the next one I appreciate and loved every last one of you watching my content but until next time have a great life and stay safe during this time of crisis [Music] you [Music]

I can code neural networks from scratch, but the moment I have to do anything even remotely artistic, like drawing pixel art for my fishing game, my mind just goes blank and I end up producing something that looks like this. But even though I'm closer to autistic than artistic, every day the singularity comes closer. So, I already tried fine-tuning the fusion models, but as Cloud Code is taking over my software engineering job, I thought maybe I could also give it a CLI tool to draw pixel art for my video games so that even if my job goes extinct, I can still keep myself busy being an indie game developer. However, I didn't find a single existing tool that allows you to draw pixel art through your CLI, which is not really surprising as I think you probably have to be a member of the NeoVim subreddit to even remotely think that this makes sense. But no worries, this is 2026 and programming has become pay to win. So, I will just bootstrap myself by implementing a RA loop. The concept of a Ralph loop was invented by George Huntley and he says that this is the essence which actually put me in a loop trying to understand what's so special about putting cloud code in a while statement. So I went and watched this YouTube video which has more resemblance to a twin peaks episode than a programming tutorial. But the essence of a Ralph loop is simple as we are just trying to build software autonomously by first refining a list of independent tasks and then creating a loop which spawns a fresh coding agent instance for each task to prevent context fraud. So I defined a series of tasks to build a CLI pixel art drawing tool in Go. For a brief moment a thought crept through my mind. You could also write it in Rust. This is an interesting way to experiment with Rust. Use Rust. However, after taking my anti-csychotics, those disturbing thoughts quickly passed. I put these tasks in a task.markdown file and I also created a prdarkdown file which uh contained a more general overview about the project and architecture and all the features I wanted to build. Then I created a shell script which contained the RA loop which instructed the agent to first find the highest priority feature from the tasks.mmarkdown file to work on and only work on that feature. He should also write tests to determine if the feature/task is done and when the task is done he needed to check it off into markdown file and make a git commit. I then put my favorite coding agent codeex 5.2 do into YOLO mode with high reasoning. Um, but as I also don't want the agent to delete my root directory to prevent some tasks from failing, I decided to create a development container with Tabian. And I just ran my coding agent in that container. So, the only thing left for me to do was sit back, watch a Psyche K episode while wondering if my CS degree was a mistake after all. However, after only a single hour of raaling, I already blew through my $20 subscription rate limit. So, if you would all just refresh this video maybe five to 10 times and just click on every ad that pops up, you would help me out a lot. However, God had hurt my prayers as a few hours later, OpenAI actually released the Codeex app, which also caused all the rate limits to reset and as a bonus, I got uh twice the usage for the remainder of this video. So I immediately put Ralphie back in his loop and not much later it exclaimed that not only all his tasks but maybe also my career as a software engineer was finished. So I tried it out and the CLI tool could correctly spawn the demon. I could then successfully draw a pixel. The pixel color was also successfully retrieved with the CLI and as a final test a PNG export was also successfully created. Like what the hell? It was actually working. I then went on to draw some beautiful pixel art to verify myself that all the methods were indeed successfully implemented. As I realized that it would also be nice to just have a live um a window that would be updated live while the AI was drawing. I also refined some tasks to visualize this with only an hour later Ralph was once again finished and I could see what was drawn live. So it was time to make the AI use this tool. I know the title says cloud codes, but I have a confession to make as we're actually going to use open code to draw the pixel art. Um, it's just simply more flexible and as all my capital is tied up in Nvidia stock, I can't um really afford the cloud max subscription at the moment. But for real, open code is simply much more flexible for defining custom agents and I could easily create an artist agent which could only use my CLI tool. I gave it some instructions about how the tool worked and some tips that it could for example um create an PNG export of the pixel art readed back in to then get a better idea of what it had drawn. So, the first thing that I asked it to do as I'm so original is um instructed it to draw a dog. And I thought it would draw it pixel per pixel, but instead it just thought for a long time and then fired off a 50 or more set pixel commands in parallel, which kind of surprised me, but the dog was actually pretty decent. As I mainly want to use this for game assets, I also gave it the instruction to draw a character from one of my favorite RPGs of all time, the frog from Chrono Trigger. And although the quality comes nowhere close to that of human artists, it was indeed able to draw a frog that looked like it belonged in a video game. So for those of you who are Clothe code fanboys and probably already reported me for clickbait, this is one for you. as I was also interested in whether maybe Cloat would be better at drawing a pixel art frog. Um, so I leave you to be the judge. Next up, I asked it to draw Goku as if it isn't obvious I'm a man of culture. Um, GPD 5.2 said it was going to draw it in Chibi style, but I think Akira Toriyama will come back from his grave if he knew that AI was drawing Goku renditions like this. Next up, some text, which works surprisingly well, and it seems like the AI has figured out an important truth you should all do. So, now maybe the most important test. Can it actually generate assets for my fishing game? And honestly, the drawings of a boss and a carp are surprisingly good and much better than I could personally draw. So, for me, this experiment was already a huge success as it showed me that drawing relatively simple assets was indeed feasible. As we are also all deep thinkers on this channel, not only interested in building AI to make ourselves insanely rich and retire at the young age of 30 somewhere in the side of Spain, but are also interested in the more philosophical aspects. I asked GPT 5.2 to visualize how it was like to be GPT 5.2. He then went on to draw this abstract pixel art with a core and all kinds of race which according to him visualized his thinking and reasoning threads. Kind of cool to be honest. The funniest pixel art was definitely when I asked it to draw Pikachu and he proceeded to draw this eggshaped creature. But hey, if anyone um if you ask anyone which Pokemon this is, you would probably recognize it. I then thought, what if I just gave it a PNG of a Pikachu sprite as reference and gave it the instruction to first read that PNG and then try to draw it. It somewhat seemed to be able to replicate it, but I wouldn't say it was very usable. I also did this technique by giving it a picture of Eisenberg, which to be honest was a pretty good pixel art rendition. And last but not least, I also asked it to draw the Giga Chat picture because why not? So in short, I find this extremely impressive as not only did the AI generate a tool to draw pixel art by itself, but it then also succeeded in using this tool to draw pixel art, which is actually usable for simple video game assets. Like, isn't this exactly what a few years ago people would have claimed to be AGI? Anyways, even though I'm literally following the AI field from up close and it's also even my job to work with it, I feel like we're still going insanely fast and we're just beginning to discover all the possibilities. By the way, if you also want to make your favorite agent tropics, the GitP of the CLI tool is all and all the code is all in the description. So, I hope you enjoyed it and I see you in the next